

---

# SOFTWARE REQUIREMENT SPECIFICATION

for

## LKFORUM WEBSITE

Version 1.0

Prepared by : Vo An Khoi (23520790)

Submitted to : Nguyen Trinh Dong  
Lecturer

May 24, 2026

# Contents

|          |                                                     |          |
|----------|-----------------------------------------------------|----------|
| <b>1</b> | <b>Introduction</b>                                 | <b>3</b> |
| 1.1      | Purpose . . . . .                                   | 3        |
| 1.2      | Intended Audience and Reading Suggestions . . . . . | 3        |
| 1.3      | Project Scope . . . . .                             | 3        |
| 1.4      | References . . . . .                                | 3        |
| <b>2</b> | <b>Overall Description</b>                          | <b>4</b> |
| 2.1      | Product Perspective . . . . .                       | 4        |
| 2.2      | Operating Environment . . . . .                     | 4        |
| 2.3      | Design and Implementation Constraints . . . . .     | 5        |
| 2.4      | Assumptions and Dependencies . . . . .              | 5        |
| <b>3</b> | <b>System Features</b>                              | <b>6</b> |
| 3.1      | Use Case Description . . . . .                      | 6        |
| 3.1.1    | UC-01 - Sign Up . . . . .                           | 6        |
| 3.1.2    | UC-02 - Sign In . . . . .                           | 9        |
| 3.1.3    | UC-03 - Forgot Password . . . . .                   | 15       |
| 3.1.4    | UC-04 - Manage Profile . . . . .                    | 19       |
| 3.1.5    | UC-05 - Change Settings . . . . .                   | 21       |
| 3.1.6    | UC-06 - View Activity History . . . . .             | 24       |
| 3.1.7    | UC-07 - Follow / Unfollow User . . . . .            | 26       |
| 3.1.8    | UC-08 - Browse Post . . . . .                       | 28       |
| 3.1.9    | UC-09 - Search Content . . . . .                    | 30       |
| 3.1.10   | UC-10 - Create Post . . . . .                       | 32       |
| 3.1.11   | UC-11 - Edit Post . . . . .                         | 36       |
| 3.1.12   | UC-12 - Delete Post . . . . .                       | 38       |
| 3.1.13   | UC-13 - Post Comment . . . . .                      | 41       |
| 3.1.14   | UC-14 - Reply to Comment . . . . .                  | 44       |
| 3.1.15   | UC-15 - Vote on Content . . . . .                   | 47       |
| 3.1.16   | UC-16 - Participate in Poll . . . . .               | 51       |
| 3.1.17   | UC-17 - Save Post . . . . .                         | 53       |
| 3.1.18   | UC-18 - Hide Post . . . . .                         | 55       |
| 3.1.19   | UC-19 - Manage Drafts . . . . .                     | 58       |
| 3.1.20   | UC-20 - Private Messaging . . . . .                 | 60       |
| 3.1.21   | UC-21 - Manage Notifications . . . . .              | 64       |
| 3.1.22   | UC-22 - Report Violation . . . . .                  | 66       |
| 3.1.23   | UC-23 - Create Community . . . . .                  | 69       |
| 3.1.24   | UC-24 - Approve / Reject Post . . . . .             | 71       |
| 3.1.25   | UC-25 - Ban / Mute User . . . . .                   | 74       |
| 3.1.26   | UC-26 - Configure Community . . . . .               | 77       |
| 3.1.27   | UC-27 - Assign Roles . . . . .                      | 80       |
| 3.1.28   | UC-28 - Handle Global Reports . . . . .             | 83       |
| 3.1.29   | UC-29 - Manage Platform . . . . .                   | 86       |

|                                                       |            |
|-------------------------------------------------------|------------|
| 3.1.30 UC-30 - View Analytics . . . . .               | 89         |
| 3.2 Data Requirements . . . . .                       | 92         |
| 3.2.1 Logical Data Model . . . . .                    | 92         |
| 3.2.2 Data Dictionary . . . . .                       | 93         |
| 3.3 Data Integrity, Retention, and Disposal . . . . . | 111        |
| <b>4 Other Nonfunctional Requirements</b>             | <b>112</b> |
| 4.1 Performance Requirements . . . . .                | 112        |
| 4.2 Security Requirements . . . . .                   | 112        |
| 4.3 Software Quality Attributes . . . . .             | 113        |
| <b>5 Other Requirements</b>                           | <b>114</b> |
| <b>6 Appendix</b>                                     | <b>115</b> |
| 6.1 Messages . . . . .                                | 115        |
| 6.2 Error Codes . . . . .                             | 118        |

# **1 Introduction**

## **1.1 Purpose**

This SRS describes the functional and nonfunctional requirements for software release 1.0 of the Community Social Network Website (LKForum). This document is intended to be used by the members of the project team who will implement and verify the correct functioning of the system. Unless otherwise noted, all requirements specified here are committed for release 1.0

## **1.2 Intended Audience and Reading Suggestions**

This SRS is for developers, project managers, users and testers. Further the discussion will provide all the internal, external, functional and also non-functional informations about "LKFORUM WEBSITE".

## **1.3 Project Scope**

This SRS describes the functional and nonfunctional requirements for software release 1.0 of the Community Social Network Website (LKForum). This document is intended to be used by the members of the project team who will implement and verify the correct functioning of the system. Unless otherwise noted, all requirements specified here are committed for release 1.0

## **1.4 References**

## 2 Overall Description

### 2.1 Product Perspective

Community Social Network LKForum is a new website that provides a comprehensive, modern solution for building and managing online communities. The context diagram in Figure 1 illustrates the external entities and system interfaces for release 1.0. The system is expected to evolve over several releases, ultimately connecting to third-party social media APIs for authentication (Google, Facebook) and expanding to include advanced content monetization tools and mobile-native applications.

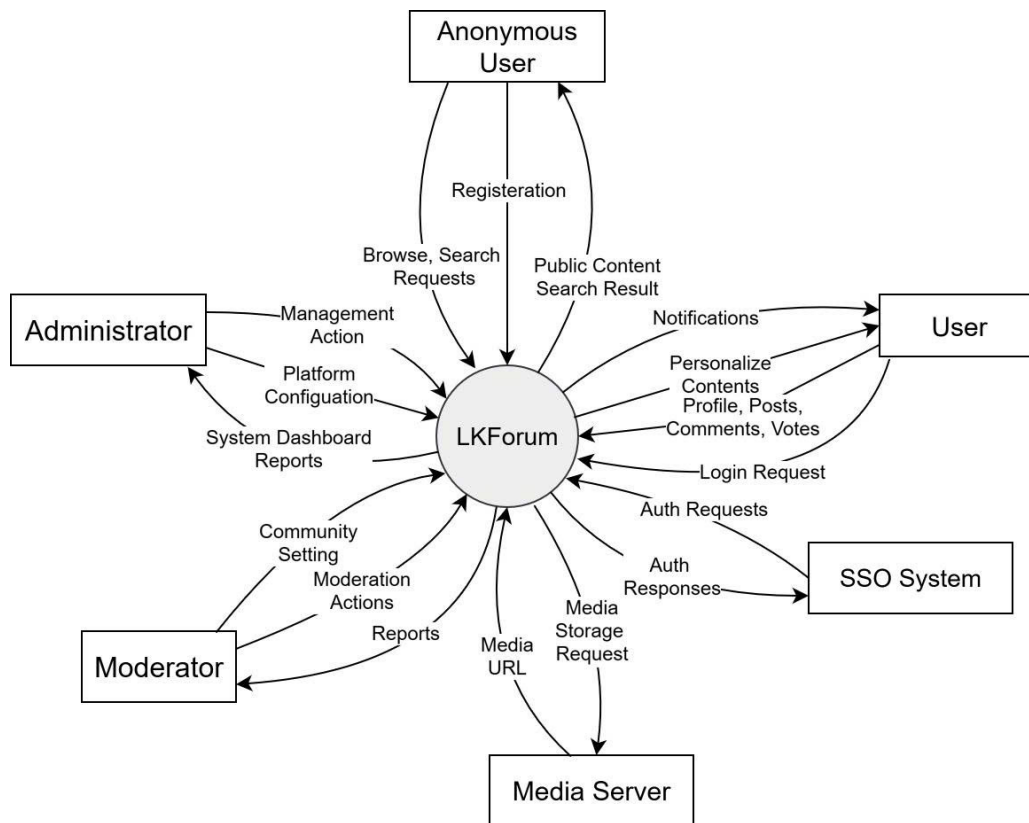


Figure 2.1: Context diagram for release 1.0 of LKForum

### 2.2 Operating Environment

**OE-1:** LKForum must operate correctly on modern web browsers including: Google Chrome (latest version), Mozilla Firefox (latest version), Apple Safari (latest version), and Microsoft Edge (latest version).

**OE-2:** Users can access LKForum via the Internet from desktop computers, laptops, and tablets. The interface must have responsive design to be compatible with various screen sizes.

## 2.3 Design and Implementation Constraints

- CO-1:** The user interface (front-end) source code must be developed using the Svelte framework.
- CO-2:** The system must use a standard MongoDB NoSQL database with a back-end developed using the Go framework.
- CO-3:** All HTML code must adhere to the HTML 5.0 standard.

## 2.4 Assumptions and Dependencies

- AS-1:** It is assumed that users have a stable Internet connection to access and use the platform's features.
- DE-1:** The social media login feature (Google, Facebook) depends on the APIs and policies of these third-party service providers. Any changes on their part may affect the operation of this feature.

## 3 System Features

The LKForum website is being built using the Go programming language for the back-end, Svelte with Vite for the front-end, MongoDB for primary data storage, and Redis for caching and session management.

Back-End – Go with the Gin web framework.

Front-End – Svelte with Vite.

Database – MongoDB.

Cache and Session Store – Redis.

### 3.1 Use Case Description

#### 3.1.1 UC-01 - Sign Up

|                       |                                                                                                     |
|-----------------------|-----------------------------------------------------------------------------------------------------|
| <b>Name</b>           | Sign Up                                                                                             |
| <b>Description</b>    | This use case describes how a new user creates an account and verifies the email address using OTP. |
| <b>Actor</b>          | User                                                                                                |
| <b>Trigger</b>        | When the user clicks the “Sign Up” button.                                                          |
| <b>Pre-condition</b>  | The user is not logged in to the system.<br>The user is on the sign up page.                        |
| <b>Post-condition</b> | The user account is created and verified.<br>The user is authenticated and can access the system.   |

Table 3.1: Use Case Specification - Sign Up

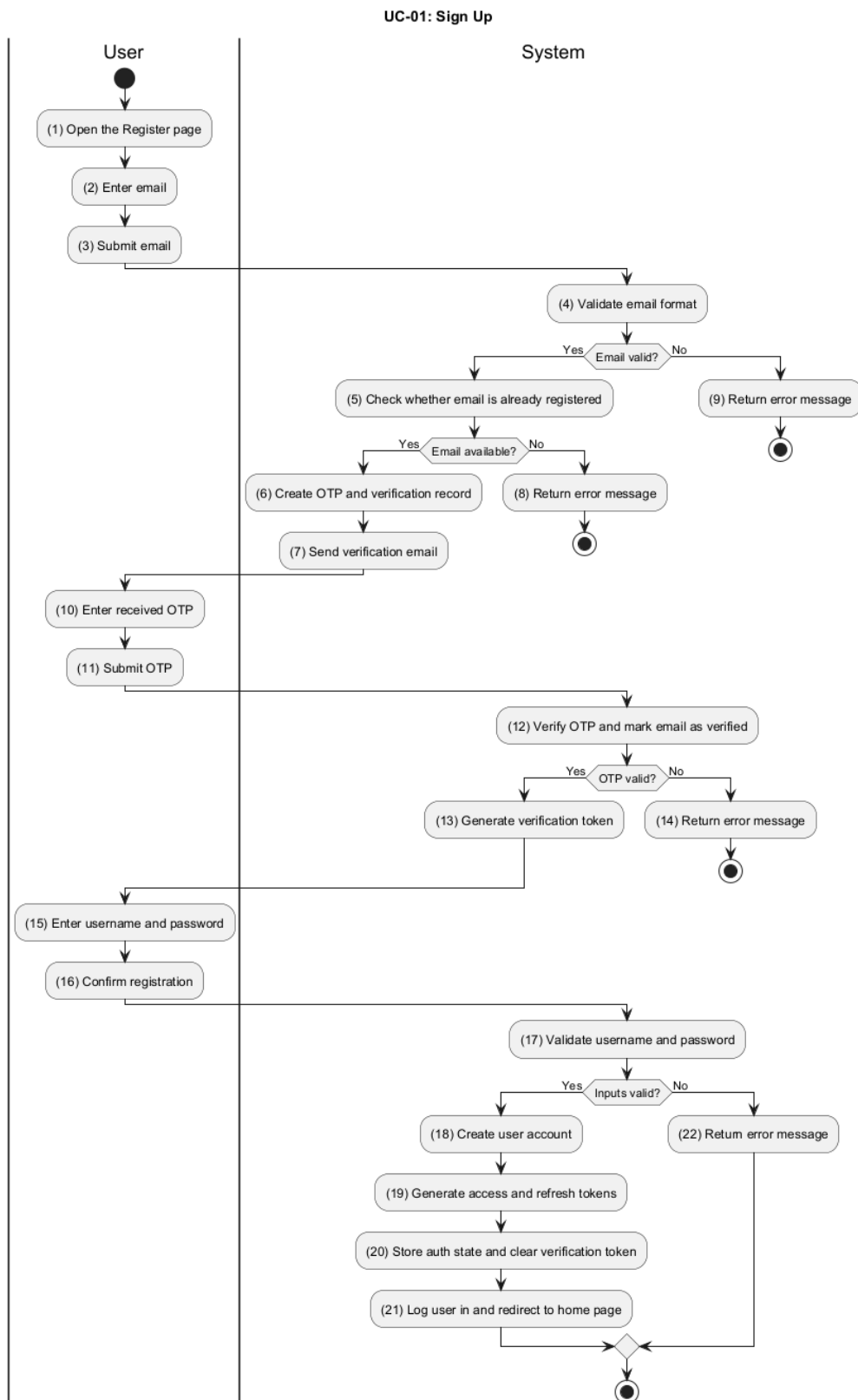


Figure 3.1: Activity Diagram for UC-01 Sign Up



| ID         | Activ-<br>ity | Rule Type        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------|---------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC01-01 | (4)           | Validation Rules | <b>Validate Email Format:</b> <ul style="list-style-type: none"> <li>The system checks that <b>email</b> is present and valid.</li> <li>If the <b>email</b> is missing or invalid, the system returns a bad-request error.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                 |
| BR-UC01-02 | (5)           | Constraint Rules | <b>Check Email Availability and Prepare Verification Record:</b> <ul style="list-style-type: none"> <li>The system checks whether the <b>email</b> already exists in the user repository.</li> <li>If the <b>email</b> already exists, the system returns an email-exists error.</li> </ul>                                                                                                                                                                                                                                                                                                                                                           |
| BR-UC01-03 | (7)           | Message Rules    | <b>Send Verification Email:</b> <ul style="list-style-type: none"> <li>Successful verification-code delivery shall use MSG-1 or MSG-9; resending shall use MSG-2 or MSG-10.</li> <li>OTP validation failures shall return ERR-17 or ERR-18.</li> <li>Completed verification and registration shall use MSG-4 or MSG-12; invalid registration data shall return ERR-30, ERR-21, ERR-22, or ERR-23.</li> </ul>                                                                                                                                                                                                                                          |
| BR-UC01-04 | (12)          | Validation Rules | <b>Verify OTP and Issue Token:</b> <ul style="list-style-type: none"> <li>The system checks that <b>email</b> and <b>otp</b> are present.</li> <li>If no verification record exists, the system returns an invalid <b>OTP</b> error.</li> <li>If the verification record is already marked verified, the system returns an email-already-verified error.</li> <li>If the <b>OTP</b> does not match or has expired, the system returns an invalid or expired <b>OTP</b> error.</li> <li>Otherwise, the system marks the verification record as verified and generates a verification token from the verified <b>email</b> and <b>nonce</b>.</li> </ul> |

Table 3.2: Business Rules for UC-01 - Sign Up

| ID         | Activ-<br>ity | Rule Type        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------|---------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC01-05 | (17)          | Validation Rules | <b>Validate Registration Fields:</b> <ul style="list-style-type: none"> <li>The system checks that <b>username</b> is between 3 and 20 characters.</li> <li>The system checks that <b>password</b> is at least 8 characters long and strong enough.</li> </ul>                                                                                                                                                                                                                                                                        |
| BR-UC01-06 | (18-20)       | Workflow Rules   | <b>Create Account and Complete Login:</b> <ul style="list-style-type: none"> <li>The system hashes the <b>password</b> with <i>GenerateFromPassword()</i> before storing it.</li> <li>The system checks that the <b>username</b> is still available and that the <b>email</b> is still not registered.</li> <li>The system creates the new local user with default settings, deletes the verification record, invalidates the <b>username</b> cache, and generates access and refresh tokens using <i>GenerateToken()</i>.</li> </ul> |
| BR-UC01-07 | (21)          | Redirect Rules   | <b>Redirect After Registration:</b> <ul style="list-style-type: none"> <li>The system logs the user in and redirects to the home page.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                     |

Table 3.2: Business Rules for UC-01 - Sign Up

### 3.1.2 UC-02 - Sign In

|                       |                                                                                                             |
|-----------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Name</b>           | Sign In                                                                                                     |
| <b>Description</b>    | This use case describes how a registered user logs into the system using local credentials or Google OAuth. |
| <b>Actor</b>          | User                                                                                                        |
| <b>Trigger</b>        | When the user clicks the “Sign In” button.                                                                  |
| <b>Pre-condition</b>  | The user is not logged in to the system.<br>The user is on the sign in page.                                |
| <b>Post-condition</b> | The user is authenticated.<br>The user is redirected to the appropriate page.                               |

Table 3.3: Use Case Specification - Sign In

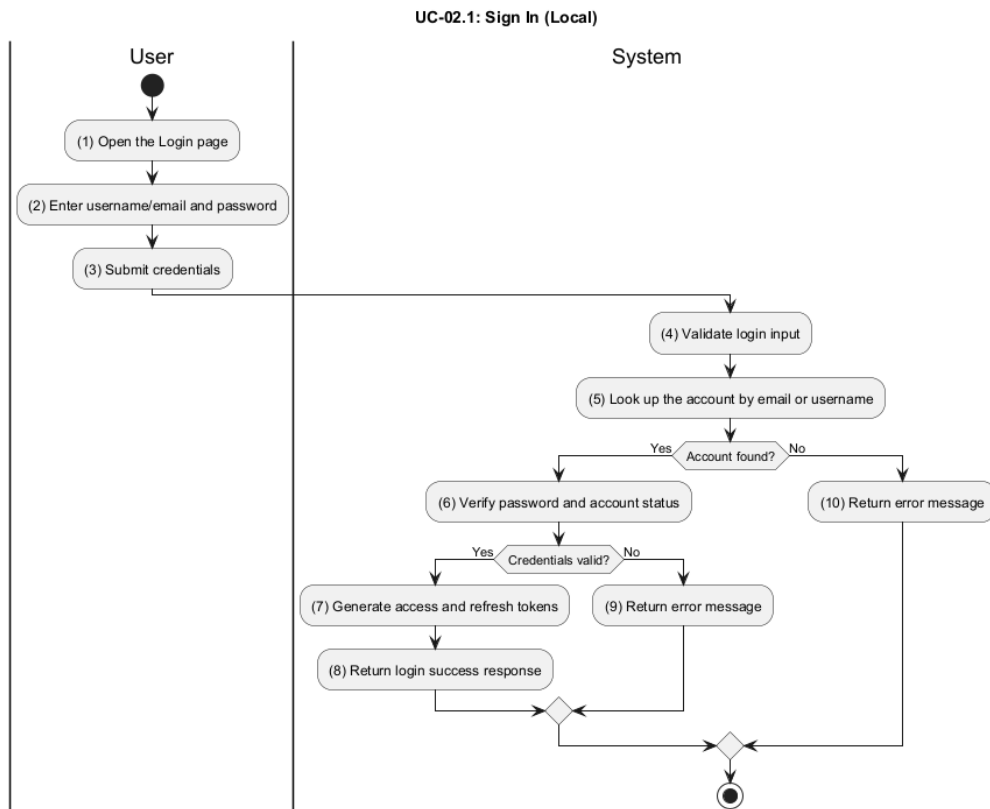


Figure 3.2: Activity Diagram for UC-02 Local Sign In

| ID         | Activ-ity | Rule Type        | Description                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------|-----------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC02-01 | (4)       | Validation Rules | <b>Validate Login Input:</b> <ul style="list-style-type: none"> <li>The controller checks the JSON body for <b>identifier</b> and <b>password</b> in <b>UserLoginRequest</b>.</li> <li>If binding fails or either field is missing, the controller returns a bad request response.</li> <li>The controller does not validate the <b>identifier</b> format here; the service decides whether to search by email or username.</li> </ul> |

Table 3.4: Business Rules for UC-02.1 - Sign In (Local)

| ID         | Activ-<br>ity | Rule Type           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------|---------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC02-02 | (5)           | Workflow Rules      | <b>Look Up Account:</b> <ul style="list-style-type: none"> <li>The service checks whether <b>identifier</b> is an email using <i>isEmail()</i>.</li> <li>The service looks up the account by email or username using the user repository.</li> <li>If no account exists, the service returns an invalid credentials error.</li> </ul>                                                                                                                                                                                                                                                           |
| BR-UC02-03 | (6)           | Authorization Rules | <b>Verify Password and Account Status:</b> <ul style="list-style-type: none"> <li>The system checks that the account uses the local provider.</li> <li>The service compares the stored password with <b>password</b> using <i>bcrypt.CompareHashAndPassword()</i>.</li> <li>If the password hash is empty or the comparison fails, the service returns an invalid credentials error.</li> <li>If <b>IsVerified</b> is false, the service returns an email-not-verified error.</li> <li>If the user is banned and the ban has not expired, the service returns a user-inactive error.</li> </ul> |
| BR-UC02-04 | (7–10)        | Message Rules       | <b>Return Authentication Response:</b> <ul style="list-style-type: none"> <li>Successful local sign-in shall use MSG-3.</li> <li>Failed local sign-in shall return ERR-1.</li> <li>Google sign-in and setup failures shall return ERR-2, ERR-19, or ERR-30 as applicable.</li> </ul>                                                                                                                                                                                                                                                                                                            |

Table 3.4: Business Rules for UC-02.1 - Sign In (Local)

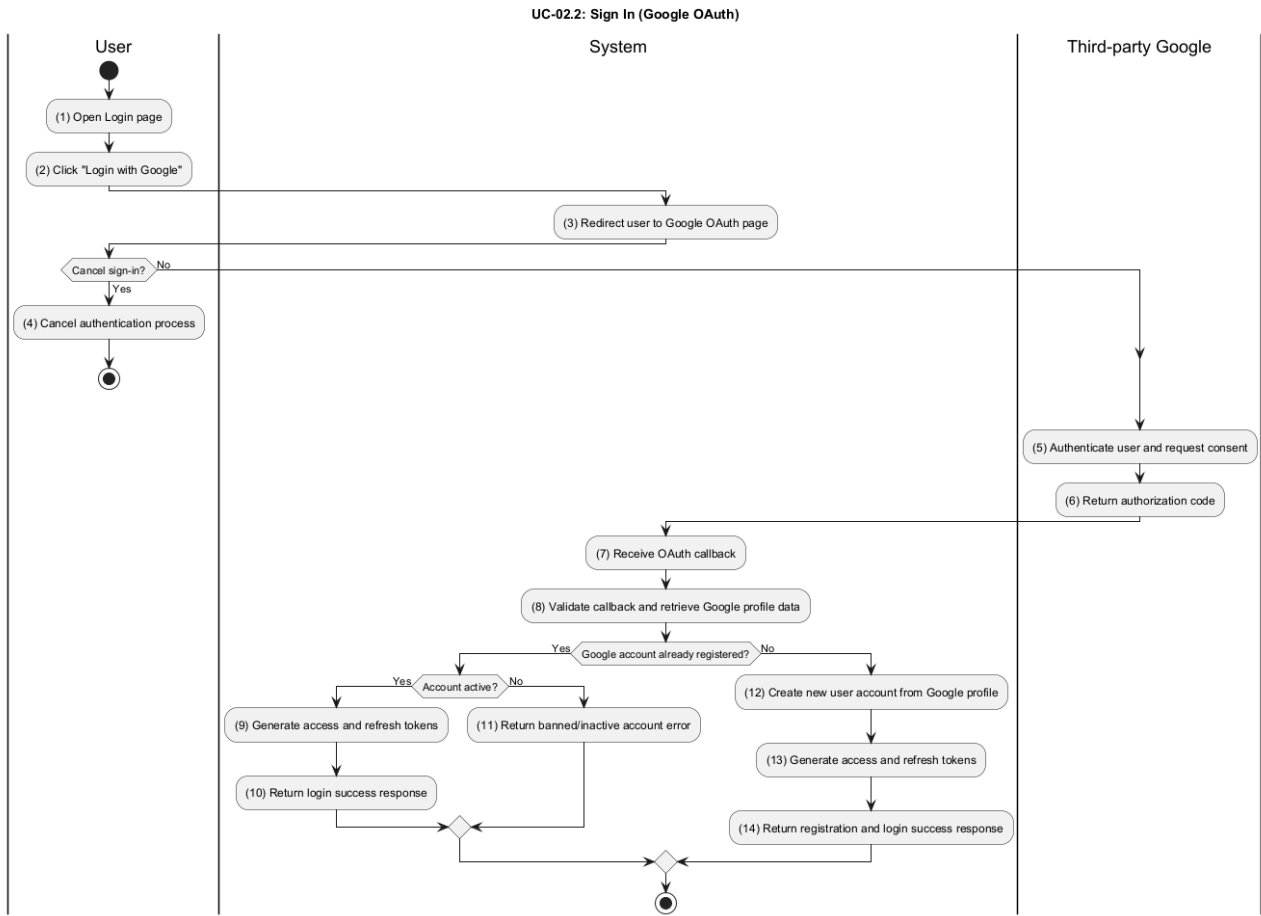


Figure 3.3: Activity Diagram for UC-02 Google Sign In

| ID         | Activ-ity | Rule Type      | Description                                                                                                                                                                                                                              |
|------------|-----------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC02-05 | (3)       | Redirect Rules | <b>Start Google OAuth Flow:</b> <ul style="list-style-type: none"> <li>The system redirects the user to Google OAuth using a generated state value.</li> <li>Google authenticates the user and returns an authorization code.</li> </ul> |

Table 3.5: Business Rules for UC-02.2 - Sign In (Google OAuth)

| ID         | Activ-ity | Rule Type           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------|-----------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC02-06 | (8)       | Validation Rules    | <p><b>Process Google Callback:</b></p> <ul style="list-style-type: none"> <li>• The controller checks that the callback contains <b>code</b>.</li> <li>• If <b>code</b> is missing, the controller redirects to the frontend error page with a missing-auth-code message.</li> <li>• The service exchanges <b>code</b> for Google profile data using <i><b>GetGoogleUserInfo()</b></i>.</li> <li>• If Google rejects the authorization code or profile lookup fails, the controller redirects to the frontend error page with the returned error message.</li> </ul>                                                      |
| BR-UC02-07 | (9–11)    | Authorization Rules | <p><b>Login or Setup Decision:</b></p> <ul style="list-style-type: none"> <li>• If an account exists and <b>provider</b> is Google, the service generates access and refresh tokens using <i><b>GenerateToken()</b></i>.</li> <li>• If the account belongs to a different login method, the service returns a login-method-mismatch error.</li> <li>• If the account is banned and the ban has not expired, the service returns a user-inactive error.</li> <li>• If no Google account exists, the service creates a setup token using <i><b>CreateSetupToken()</b></i> and returns the setup-required result.</li> </ul> |

Table 3.5: Business Rules for UC-02.2 - Sign In (Google OAuth) (Continued)

| ID         | Activ-<br>ity | Rule Type        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------|---------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC02-08 | (12)          | Validation Rules | <b>Validate Google Setup Input:</b> <ul style="list-style-type: none"> <li>The controller checks the JSON body for <b>setup_token</b> and <b>username</b> in <code>CompleteGoogleSetupRequest</code>.</li> <li>If binding fails or <b>setup_token</b> is missing, the controller returns a bad request response.</li> <li>The service validates that <b>username</b> is at least 3 characters and no more than 20 characters.</li> <li>If <b>username</b> already exists, the service returns a username-exists error.</li> <li>If the Google email is already taken, the service returns an email-exists error.</li> </ul> |
| BR-UC02-09 | (13)          | Workflow Rules   | <b>Create Google Account:</b> <ul style="list-style-type: none"> <li>The system parses <b>setup_token</b> to recover the Google claims.</li> <li>The system creates the user with <b>provider</b> set to Google, stores <b>provider_id</b>, applies default settings, and copies the Google avatar URL when available.</li> <li>The system generates access and refresh tokens after the user is created.</li> </ul>                                                                                                                                                                                                        |
| BR-UC02-10 | (14)          | Message Rules    | <b>Return Setup Success:</b> <ul style="list-style-type: none"> <li>Successful Google-account completion shall use MSG-3 after the account and tokens are issued.</li> <li>Validation failures during setup shall return ERR-30, ERR-21, ERR-22, or ERR-23 when relevant.</li> </ul>                                                                                                                                                                                                                                                                                                                                        |

Table 3.5: Business Rules for UC-02.2 - Sign In (Google OAuth)

### 3.1.3 UC-03 - Forgot Password

|                       |                                                                                                                           |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>           | Forgot Password                                                                                                           |
| <b>Description</b>    | This use case describes how a user resets a forgotten password through email verification and OTP.                        |
| <b>Actor</b>          | User                                                                                                                      |
| <b>Trigger</b>        | When the user clicks the “Forgot Password” link.                                                                          |
| <b>Pre-condition</b>  | The user has an existing account and can access the registered email address.<br>The user is not logged in to the system. |
| <b>Post-condition</b> | The user password is reset successfully.<br>The user can sign in again with the new password.                             |

Table 3.6: Use Case Specification - Forgot Password

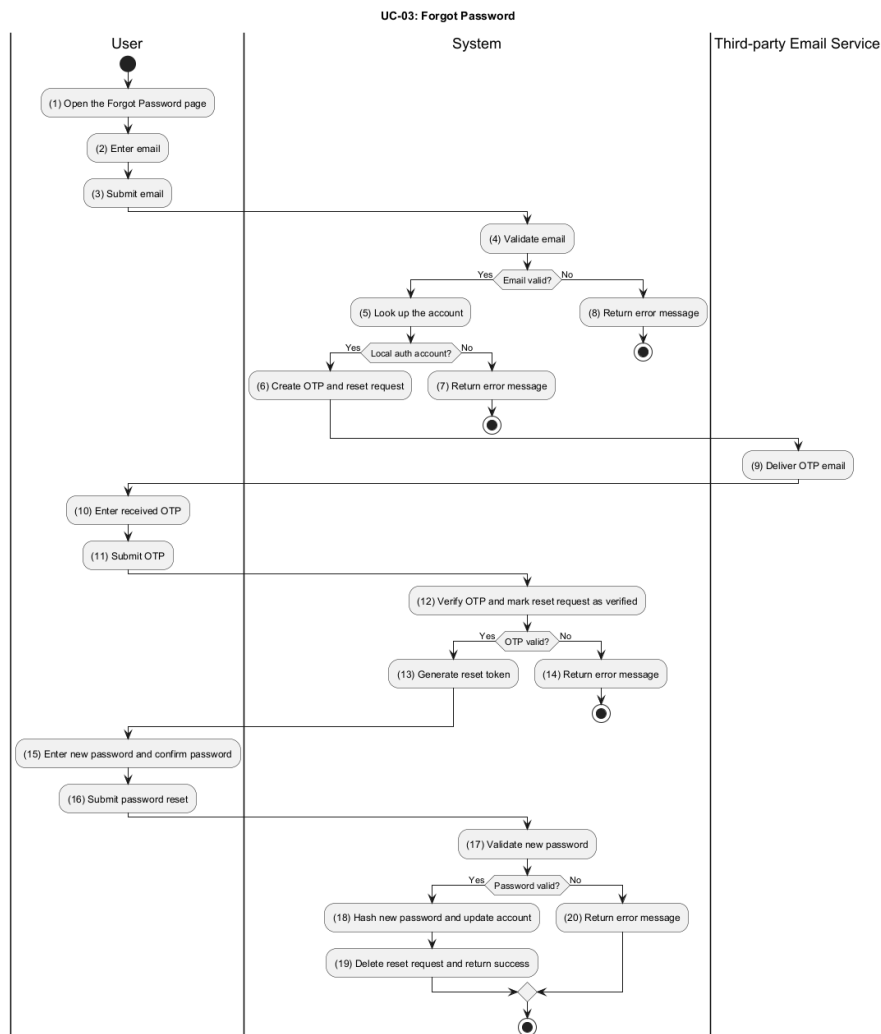


Figure 3.4: Activity Diagram for UC-03



| ID         | Activ-<br>ity | Rule Type           | Description                                                                                                                                                                                                                                                                                                                                                                                               |
|------------|---------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC03-01 | (4)           | Validation Rules    | <b>Validate Reset Request:</b> <ul style="list-style-type: none"> <li>The controller checks <b>email</b> in <code>ForgotPasswordRequest</code>.</li> <li>If binding fails or <b>email</b> is missing or invalid, the controller returns a bad request response.</li> </ul>                                                                                                                                |
| BR-UC03-02 | (5)           | Authorization Rules | <b>Validate Account Ownership:</b> <ul style="list-style-type: none"> <li>The service looks up the account by <b>email</b> using the user repository.</li> <li>If the account does not exist, the service returns an email-not-registered error.</li> <li>If the account exists but the <b>provider</b> is not local, the service returns a login-method-mismatch error.</li> </ul>                       |
| BR-UC03-03 | (6)           | Workflow Rules      | <b>Create Password Reset Request:</b> <ul style="list-style-type: none"> <li>The system deletes any existing reset record for the same <b>email</b>.</li> <li>The system generates a 6-digit <b>OTP</b>, a <b>nonce</b>, and an expiry timestamp using <i>generateOTP()</i> and <i>generateNonce()</i>.</li> <li>The system stores a new password-reset record with <b>IsVerified = false</b>.</li> </ul> |
| BR-UC03-04 | (7)           | Message Rules       | <b>Return Reset Request Result:</b> <ul style="list-style-type: none"> <li>Successful reset-code delivery shall use MSG-6.</li> <li>If the account is not local, the service shall return ERR-19; unknown local accounts shall return ERR-20.</li> </ul>                                                                                                                                                  |

Table 3.7: Business Rules for UC-03 - Forgot Password

| ID         | Activ-<br>ity | Rule Type        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------|---------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC03-05 | (12)          | Validation Rules | <b>Validate Reset OTP:</b> <ul style="list-style-type: none"> <li>The controller checks <b>email</b> and <b>otp</b> in <code>VerifyResetPasswordOTPRequest</code>.</li> <li>If binding fails, the controller returns a bad request response.</li> <li>If no reset record exists for the <b>email</b>, the service returns an invalid <b>OTP</b> error.</li> <li>If the reset record is already verified, the service returns an email-already-verified error.</li> <li>If the <b>OTP</b> does not match or has expired, the service returns an invalid or expired <b>OTP</b> error.</li> </ul> |
| BR-UC03-06 | (12)          | Workflow Rules   | <b>Verify OTP and Generate Reset Token:</b> <ul style="list-style-type: none"> <li>The service marks the reset record as verified.</li> <li>The service generates a reset token from the verified <b>email</b> and <b>nonce</b> using <code>CreateVerificationToken()</code>.</li> </ul>                                                                                                                                                                                                                                                                                                       |
| BR-UC03-07 | (14)          | Message Rules    | <b>Return OTP Error:</b> <ul style="list-style-type: none"> <li>Successful reset-OTP verification shall use MSG-7.</li> <li>Invalid or expired reset OTP values shall return ERR-17 or ERR-18.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                      |

Table 3.7: Business Rules for UC-03 - Forgot Password (Continued)

| ID         | Activ-<br>ity | Rule Type        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------|---------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC03-08 | (17)          | Validation Rules | <b>Validate New Password:</b> <ul style="list-style-type: none"> <li>The controller checks <b>reset_token</b> and <b>new_password</b> in <b>ResetPasswordRequest</b>.</li> <li>If binding fails or <b>reset_token</b> is missing, the controller returns a bad request response.</li> <li>The DTO binding requires <b>new_password</b> to be present and at least 6 characters long.</li> <li>The service requires <b>new_password</b> to be at least 8 characters long.</li> <li>The service requires <b>new_password</b> to contain uppercase, lowercase, numeric, and special characters.</li> </ul> |
| BR-UC03-09 | (18)          | Workflow Rules   | <b>Hash and Update Password:</b> <ul style="list-style-type: none"> <li>The service parses the reset token and verifies the stored <b>nonce</b>.</li> <li>The service looks up the user by <b>email</b>.</li> <li>The service hashes <b>new_password</b> with <i><b>GenerateFromPassword()</b></i>.</li> <li>The service updates the user password in the repository.</li> </ul>                                                                                                                                                                                                                        |
| BR-UC03-10 | (19)          | Workflow Rules   | <b>Clean Up Reset Record:</b> <ul style="list-style-type: none"> <li>The service deletes the reset record after a successful password change.</li> <li>The controller returns a success response telling the user they can log in with the new password.</li> </ul>                                                                                                                                                                                                                                                                                                                                     |
| BR-UC03-11 | (20)          | Message Rules    | <b>Return Password Error:</b> <ul style="list-style-type: none"> <li>Successful password reset shall use MSG-8 or MSG-13.</li> <li>Invalid replacement passwords shall return ERR-22 or ERR-23.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                              |

Table 3.7: Business Rules for UC-03 - Forgot Password

### 3.1.4 UC-04 - Manage Profile

|                       |                                                                                                      |
|-----------------------|------------------------------------------------------------------------------------------------------|
| <b>Name</b>           | Manage Profile                                                                                       |
| <b>Description</b>    | This use case describes how a user updates profile information such as bio, avatar, and cover image. |
| <b>Actor</b>          | User                                                                                                 |
| <b>Trigger</b>        | When the user opens the profile edit page.                                                           |
| <b>Pre-condition</b>  | The user is logged in to the system.<br>The user has permission to edit the profile.                 |
| <b>Post-condition</b> | The profile information is updated and stored.<br>The new profile data is visible across the system. |

Table 3.8: Use Case Specification - Manage Profile

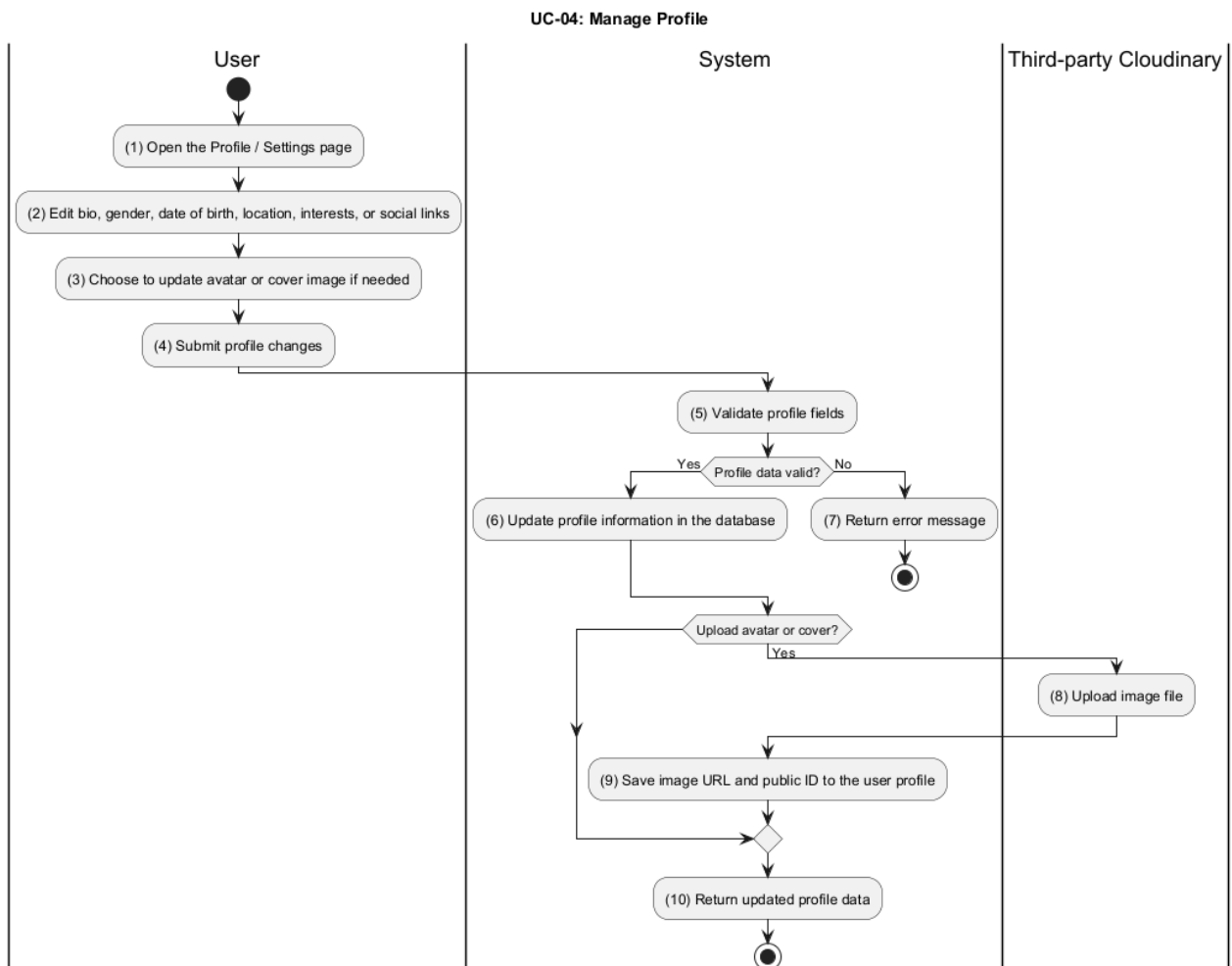


Figure 3.5: Activity Diagram for UC-04

| ID         | Activ-ity | Rule Type        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------|-----------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC04-01 | (5)       | Validation Rules | <b>Validate Profile Fields:</b> <ul style="list-style-type: none"> <li>The controller checks the JSON body for <b>UserProfileUpdateRequest</b>.</li> <li>If binding fails, the controller returns a bad request response.</li> <li><b>bio</b> is limited by DTO validation to 500 characters.</li> <li><b>social_links.website</b> must be a valid URL when provided.</li> </ul>                                                                                                                                                                                                                                                                                                            |
| BR-UC04-02 | (6)       | Workflow Rules   | <b>Update Profile Data:</b> <ul style="list-style-type: none"> <li>The service loads the current user and updates <b>RoleContent.AsUser</b>.</li> <li>Empty string values clear the corresponding profile field.</li> <li>The service accepts updates for <b>gender</b>, <b>date_of_birth</b>, <b>location</b>, <b>interests</b>, and <b>social_links</b>.</li> </ul>                                                                                                                                                                                                                                                                                                                       |
| BR-UC04-03 | (7)       | Validation Rules | <b>Reject Invalid Profile Values:</b> <ul style="list-style-type: none"> <li>If <b>gender</b> is not valid, the service returns an invalid gender error.</li> <li>If <b>date_of_birth</b> cannot be parsed as 2006-01-02, the service returns an invalid date format error.</li> <li>If the calculated age is below 13, the service returns an age-too-young error.</li> <li>If the calculated age is above 150, the service returns an invalid birth date error.</li> <li>If <b>location</b> is not a valid province, the service returns an invalid province error.</li> <li>If more than 10 interests are submitted or any interest is invalid, the service returns an error.</li> </ul> |

Table 3.9: Business Rules for UC-04 - Manage Profile

| ID         | Activ-ity | Rule Type      | Description                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------|-----------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC04-04 | (8)       | Workflow Rules | <b>Upload Profile Image:</b> <ul style="list-style-type: none"> <li>When the user uploads an avatar or cover image, the controller sends the file to <i>cloudinary.UploadImages()</i>.</li> <li>The service stores the returned image <b>URL</b> and <b>PublicID</b> in the user profile.</li> <li>If an older image exists, the service deletes the old asset asynchronously using <i>cloudinary.Delete()</i>.</li> </ul> |
| BR-UC04-05 | (9)       | Workflow Rules | <b>Save Image Metadata:</b> <ul style="list-style-type: none"> <li>The service updates <b>Avatar</b> or <b>Cover</b> with the new image metadata.</li> <li>The service persists the updated user profile in the repository.</li> </ul>                                                                                                                                                                                     |
| BR-UC04-06 | (10)      | Message Rules  | <b>Return Updated Profile:</b> <ul style="list-style-type: none"> <li>Successful profile and avatar actions shall use MSG-48, MSG-53, MSG-55, and the confirmation prompt MSG-57.</li> <li>Failed profile updates shall use MSG-49, MSG-54, or MSG-56.</li> <li>Validation failures shall return ERR-28, ERR-32, ERR-33, ERR-35, ERR-36, ERR-37, or ERR-38.</li> </ul>                                                     |

Table 3.9: Business Rules for UC-04 - Manage Profile

### 3.1.5 UC-05 - Change Settings

|                       |                                                                                                           |
|-----------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Name</b>           | Change Settings                                                                                           |
| <b>Description</b>    | This use case describes how a user changes account preferences such as privacy, notifications, and theme. |
| <b>Actor</b>          | User                                                                                                      |
| <b>Trigger</b>        | When the user opens the settings page.                                                                    |
| <b>Pre-condition</b>  | The user is logged in to the system.<br>The settings page is available to the user.                       |
| <b>Post-condition</b> | The selected preferences are saved.<br>The account settings are updated for future sessions.              |

Table 3.10: Use Case Specification - Change Settings

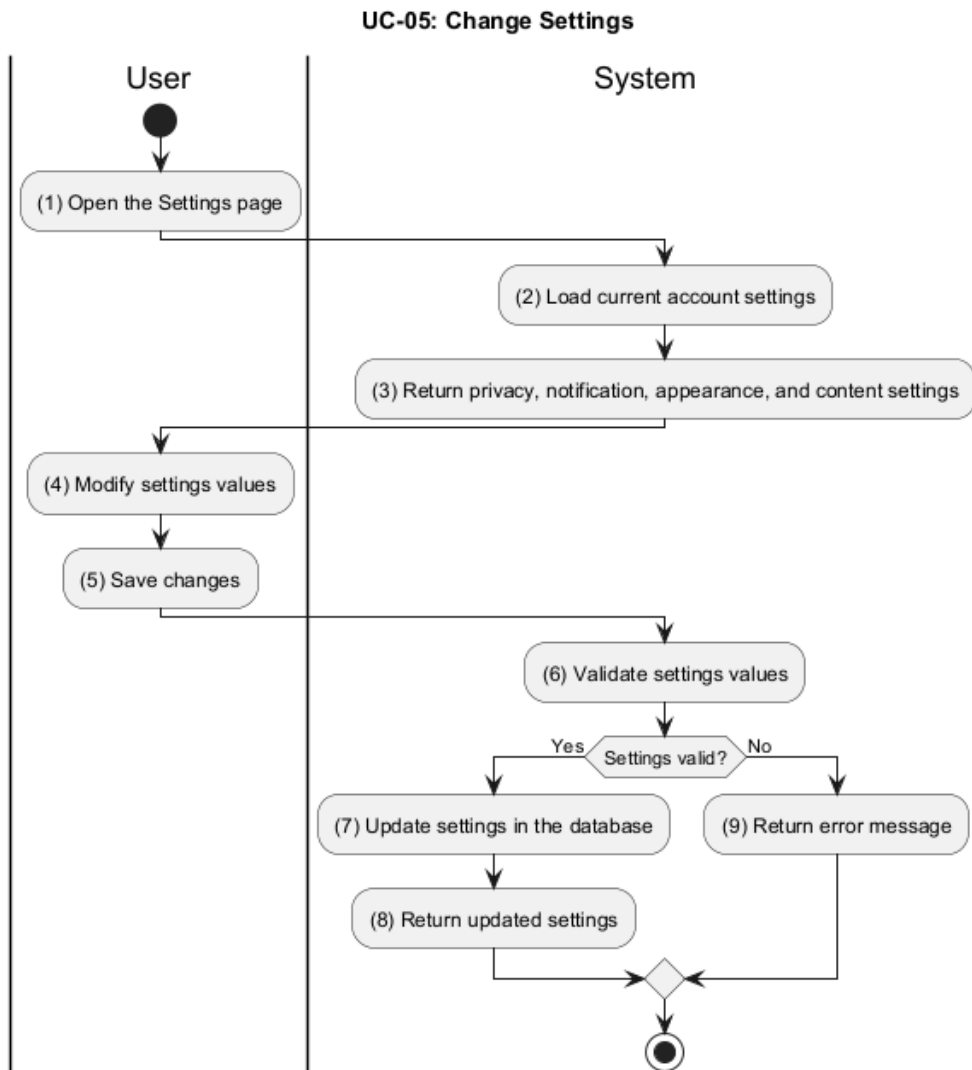


Figure 3.6: Activity Diagram for UC-05

| ID         | Activ-ity | Rule Type      | Description                                                                                                                                                                                                                                                                                 |
|------------|-----------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC05-01 | (2)       | Workflow Rules | <b>Load Current Settings:</b> <ul style="list-style-type: none"> <li>The system loads the current user's settings from the database using <i>GetSettings()</i>.</li> <li>If the user has no settings record, the service returns default settings through <i>FromSettings()</i>.</li> </ul> |

Table 3.11: Business Rules for UC-05 - Change Settings

| ID         | Activ-ity | Rule Type        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------|-----------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC05-02 | (6)       | Validation Rules | <b>Validate Settings Values:</b> <ul style="list-style-type: none"> <li>The system checks the JSON body for <code>UpdateSettingsRequest</code>.</li> <li>If binding fails, the controller returns a bad request response.</li> <li><b>theme</b> must be one of <code>light</code>, <code>dark</code>, or <code>auto</code>.</li> <li><b>font_size</b> must be one of <code>small</code>, <code>medium</code>, or <code>large</code>.</li> </ul> |
| BR-UC05-03 | (7)       | Workflow Rules   | <b>Update Settings in Database:</b> <ul style="list-style-type: none"> <li>The service loads the current user and initializes <b>Settings</b> when it is nil.</li> <li>The service updates only the settings fields provided in the request.</li> <li>The service persists the updated user record using the user repository.</li> </ul>                                                                                                        |
| BR-UC05-04 | (8)       | Message Rules    | <b>Return Updated Settings:</b> <ul style="list-style-type: none"> <li>Successful account-setting changes shall use MSG-51 or MSG-13.</li> <li>The response includes the updated <b>appearance</b>, <b>notifications</b>, <b>privacy</b>, and <b>content</b> values.</li> </ul>                                                                                                                                                                 |
| BR-UC05-05 | (9)       | Message Rules    | <b>Return Error Message:</b> <ul style="list-style-type: none"> <li>Failed setting updates shall use MSG-52 or MSG-50.</li> <li>Invalid input or authentication failures shall return ERR-14, ERR-2, ERR-10, or ERR-25.</li> </ul>                                                                                                                                                                                                              |

Table 3.11: Business Rules for UC-05 - Change Settings



### 3.1.6 UC-06 - View Activity History

|                       |                                                                                       |
|-----------------------|---------------------------------------------------------------------------------------|
| <b>Name</b>           | View Activity History                                                                 |
| <b>Description</b>    | This use case describes how a user reviews past posts, comments, and interactions.    |
| <b>Actor</b>          | User                                                                                  |
| <b>Trigger</b>        | When the user opens the activity history page.                                        |
| <b>Pre-condition</b>  | The user is logged in to the system.<br>The user has activity records in the system.  |
| <b>Post-condition</b> | The activity history is displayed to the user.<br>The user can review prior activity. |

Table 3.12: Use Case Specification - View Activity History

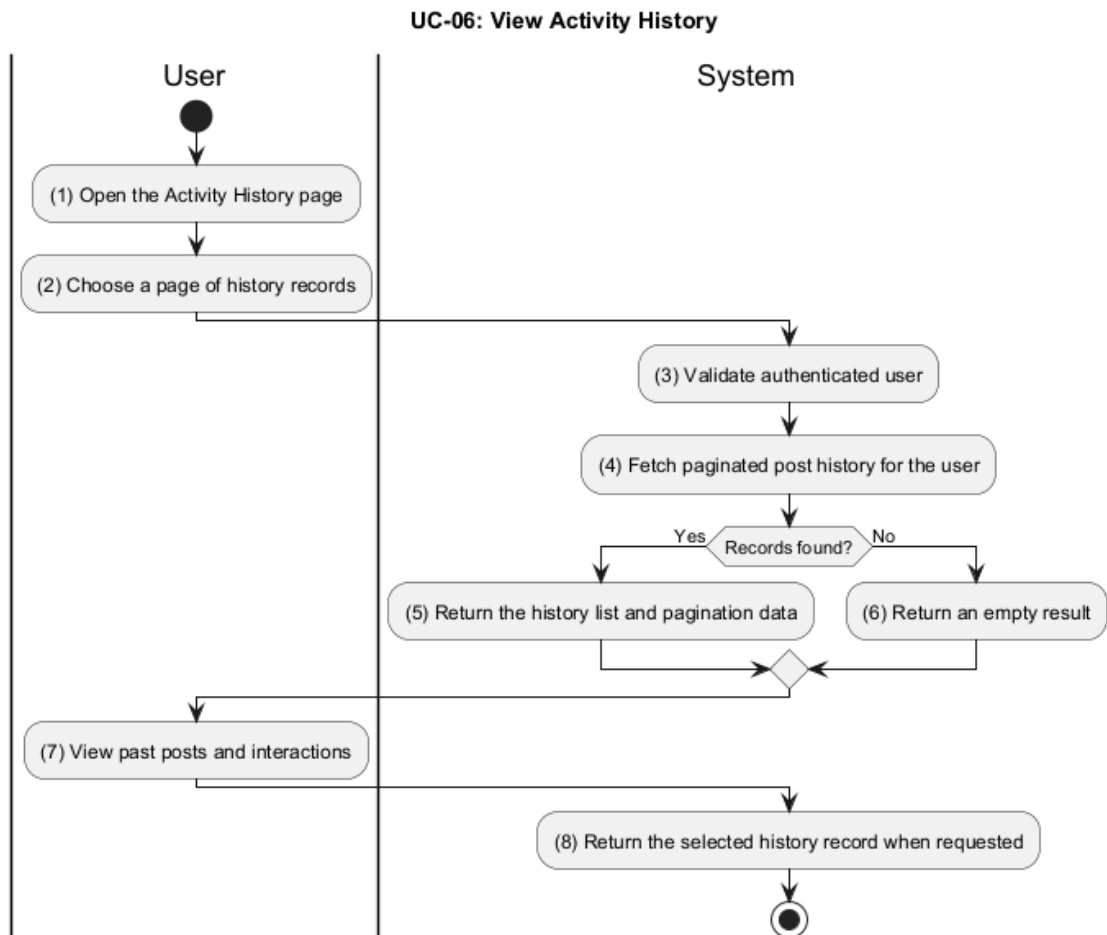


Figure 3.7: Activity Diagram for UC-06

| ID         | Activ-<br>ity | Rule Type           | Description                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------|---------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC06-01 | (3)           | Authorization Rules | <b>Validate Authenticated User:</b> <ul style="list-style-type: none"> <li>if <b>authUser</b> is missing, then the controller returns a forbidden response.</li> <li>if <b>authUser</b> exists, then the system continues to fetch the activity history for that user.</li> </ul>                                                                                                                                                |
| BR-UC06-02 | (4)           | Workflow Rules      | <b>Fetch Paginated History:</b> <ul style="list-style-type: none"> <li>if <b>user__id</b> does not match the authenticated user, then the service returns a forbidden error.</li> <li>if the request is valid, then the service calls <i><b>GetPostHistoryByUserID()</b></i> to fetch the user's post history.</li> <li>if <b>page</b> or <b>page_size</b> is invalid, then the controller uses defaults of 1 and 10.</li> </ul> |
| BR-UC06-03 | (5–6)         | Message Rules       | <b>Return History List or Empty Result:</b> <ul style="list-style-type: none"> <li>This use case has no dedicated success toast; successful history retrieval returns the activity list or an empty result directly in the UI.</li> <li>History-loading failures shall return <a href="#">ERR-10</a>, <a href="#">ERR-13</a>, <a href="#">ERR-26</a>, or <a href="#">ERR-27</a>.</li> </ul>                                      |
| BR-UC06-04 | (7)           | Authorization Rules | <b>Check Record Ownership:</b> <ul style="list-style-type: none"> <li>if the requester opens a selected history record, then the service checks that <b>requesterID</b> matches the stored <b>UserID</b>.</li> <li>if the requester does not own the record, then the service returns a forbidden error.</li> </ul>                                                                                                              |

Table 3.13: Business Rules for UC-06 - View Activity History

| ID         | Activ-ity | Rule Type     | Description                                                                                                                                                                                                                                                                                                       |
|------------|-----------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC06-05 | (8)       | Message Rules | <b>Return Selected History Record:</b> <ul style="list-style-type: none"> <li>This use case has no dedicated success toast; the selected history item is returned inline after conversion with <i>FromPostHistory()</i>.</li> <li>Ownership or lookup failures shall return ERR-13, ERR-26, or ERR-53.</li> </ul> |

Table 3.13: Business Rules for UC-06 - View Activity History

### 3.1.7 UC-07 - Follow / Unfollow User

|                       |                                                                                                     |
|-----------------------|-----------------------------------------------------------------------------------------------------|
| <b>Name</b>           | Follow / Unfollow User                                                                              |
| <b>Description</b>    | This use case describes how a user subscribes to or unsubscribes from another user's activity feed. |
| <b>Actor</b>          | User                                                                                                |
| <b>Trigger</b>        | When the user clicks the follow or unfollow action on another profile.                              |
| <b>Pre-condition</b>  | The user is logged in to the system.<br>The target profile is available for follow actions.         |
| <b>Post-condition</b> | The follow relationship is created or removed.<br>The user's feed and follower lists are updated.   |

Table 3.14: Use Case Specification - Follow / Unfollow User

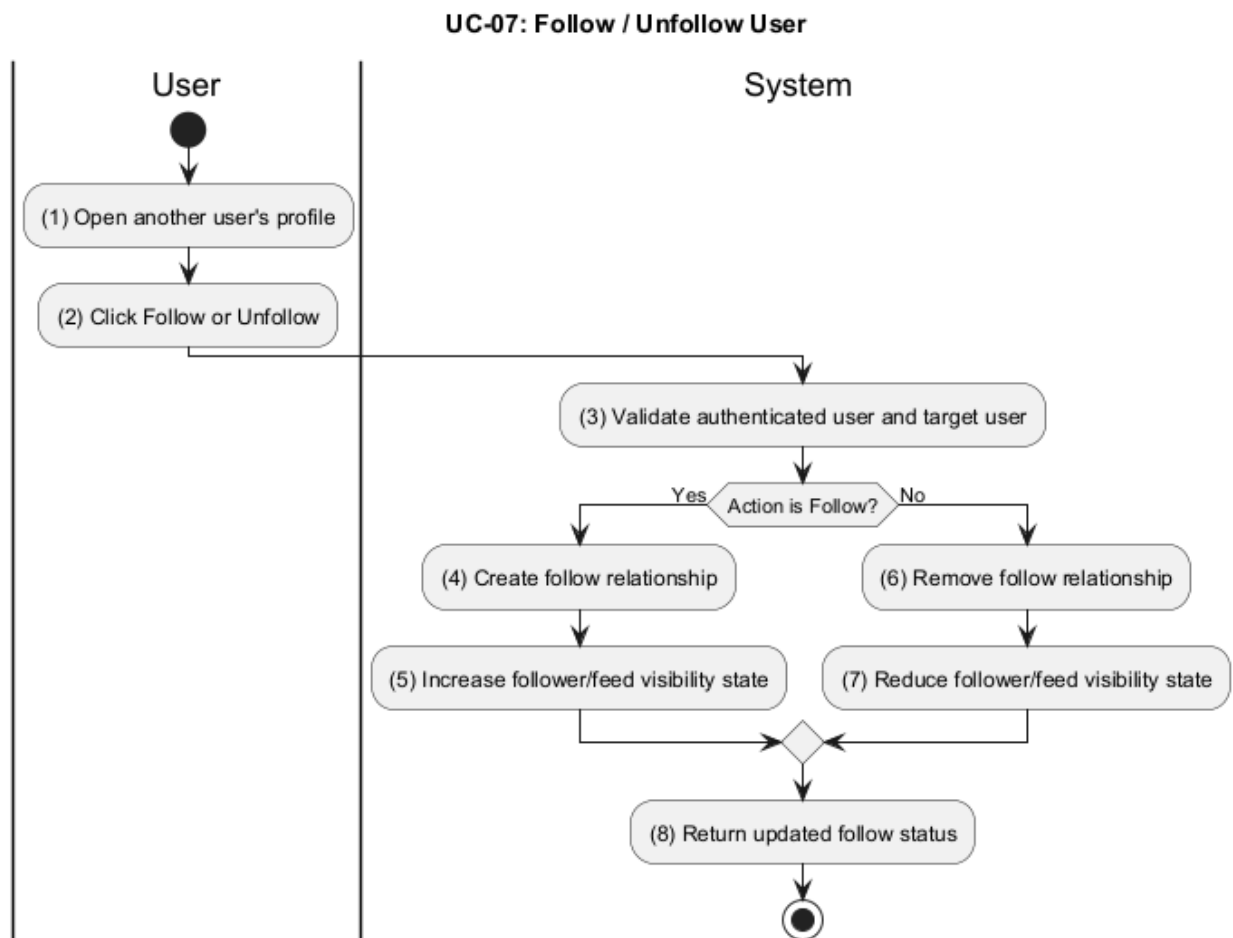


Figure 3.8: Activity Diagram for UC-07

| Activ-ity | BR Code | Description                                                                                                                                                                                                                                                                                                                                                |
|-----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (3–6)     | BR1     | <b>Backend Rule Status:</b> <ul style="list-style-type: none"> <li>No follow or unfollow route, controller, or service is present in the backend code.</li> <li>The repository currently has no server-side business rules to define for this use case.</li> <li>This use case remains documented-only until a backend implementation is added.</li> </ul> |

Table 3.15: Business Rules for UC-07 - Follow / Unfollow User

### 3.1.8 UC-08 - Browse Post

|                       |                                                                                                 |
|-----------------------|-------------------------------------------------------------------------------------------------|
| <b>Name</b>           | Browse Content                                                                                  |
| <b>Description</b>    | This use case describes how a visitor or user explores communities and public posts.            |
| <b>Actor</b>          | Guest User / User                                                                               |
| <b>Trigger</b>        | When the user opens the forum homepage or a community page.                                     |
| <b>Pre-condition</b>  | The browsing page is accessible.<br>The user may be logged in or not logged in.                 |
| <b>Post-condition</b> | Public posts and community content are shown.<br>The user can continue browsing or open a post. |

Table 3.16: Use Case Specification - Browse Post

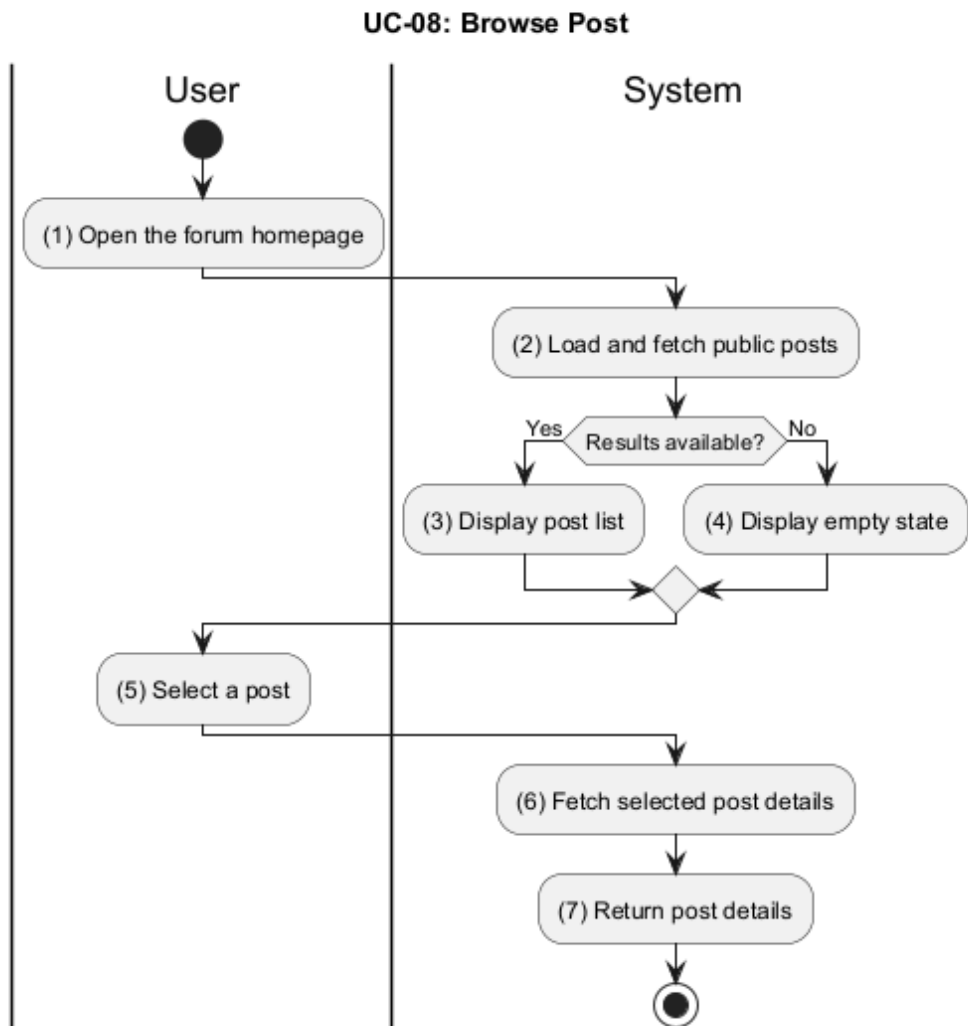


Figure 3.9: Activity Diagram for UC-08

| ID         | Activ-ity | Rule Type        | Description                                                                                                                                                                                                                                                                                                                                                              |
|------------|-----------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC08-01 | (2)       | Workflow Rules   | <b>Fetch Posts from Database:</b> <ul style="list-style-type: none"> <li>The system calls <i>GetPosts()</i> to fetch public posts from the database.</li> <li>The system fetches authors and communities to enrich the post response.</li> <li>if <b>totalPosts</b> == 0, then return an empty state instead of an error.</li> </ul>                                     |
| BR-UC08-02 | (3–4)     | Message Rules    | <b>Return Content List or Empty State:</b> <ul style="list-style-type: none"> <li>This use case has no dedicated success toast; successful browsing returns content lists or empty states directly in the UI.</li> <li>Content-loading failures shall use MSG-58, MSG-84, or MSG-83.</li> <li>Lookup failures shall return ERR-53, ERR-39, ERR-26, or ERR-27.</li> </ul> |
| BR-UC08-03 | (6)       | Workflow Rules   | <b>View Selected Post:</b> <ul style="list-style-type: none"> <li>if the user selects a post, then the system loads the post details using <i>GetPostById()</i>.</li> <li>if <b>userID</b> != null, then the system also loads <b>userVote</b> and <b>userPollVoteIDs</b>.</li> </ul>                                                                                    |
| BR-UC08-04 | (6)       | Validation Rules | <b>Validate Selected Post Visibility:</b> <ul style="list-style-type: none"> <li>if the selected post is hidden and <b>post.AuthorID</b> != <b>userID</b>, then return post-not-found.</li> <li>if the selected post is hidden and <b>post.AuthorID</b> == <b>userID</b>, then allow the request to continue.</li> </ul>                                                 |
| BR-UC08-06 | (7)       | Message Rules    | <b>Return Selected Post:</b> <ul style="list-style-type: none"> <li>The selected post is returned inline when found; this use case does not define a dedicated success toast.</li> <li>Post-loading failures shall use MSG-84 and return ERR-53 or ERR-26 when applicable.</li> </ul>                                                                                    |

Table 3.17: Business Rules for UC-08 - Browse Post

### 3.1.9 UC-09 - Search Content

|                       |                                                                                                |
|-----------------------|------------------------------------------------------------------------------------------------|
| <b>Name</b>           | Search Content                                                                                 |
| <b>Description</b>    | This use case describes how a user searches for posts, users, or communities by keyword.       |
| <b>Actor</b>          | User                                                                                           |
| <b>Trigger</b>        | When the user enters a search term and submits the search.                                     |
| <b>Pre-condition</b>  | The search page is available.<br>The system has searchable content.                            |
| <b>Post-condition</b> | Matching results are returned to the user.<br>The user can open a result or refine the search. |

Table 3.18: Use Case Specification - Search Content

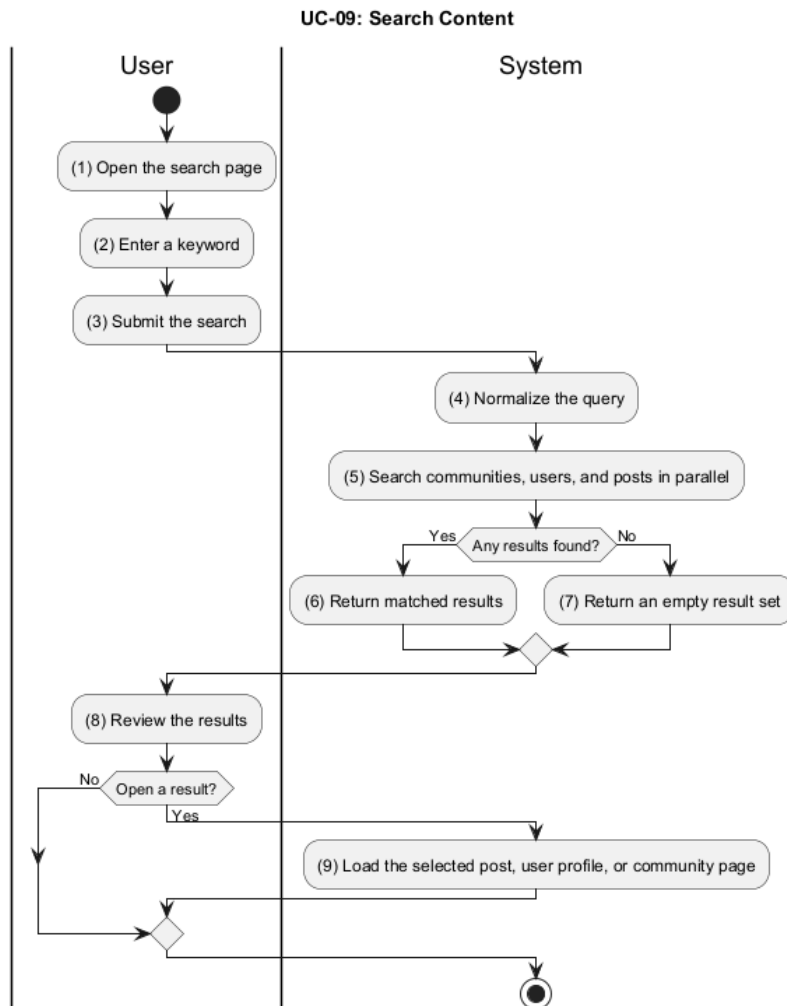


Figure 3.10: Activity Diagram for UC-09

| ID         | Activ-<br>ity | Rule Type        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------|---------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC09-01 | (4)           | Validation Rules | <b>Normalize the Query:</b> <ul style="list-style-type: none"> <li>The system decodes the query and removes optional prefixes such as <code>c/</code>, <code>lk/</code>, and <code>u/</code>.</li> <li>if <code>cleanQuery == null</code> or <code>cleanQuery.trim() == ""</code>, then the system keeps the result sets empty.</li> </ul>                                                                                                                     |
| BR-UC09-02 | (5)           | Workflow Rules   | <b>Search Communities, Users, and Posts in Parallel:</b> <ul style="list-style-type: none"> <li>The system calls <i><b>GetCommunitiesFilter()</b></i>, <i><b>GetUsers()</b></i>, and <i><b>GetPosts()</b></i> through the frontend service layer.</li> <li>The three requests run in parallel with the same cleaned query.</li> <li>if any request fails, then the search page clears the result lists and leaves the page in the empty-state path.</li> </ul> |
| BR-UC09-03 | (6–7)         | Message Rules    | <b>Return Matched Results or Empty State:</b> <ul style="list-style-type: none"> <li>This use case has no dedicated success toast; matched results or the empty state are rendered directly in the UI.</li> <li>Search-request failures shall return <a href="#">ERR-14</a>, <a href="#">ERR-27</a>, or <a href="#">ERR-26</a>.</li> </ul>                                                                                                                     |
| BR-UC09-04 | (9)           | Redirect Rules   | <b>Open Selected Result:</b> <ul style="list-style-type: none"> <li>if the user selects a post, then the system opens the generated post URL.</li> <li>if the user selects a community, then the system navigates to the community page using <code>/lk/{communityName}</code>.</li> <li>if the user selects a user, then the system navigates to the profile page using <code>/profile/{username}</code>.</li> </ul>                                          |

Table 3.19: Business Rules for UC-09 - Search Content



### 3.1.10 UC-10 - Create Post

|                       |                                                                                                         |
|-----------------------|---------------------------------------------------------------------------------------------------------|
| <b>Name</b>           | Create Post                                                                                             |
| <b>Description</b>    | This use case describes how a member creates and submits a new post in a community.                     |
| <b>Actor</b>          | User                                                                                                    |
| <b>Trigger</b>        | When the user clicks the “Create Post” action.                                                          |
| <b>Pre-condition</b>  | The user is logged in to the system.<br>The user is a member of the selected community.                 |
| <b>Post-condition</b> | The post is stored successfully.<br>The post appears in the community or enters moderation if required. |

Table 3.20: Use Case Specification - Create Post

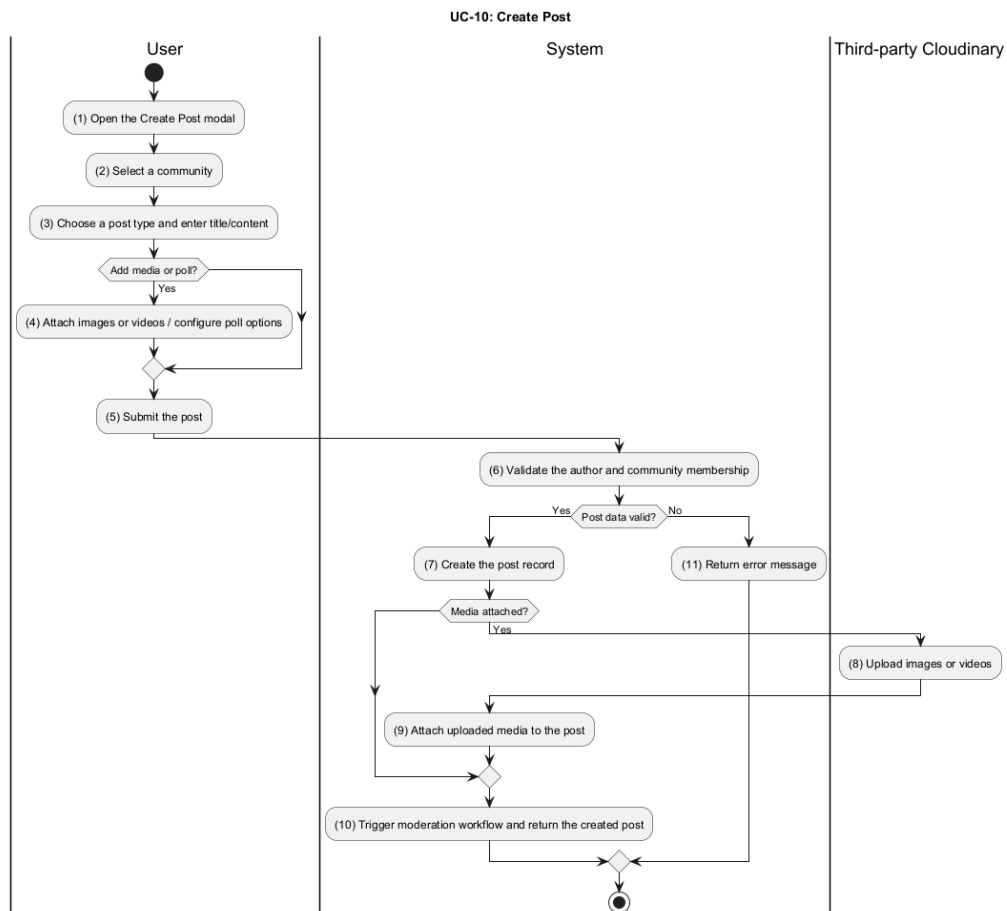


Figure 3.11: Activity Diagram for UC-10

| ID         | Activ-<br>ity | Rule Type           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------|---------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC10-01 | (6)           | Validation Rules    | <b>Validate Create Post Request:</b> <ul style="list-style-type: none"> <li>if <b>authUser</b> == <b>null</b>, then the controller returns a forbidden response.</li> <li>if JSON binding fails for <b>CreatePostRequest</b>, then the controller returns a bad request response.</li> <li>the DTO requires <b>community_id</b>, <b>title</b>, and <b>type</b>; if any required field is missing, binding fails.</li> <li><b>type</b> must be one of <b>text</b>, <b>poll</b>, <b>image</b>, or <b>video</b>.</li> </ul>                |
| BR-UC10-02 | (6)           | Authorization Rules | <b>Validate Author and Community Membership:</b> <ul style="list-style-type: none"> <li>if <b>communityID</b> == <b>null</b> or cannot be parsed to an <b>ObjectID</b>, then the service returns an invalid ID error.</li> <li>if the community does not exist, then the service returns the repository error.</li> <li>if the author does not exist, then the service returns the repository error.</li> <li>if <b>communityRepo.IsUserBanned()</b> == <b>true</b>, then the service returns a banned-from-community error.</li> </ul> |

Table 3.21: Business Rules for UC-10 - Create Post

| ID         | Activ-<br>ity | Rule Type      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------|---------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC10-03 | (6)           | Workflow Rules | <p><b>Determine Moderation State:</b></p> <ul style="list-style-type: none"> <li>• if <b>author.Role == AdminRole</b>, then set <b>initialStatus = ModerationSkipped</b>.</li> <li>• if <b>community.CreateByID == userID</b>, then set <b>initialStatus = ModerationSkipped</b>.</li> <li>• if any community moderator has <b>mod.UserID == userID</b>, then set <b>initialStatus = ModerationSkipped</b>.</li> <li>• if <b>community.Setting.PostRequireApproval == true</b> and the author is not admin/mod/creator, then set <b>initialStatus = ModerationPending</b>.</li> <li>• otherwise, set <b>initialStatus = ModerationPending</b> for AI moderation.</li> </ul> |
| BR-UC10-04 | (7)           | Workflow Rules | <p><b>Create Post Record:</b></p> <ul style="list-style-type: none"> <li>• The system creates a <b>Post</b> with <b>AuthorID</b>, <b>CommunityID</b>, <b>Title</b>, <b>Type</b>, <b>Content.text</b>, <b>Tags</b>, and <b>ModerationStatus</b>.</li> <li>• The system initializes <b>VotesCount</b>, <b>CommentsCount</b>, <b>HotScore</b>, and flag fields such as <b>IsDeleted</b>, <b>IsHidden</b>, <b>IsDraft</b>, and <b>IsBan</b>.</li> <li>• if <b>type == poll</b> and <b>req.Poll != null</b>, then the system creates poll options and attaches poll data to the post content.</li> <li>• The system persists the post through the post repository.</li> </ul>    |

Table 3.21: Business Rules for UC-10 - Create Post (Continued)

| ID         | Activity | Rule Type      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------|----------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC10-05 | (8–9)    | Workflow Rules | <b>Attach Media to Post:</b> <ul style="list-style-type: none"> <li>if images are attached, then the system uploads them using <i>UploadImages()</i> and pushes them into <b>content.images</b>.</li> <li>if videos are attached, then the system uploads them using <i>UploadVideos()</i> and pushes them into <b>content.videos</b>.</li> <li>if media upload fails, then the system returns the upload error and does not complete the attach step.</li> </ul> |
| BR-UC10-06 | (10)     | Workflow Rules | <b>Trigger Moderation and Return Post:</b> <ul style="list-style-type: none"> <li>The system increments the author's post count after the post is created.</li> <li>The system auto-upvotes the post for the creator using <i>VoteOnTargetByAuthor()</i>.</li> <li>The system publishes a post-created event to trigger moderation workflow.</li> <li>The system returns the created post using <i>GetPostById()</i>.</li> </ul>                                  |
| BR-UC10-07 | (11)     | Message Rules  | <b>Return Error Message:</b> <ul style="list-style-type: none"> <li>This use case has no dedicated success toast in the current code; successful post creation returns the created post payload inline.</li> <li>Validation, membership, or authorization failures shall return ERR-14, ERR-39, ERR-46, ERR-45, ERR-26, or ERR-10.</li> </ul>                                                                                                                     |

Table 3.21: Business Rules for UC-10 - Create Post

### 3.1.11 UC-11 - Edit Post

|                       |                                                                                            |
|-----------------------|--------------------------------------------------------------------------------------------|
| <b>Name</b>           | Edit Post                                                                                  |
| <b>Description</b>    | This use case describes how a member modifies an existing post and preserves edit history. |
| <b>Actor</b>          | User                                                                                       |
| <b>Trigger</b>        | When the user opens an existing post and chooses “Edit”.                                   |
| <b>Pre-condition</b>  | The user is logged in to the system.<br>The user owns the post or has edit permission.     |
| <b>Post-condition</b> | The edited post is saved successfully.<br>The edit history is retained.                    |

Table 3.22: Use Case Specification - Edit Post

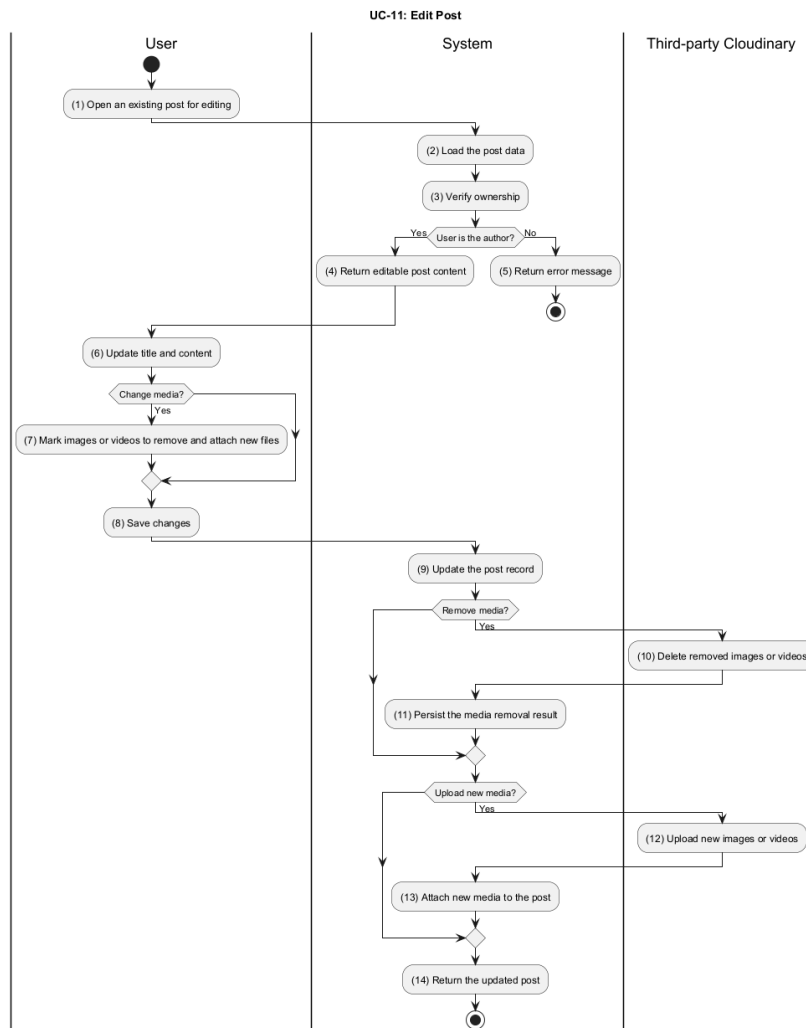


Figure 3.12: Activity Diagram for UC-11

| ID         | Activ-ity | Rule Type           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------|-----------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC11-01 | (4)       | Authorization Rules | <b>Load Post and Verify Ownership:</b> <ul style="list-style-type: none"> <li>The controller requires an authenticated user for the edit request.</li> <li>if <b>authUser == null</b>, then the controller returns a forbidden response.</li> <li>if JSON binding fails for <b>UpdatePostRequest</b>, then the controller returns a bad-request response.</li> <li>The service loads the post by ID and if <b>post.AuthorID != userID</b>, then the service returns a forbidden error.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                 |
| BR-UC11-02 | (9)       | Workflow Rules      | <b>Handle Media Changes Through Post Media Endpoints:</b> <ul style="list-style-type: none"> <li>if the user removes images, then the system calls <b><i>RemoveImagesFromPost()</i></b> after validating <b>public_ids</b> and author ownership.</li> <li>if the user adds images, then the system calls <b><i>AddImagesToPost()</i></b> and uploads the files with <b><i>UploadImages()</i></b> before pushing them into <b>content.images</b>.</li> <li>if the user removes videos, then the system calls <b><i>RemoveVideosFromPost()</i></b> and deletes the matched Cloudinary assets after the database update.</li> <li>if the user adds videos, then the system calls <b><i>AddVideosToPost()</i></b> and uploads the files with <b><i>UploadVideos()</i></b> before pushing them into <b>content.videos</b>.</li> <li>if new videos are added, then the system also sets <b>type = video</b>.</li> </ul> |

Table 3.23: Business Rules for UC-11 - Edit Post

| ID         | Activ-ity | Rule Type      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------|-----------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC11-04 | (10–13)   | Workflow Rules | <b>Persist Edit Result and Media Side Effects:</b> <ul style="list-style-type: none"> <li>The system updates the post record with the new content and <b>updated_at</b> timestamp.</li> <li>The system publishes a <b>PostUpdatedEvent</b> for moderation.</li> <li>if image or video upload fails, then the system returns the error and does not finish the edit flow.</li> <li>if video removal is requested, then the system only removes IDs that already exist in the post.</li> </ul> |
| BR-UC11-05 | (14)      | Message Rules  | <b>Return the Updated Post:</b> <ul style="list-style-type: none"> <li>Successful post updates shall use MSG-29; cancellation of unsaved edits shall use MSG-20.</li> <li>Failed updates shall use MSG-30.</li> <li>Authorization or content-state failures shall return ERR-53, ERR-13, ERR-56, or ERR-26.</li> </ul>                                                                                                                                                                       |

Table 3.23: Business Rules for UC-11 - Edit Post

### 3.1.12 UC-12 - Delete Post

|                       |                                                                                                      |
|-----------------------|------------------------------------------------------------------------------------------------------|
| <b>Name</b>           | Delete Post                                                                                          |
| <b>Description</b>    | This use case describes how a member or moderator removes a post from public view using soft delete. |
| <b>Actor</b>          | User / Moderator                                                                                     |
| <b>Trigger</b>        | When the user clicks the “Delete” action on a post.                                                  |
| <b>Pre-condition</b>  | The user is logged in to the system.<br>The user has permission to delete the post.                  |
| <b>Post-condition</b> | The post is marked as deleted.<br>The post is hidden from public view.                               |

Table 3.24: Use Case Specification - Delete Post

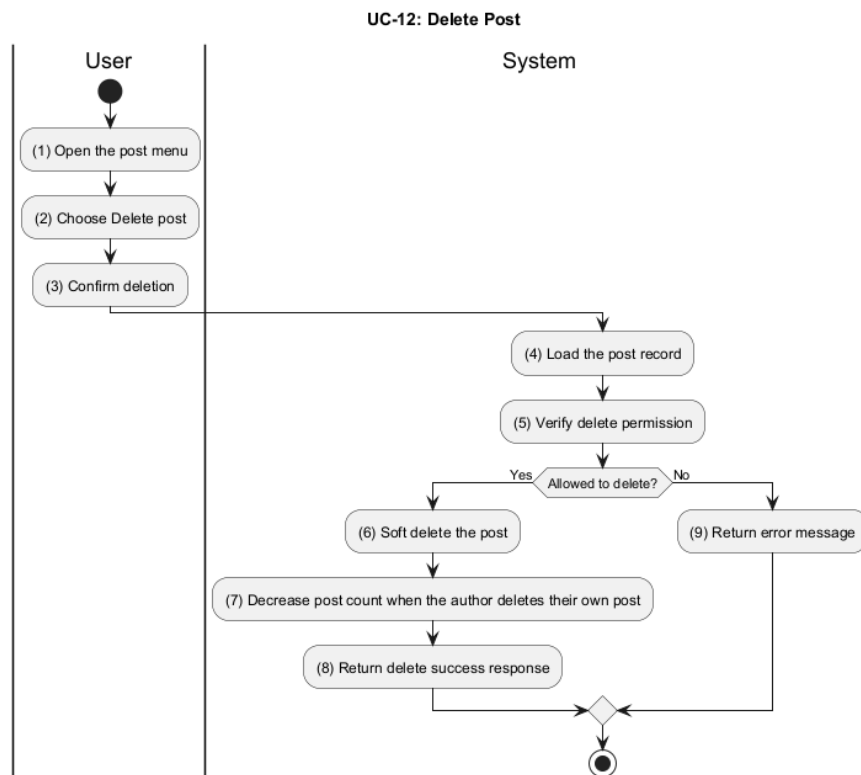


Figure 3.13: Activity Diagram for UC-12



| ID         | Activ-<br>ity | Rule Type           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------|---------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC12-01 | (4)           | Authorization Rules | <b>Verify Delete Permission:</b> <ul style="list-style-type: none"> <li>The system requires an authenticated user for the delete request.</li> <li>if <b>authUser == null</b>, then the controller returns an unauthorized response.</li> <li>The service loads the post by ID and if the post no longer exists, then it returns a post-not-found error.</li> <li>if <b>post.AuthorID == userID</b>, then the requester may delete their own post.</li> <li>if <b>post.AuthorID != userID</b>, then the service checks whether the user is an admin, the community creator, or a community moderator.</li> <li>if none of those permissions exist, then the service returns a forbidden error.</li> </ul> |
| BR-UC12-02 | (6-7)         | Workflow Rules      | <b>Soft Delete the Post Record and Adjust Counters:</b> <ul style="list-style-type: none"> <li>The system marks the post as deleted in storage through <i><b>SoftDelete()</b></i>.</li> <li>if the author deletes their own post, then the system decreases the author's post count by one.</li> <li>if the requester is not the author, then the system does not change the author's post count.</li> </ul>                                                                                                                                                                                                                                                                                              |
| BR-UC12-03 | (8)           | Message Rules       | <b>Return Success or Error Response:</b> <ul style="list-style-type: none"> <li>Delete confirmation shall use MSG-14; successful deletion shall use MSG-87.</li> <li>Failed deletion shall use MSG-27.</li> <li>Permission or lookup failures shall return ERR-53, ERR-13, or ERR-26.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                          |

Table 3.25: Business Rules for UC-12 - Soft Delete Post

| ID         | Activ-<br>ity | Rule Type     | Description                                                                                                                                                                                                                                                    |
|------------|---------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC12-04 | (9)           | Message Rules | <b>Return Success or Error Response:</b> <ul style="list-style-type: none"> <li>The same delete outcome shall reuse MSG-87 on success and MSG-27 on failure.</li> <li>Authorization or persistence failures shall return ERR-53, ERR-13, or ERR-26.</li> </ul> |

Table 3.25: Business Rules for UC-12 - Soft Delete Post

### 3.1.13 UC-13 - Post Comment

|                       |                                                                                     |
|-----------------------|-------------------------------------------------------------------------------------|
| <b>Name</b>           | Post Comment                                                                        |
| <b>Description</b>    | This use case describes how a user adds a direct comment to a post.                 |
| <b>Actor</b>          | User                                                                                |
| <b>Trigger</b>        | When the user submits a comment on a post.                                          |
| <b>Pre-condition</b>  | The user is logged in to the system.<br>The target post is visible and commentable. |
| <b>Post-condition</b> | The comment is created and attached to the post.<br>The comment count is updated.   |

Table 3.26: Use Case Specification - Post Comment

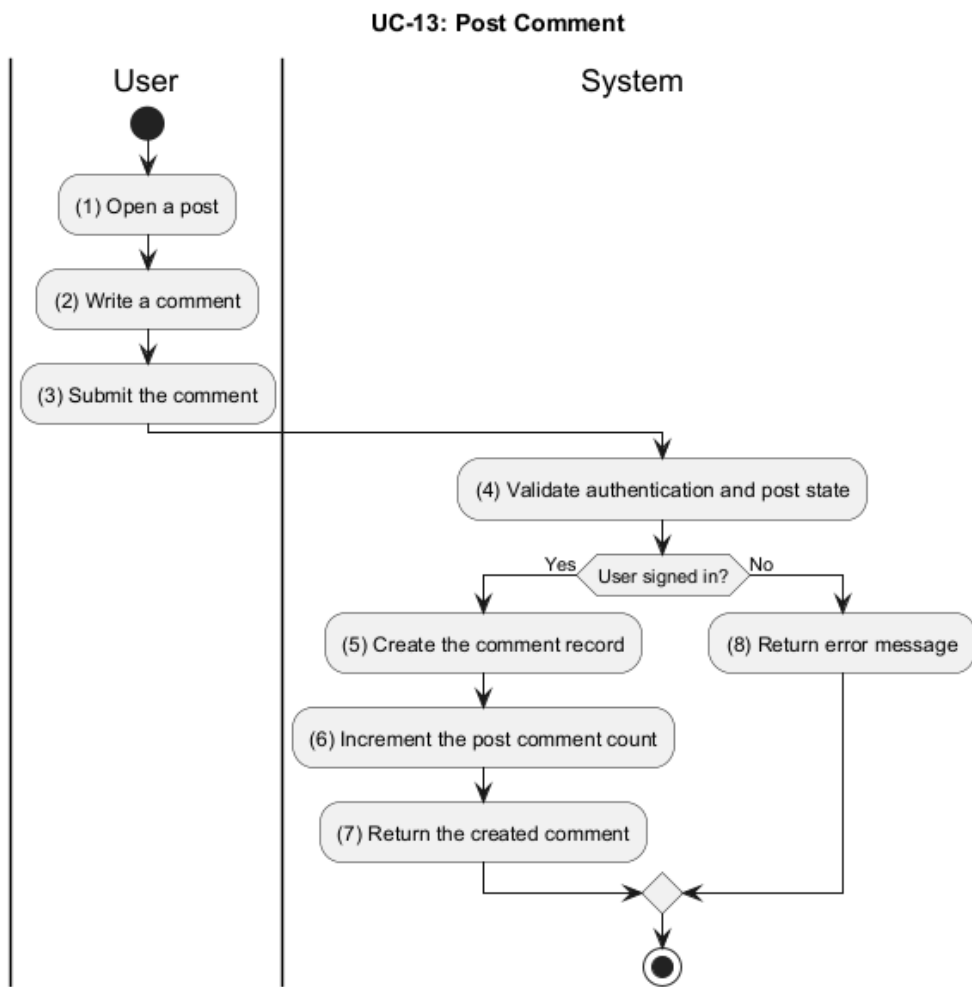


Figure 3.14: Activity Diagram for UC-13

| ID         | Activ-<br>ity | Rule Type        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------|---------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC13-01 | (4)           | Validation Rules | <b>Validate Request, Post State, and Access:</b> <ul style="list-style-type: none"> <li>The controller binds the request to <code>CreateCommentRequest</code>.</li> <li>if JSON binding fails, then the controller returns a bad-request response.</li> <li><b>post_id</b> and <b>content</b> are required by the request DTO.</li> <li>The service loads the target post by <b>post_id</b>; if the post cannot be found, then the comment flow stops.</li> <li>The system checks whether the user is muted or banned in the post's community.</li> <li>if <b>authUser == null</b>, then the controller returns a forbidden response.</li> <li>if the user is muted or banned, then the service returns the corresponding access error.</li> </ul> |
| BR-UC13-02 | (4)           | Message Rules    | <b>Return Bad-Request Message:</b> <ul style="list-style-type: none"> <li>Invalid comment payloads shall return <a href="#">ERR-14</a>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| BR-UC13-03 | (5–6)         | Workflow Rules   | <b>Create Comment and Update Counters:</b> <ul style="list-style-type: none"> <li>The system creates the comment with the current user as author, the selected post as target, and the optional parent comment.</li> <li>The system stores the new comment with <b>IsDeleted = false</b>.</li> <li>The system increments the post comment count and the user's comment count.</li> <li>The system publishes the comment-created event for downstream processing.</li> </ul>                                                                                                                                                                                                                                                                        |

Table 3.27: Business Rules for UC-13 - Post Comment

| ID         | Activ-<br>ity | Rule Type     | Description                                                                                                                                                                                                                              |
|------------|---------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC13-04 | (7)           | Message Rules | <b>Return Created Comment Message:</b> <ul style="list-style-type: none"> <li>Failed comment submission shall use MSG-64.</li> <li>Validation and permission failures shall return ERR-53, ERR-61, ERR-45, ERR-10, or ERR-26.</li> </ul> |

Table 3.27: Business Rules for UC-13 - Post Comment

### 3.1.14 UC-14 - Reply to Comment

|                       |                                                                                                  |
|-----------------------|--------------------------------------------------------------------------------------------------|
| <b>Name</b>           | Reply to Comment                                                                                 |
| <b>Description</b>    | This use case describes how a user replies to an existing comment in a nested discussion thread. |
| <b>Actor</b>          | User                                                                                             |
| <b>Trigger</b>        | When the user submits a reply to a comment.                                                      |
| <b>Pre-condition</b>  | The user is logged in to the system.<br>The parent comment exists and can receive replies.       |
| <b>Post-condition</b> | The reply is created as a nested comment.<br>The discussion thread is updated.                   |

Table 3.28: Use Case Specification - Reply to Comment

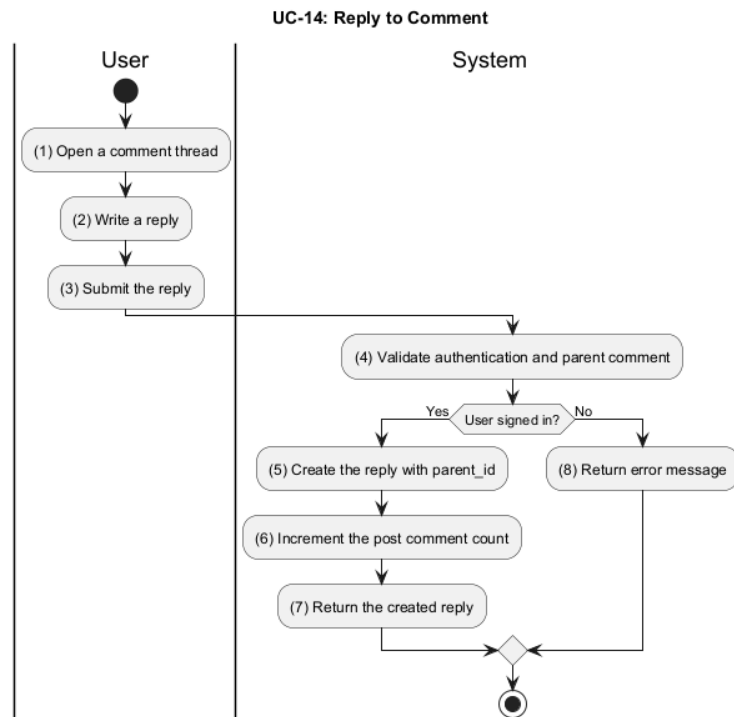


Figure 3.15: Activity Diagram for UC-14

| ID | Activ-ity | Rule Type | Description |
|----|-----------|-----------|-------------|
|----|-----------|-----------|-------------|

Table 3.29: Business Rules for UC-14 - Reply to Comment

|            |     |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------|-----|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC14-01 | (4) | Validation Rules | <b>Validate Parent Reply Request and Access:</b> <ul style="list-style-type: none"> <li>• The controller binds the request to <b>CreateCommentRequest</b>.</li> <li>• if JSON binding fails, then the controller returns a bad-request response.</li> <li>• <b>post_id</b> and <b>content</b> are required by the request DTO.</li> <li>• The controller requires an authenticated user before creating the reply.</li> <li>• if <b>authUser == null</b>, then the controller returns a forbidden response.</li> <li>• The service loads the target post by <b>post_id</b>; if the post cannot be found, then the reply flow stops.</li> <li>• if <b>parent_id</b> is provided, then the service parses it and uses it as the reply parent when valid.</li> <li>• The system checks whether the user is muted or banned in the post's community.</li> <li>• if the user is muted or banned, then the service returns the corresponding access error.</li> </ul> |
| BR-UC14-02 | (4) | Message Rules    | <b>Return Bad-Request Message:</b> <ul style="list-style-type: none"> <li>• Invalid reply payloads shall return <b>ERR-14</b>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

Table 3.29: Business Rules for UC-14 - Reply to Comment (Continued)

|            |       |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------|-------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC14-03 | (5–6) | Workflow Rules | <b>Create Reply and Update Counters:</b> <ul style="list-style-type: none"> <li>• The system creates the reply with the current user as author, the selected post as target, and the optional parent comment.</li> <li>• The system stores the new reply with <b>IsDeleted = false</b>.</li> <li>• The system increments the post comment count and the user's comment count.</li> <li>• The system publishes the comment-created event for downstream processing.</li> </ul> |
| BR-UC14-04 | (7)   | Message Rules  | <b>Return Created Reply Message:</b> <ul style="list-style-type: none"> <li>• Failed reply submission shall use MSG-65.</li> <li>• Validation and permission failures shall return ERR-59, ERR-53, ERR-61, ERR-60, or ERR-10.</li> </ul>                                                                                                                                                                                                                                      |

Table 3.29: Business Rules for UC-14 - Reply to Comment

### 3.1.15 UC-15 - Vote on Content

|                       |                                                                                     |
|-----------------------|-------------------------------------------------------------------------------------|
| <b>Name</b>           | Vote on Content                                                                     |
| <b>Description</b>    | This use case describes how a user upvotes or downvotes a post or comment.          |
| <b>Actor</b>          | User                                                                                |
| <b>Trigger</b>        | When the user selects an upvote or downvote action.                                 |
| <b>Pre-condition</b>  | The user is logged in to the system.<br>The target content is available for voting. |
| <b>Post-condition</b> | The vote is recorded or updated.<br>The content score is recalculated.              |

Table 3.30: Use Case Specification - Vote on Content



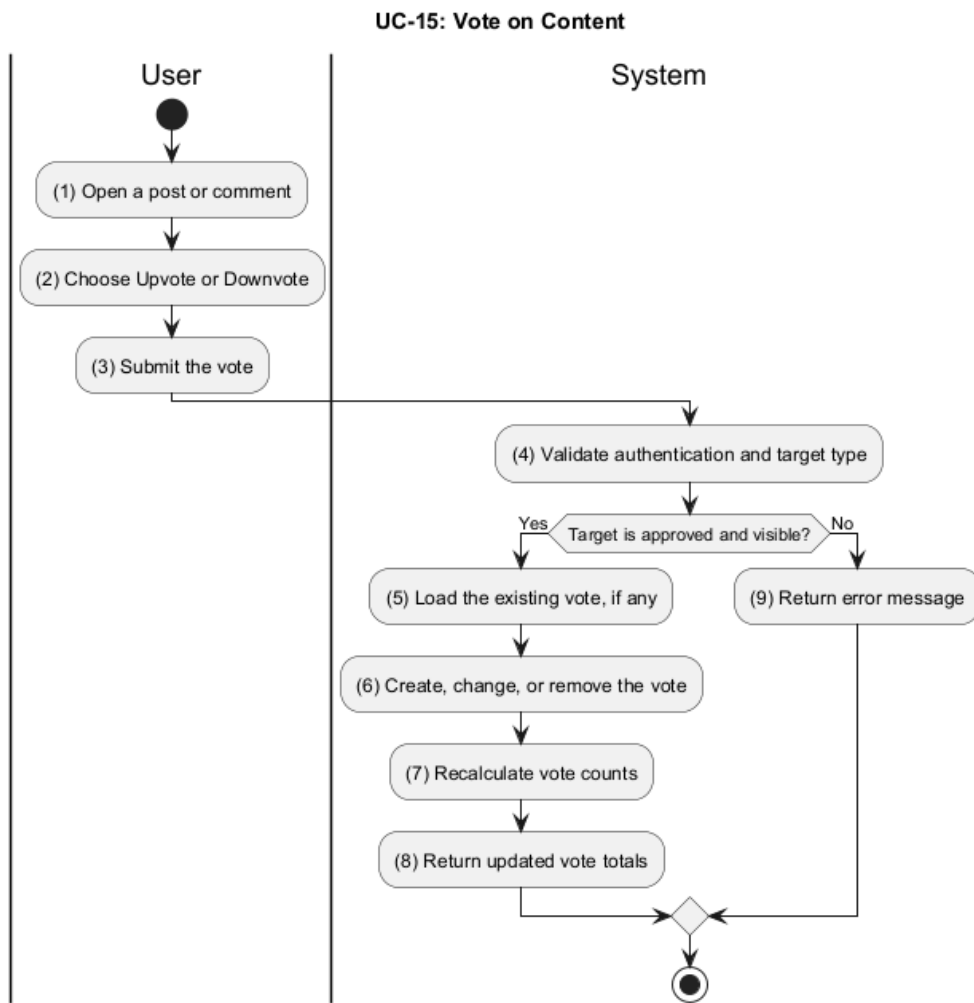


Figure 3.16: Activity Diagram for UC-15

| ID         | Activ-<br>ity | Rule Type        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------|---------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC15-01 | (4)           | Validation Rules | <b>Validate Authentication, Target Type, and Vote Payload:</b> <ul style="list-style-type: none"> <li>The controller requires an authenticated user before casting or removing a vote.</li> <li>if <b>authUser</b> == <b>null</b>, then the controller returns an unauthorized response.</li> <li>The controller accepts only <b>post</b> or <b>comment</b> as <b>target_type</b>.</li> <li>if the target type is invalid, then the controller returns a bad-request response.</li> <li>The controller binds the request body to <b>PostVoteRequest</b>.</li> <li>if JSON binding fails or <b>value</b> == <b>null</b>, then the controller returns a bad-request response.</li> </ul> |
| BR-UC15-02 | (4)           | Workflow Rules   | <b>Check Target Visibility and Load Existing Vote:</b> <ul style="list-style-type: none"> <li>For post voting, the system loads the post and allows voting only when the moderation status is <b>approved</b> or <b>skipped</b>.</li> <li>For comment voting, the system loads the comment and does not apply a moderation gate.</li> <li>The system loads the existing vote, if any, before applying the new vote action.</li> </ul>                                                                                                                                                                                                                                                  |
| BR-UC15-03 | (4)           | Message Rules    | <b>Return Bad-Request Message:</b> <ul style="list-style-type: none"> <li>Invalid vote targets or malformed vote payloads shall return ERR-14.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

Table 3.31: Business Rules for UC-15 - Vote on Content

| ID         | Activ-<br>ity | Rule Type      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------|---------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC15-04 | (5–7)         | Workflow Rules | <b>Create, Change, or Remove the Vote:</b> <ul style="list-style-type: none"> <li>• if no previous vote exists, then the system creates a new vote record.</li> <li>• if the new value matches the previous value, then the system removes the vote.</li> <li>• if the new value differs from the previous value, then the system updates the existing vote value.</li> <li>• The vote repository updates the target’s vote counters inside a transaction.</li> <li>• For post votes, the repository also recalculates <b>hot_score</b> from the new totals.</li> </ul> |
| BR-UC15-05 | (8)           | Message Rules  | <b>Return Updated Vote Totals:</b> <ul style="list-style-type: none"> <li>• This use case has no dedicated success toast; the updated upvote, downvote, and score totals are returned inline after the transaction completes.</li> </ul>                                                                                                                                                                                                                                                                                                                                |
| BR-UC15-06 | (9)           | Message Rules  | <b>Return Vote Error Message:</b> <ul style="list-style-type: none"> <li>• Poll-vote failures shall use MSG-25 or MSG-26.</li> <li>• Poll-state failures shall return ERR-53, ERR-54, ERR-55, ERR-56, or ERR-10.</li> </ul>                                                                                                                                                                                                                                                                                                                                             |

Table 3.31: Business Rules for UC-15 - Vote on Content

### 3.1.16 UC-16 - Participate in Poll

|                       |                                                                        |
|-----------------------|------------------------------------------------------------------------|
| <b>Name</b>           | Participate in Poll                                                    |
| <b>Description</b>    | This use case describes how a user casts a vote in a community poll.   |
| <b>Actor</b>          | User                                                                   |
| <b>Trigger</b>        | When the user submits a poll vote.                                     |
| <b>Pre-condition</b>  | The user is logged in to the system.<br>The poll is open for voting.   |
| <b>Post-condition</b> | The selected poll option is recorded.<br>The poll results are updated. |

Table 3.32: Use Case Specification - Participate in Poll

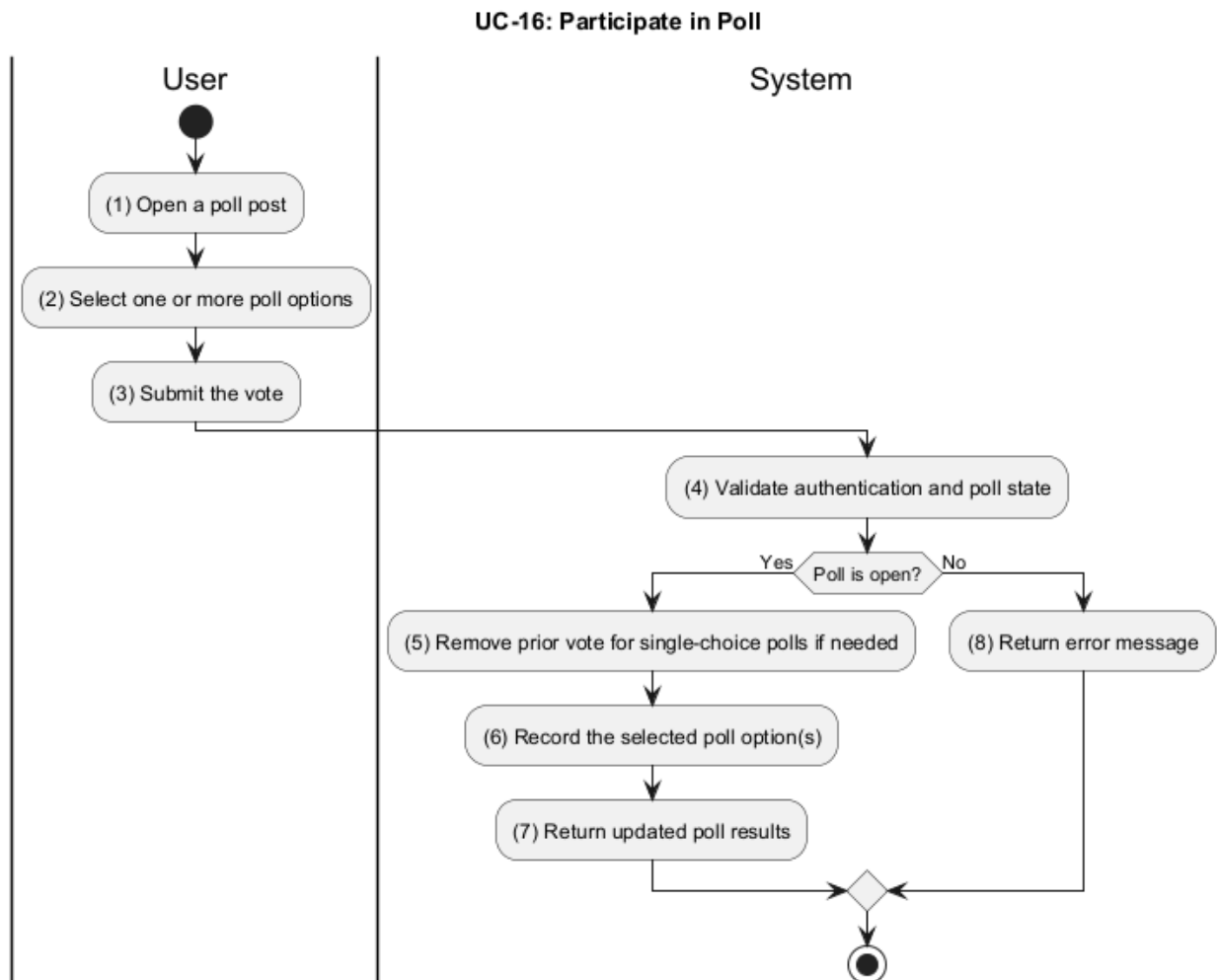


Figure 3.17: Activity Diagram for UC-16

| ID         | Activ-<br>ity | Rule Type        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------|---------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC16-01 | (4)           | Validation Rules | <b>Validate Authentication, Poll State, and Vote Payload:</b> <ul style="list-style-type: none"> <li>The controller requires an authenticated user before recording a poll vote.</li> <li>if <b>authUser</b> == <b>null</b>, then the controller returns an unauthorized response.</li> <li>The controller binds the request body to <b>PollVoteRequest</b>.</li> <li>if JSON binding fails or <b>option_id</b> is missing, then the controller returns a bad-request response.</li> <li>The service loads the target post and requires it to contain poll content.</li> <li>if the post cannot be found, is not a poll, or does not contain poll data, then the service returns an error.</li> </ul> |
| BR-UC16-02 | (4)           | Message Rules    | <b>Return Bad-Request Message:</b> <ul style="list-style-type: none"> <li>Invalid poll-vote payloads shall return <b>ERR-14</b>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| BR-UC16-03 | (5–6)         | Workflow Rules   | <b>Record the Poll Vote:</b> <ul style="list-style-type: none"> <li>The repository loads the user's existing poll votes for the post.</li> <li>if the poll is single-choice and the user already voted another option, then the repository removes the previous vote before adding the new one.</li> <li>if the selected option was already voted by the user, then the repository removes that specific vote instead of adding a duplicate.</li> <li>The repository stores the new poll vote and increments the selected option vote count plus the poll total votes.</li> </ul>                                                                                                                     |

Table 3.33: Business Rules for UC-16 - Participate in Poll

| ID         | Activity | Rule Type      | Description                                                                                                                                                                                                                                                          |
|------------|----------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC16-04 | (7)      | Workflow Rules | <b>Return Updated Poll Results:</b> <ul style="list-style-type: none"> <li>The service reloads the poll and user vote IDs after the transaction completes.</li> <li>The system returns the refreshed poll response with updated totals and user vote IDs.</li> </ul> |
| BR-UC16-05 | (7)      | Message Rules  | <b>Return Created Poll Vote Message:</b> <ul style="list-style-type: none"> <li>This use case has no dedicated success toast in the current code; the updated poll payload is returned inline after a successful vote.</li> </ul>                                    |
| BR-UC16-06 | (8)      | Message Rules  | <b>Return Poll Vote Error Message:</b> <ul style="list-style-type: none"> <li>Poll-vote failures shall use MSG-25 or MSG-26.</li> <li>Poll-state failures shall return ERR-53, ERR-54, ERR-55, ERR-56, or ERR-10.</li> </ul>                                         |

Table 3.33: Business Rules for UC-16 - Participate in Poll

### 3.1.17 UC-17 - Save Post

|                       |                                                                                             |
|-----------------------|---------------------------------------------------------------------------------------------|
| <b>Name</b>           | Save Post                                                                                   |
| <b>Description</b>    | This use case describes how a user bookmarks a post for later viewing.                      |
| <b>Actor</b>          | User                                                                                        |
| <b>Trigger</b>        | When the user clicks the “Save” action on a post.                                           |
| <b>Pre-condition</b>  | The user is logged in to the system.<br>The target post exists and is visible to the user.  |
| <b>Post-condition</b> | The post is added to the user’s saved list.<br>The saved state is available in the account. |

Table 3.34: Use Case Specification - Save Post

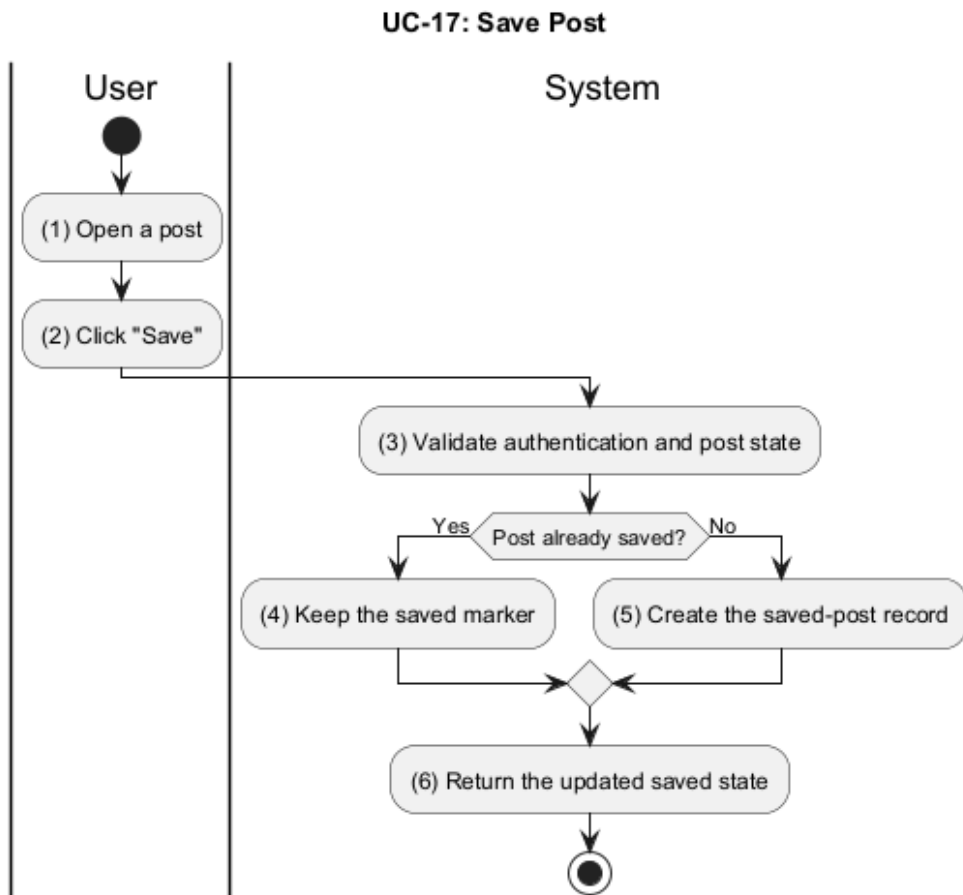


Figure 3.18: Activity Diagram for UC-17

| ID         | Activ-ity | Rule Type        | Description                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------|-----------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC17-01 | (3)       | Validation Rules | <b>Validate Authentication and Post ID:</b> <ul style="list-style-type: none"> <li>The controller requires an authenticated user before saving a post.</li> <li>if <b>authUser == null</b>, then the controller returns a forbidden response.</li> <li>The service parses the user ID and post ID to ObjectIDs.</li> <li>if either ID cannot be parsed, then the service returns an invalid-ID error.</li> </ul> |

Table 3.35: Business Rules for UC-17 - Save Post

| ID         | Activ-ity | Rule Type      | Description                                                                                                                                                                                                                                                                                                                                                             |
|------------|-----------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC17-02 | (3)       | Workflow Rules | <b>Check Post State and Save Relation:</b> <ul style="list-style-type: none"> <li>The service loads the post before saving it.</li> <li>if the post cannot be found, then the service returns the post lookup error.</li> <li>The saved-post repository stores the relation with upsert semantics so an existing saved marker is kept instead of duplicated.</li> </ul> |
| BR-UC17-03 | (4–5)     | Workflow Rules | <b>Create or Keep the Saved Marker:</b> <ul style="list-style-type: none"> <li>if the post is already saved by the user, then the system keeps the saved marker unchanged.</li> <li>if the post is not yet saved, then the system creates the saved-post record.</li> </ul>                                                                                             |
| BR-UC17-04 | (6)       | Message Rules  | <b>Return Updated Saved State:</b> <ul style="list-style-type: none"> <li>This use case has no dedicated success toast in the current code; the updated saved state is returned inline after the relation is stored.</li> </ul>                                                                                                                                         |
| BR-UC17-05 | (6)       | Message Rules  | <b>Return Save Error Message:</b> <ul style="list-style-type: none"> <li>Save-post failures shall return <a href="#">ERR-10</a>, <a href="#">ERR-26</a>, <a href="#">ERR-53</a>, or <a href="#">ERR-13</a>.</li> </ul>                                                                                                                                                  |

Table 3.35: Business Rules for UC-17 - Save Post

### 3.1.18 UC-18 - Hide Post

|                       |                                                                                               |
|-----------------------|-----------------------------------------------------------------------------------------------|
| <b>Name</b>           | Hide Post                                                                                     |
| <b>Description</b>    | This use case describes how a user hides a post from their personal feed.                     |
| <b>Actor</b>          | User                                                                                          |
| <b>Trigger</b>        | When the user clicks the “Hide” action on a post.                                             |
| <b>Pre-condition</b>  | The user is logged in to the system.<br>The target post is visible in the user’s feed.        |
| <b>Post-condition</b> | The post is hidden from the user’s feed.<br>The hidden-post record is stored for the account. |

Table 3.36: Use Case Specification - Hide Post



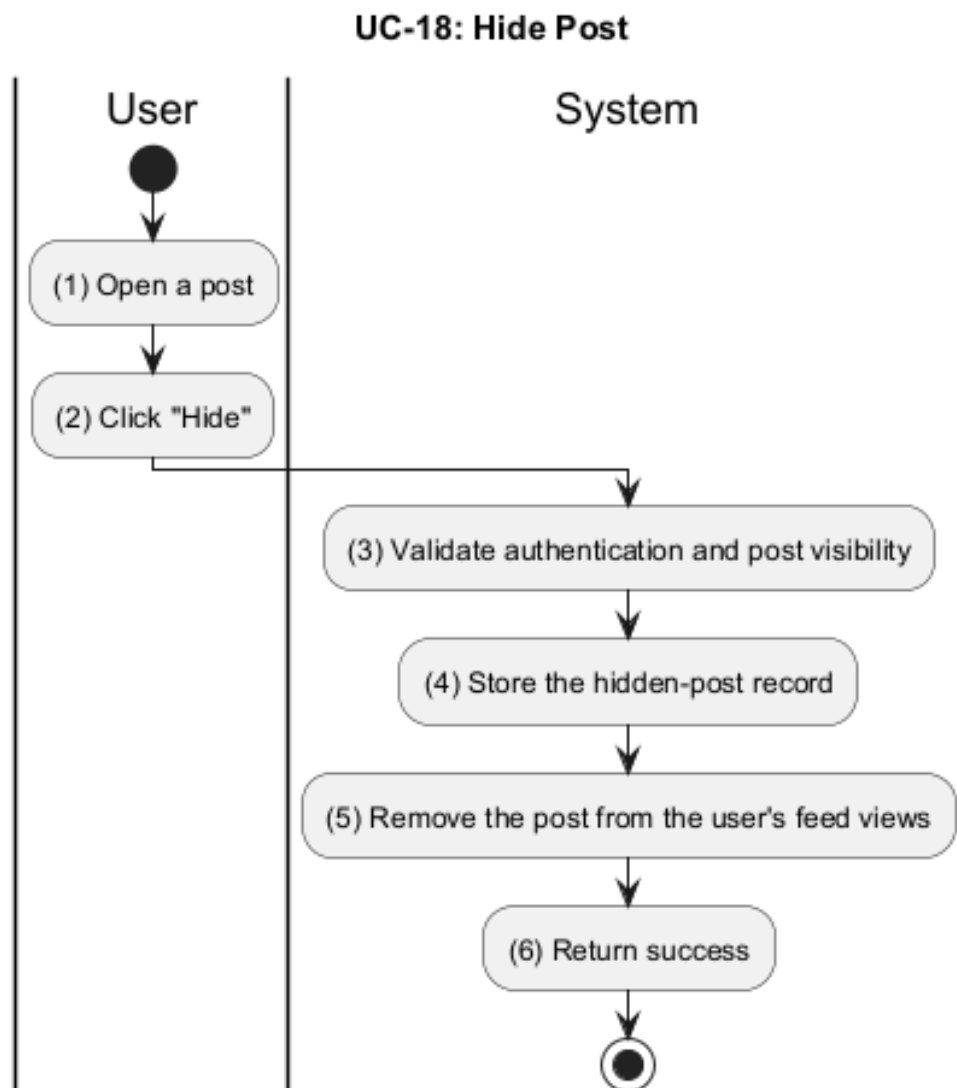


Figure 3.19: Activity Diagram for UC-18

| ID         | Activ-<br>ity | Rule Type        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------|---------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC18-01 | (3)           | Validation Rules | <b>Validate Authentication and Post Ownership:</b> <ul style="list-style-type: none"> <li>The controller requires an authenticated user before hiding a post.</li> <li>if <b>authUser == null</b>, then the controller returns a forbidden response.</li> <li>The service loads the post by ID before hiding it.</li> <li>if the post cannot be found, then the service returns the post lookup error.</li> <li>if the authenticated user is not the post author, then the service returns a forbidden error.</li> </ul> |
| BR-UC18-02 | (4-5)         | Workflow Rules   | <b>Mark the Post Hidden and Remove It From Visible Feeds:</b> <ul style="list-style-type: none"> <li>The service updates the post record by setting <b>is_hidden = true</b>.</li> <li>The hidden marker keeps the post out of normal feed queries because feed filters exclude hidden posts.</li> </ul>                                                                                                                                                                                                                  |
| BR-UC18-03 | (6)           | Message Rules    | <b>Return Hide Success Message:</b> <ul style="list-style-type: none"> <li>This use case has no dedicated success toast in the current code; the updated hidden state is returned inline after storage.</li> </ul>                                                                                                                                                                                                                                                                                                       |
| BR-UC18-04 | (6)           | Message Rules    | <b>Return Hide Error Message:</b> <ul style="list-style-type: none"> <li>Hide-post failures shall return <b>ERR-10</b>, <b>ERR-13</b>, <b>ERR-26</b>, or <b>ERR-53</b>.</li> </ul>                                                                                                                                                                                                                                                                                                                                       |

Table 3.37: Business Rules for UC-18 - Hide Post

### 3.1.19 UC-19 - Manage Drafts

|                       |                                                                                                       |
|-----------------------|-------------------------------------------------------------------------------------------------------|
| <b>Name</b>           | Manage Drafts                                                                                         |
| <b>Description</b>    | This use case describes how a user saves unfinished posts as drafts and publishes them later.         |
| <b>Actor</b>          | User                                                                                                  |
| <b>Trigger</b>        | When the user saves, edits, opens, or publishes a draft.                                              |
| <b>Pre-condition</b>  | The user is logged in to the system.<br>The user has draft access.                                    |
| <b>Post-condition</b> | The draft is saved, updated, or deleted after publishing.<br>A published draft becomes a normal post. |

Table 3.38: Use Case Specification - Manage Drafts

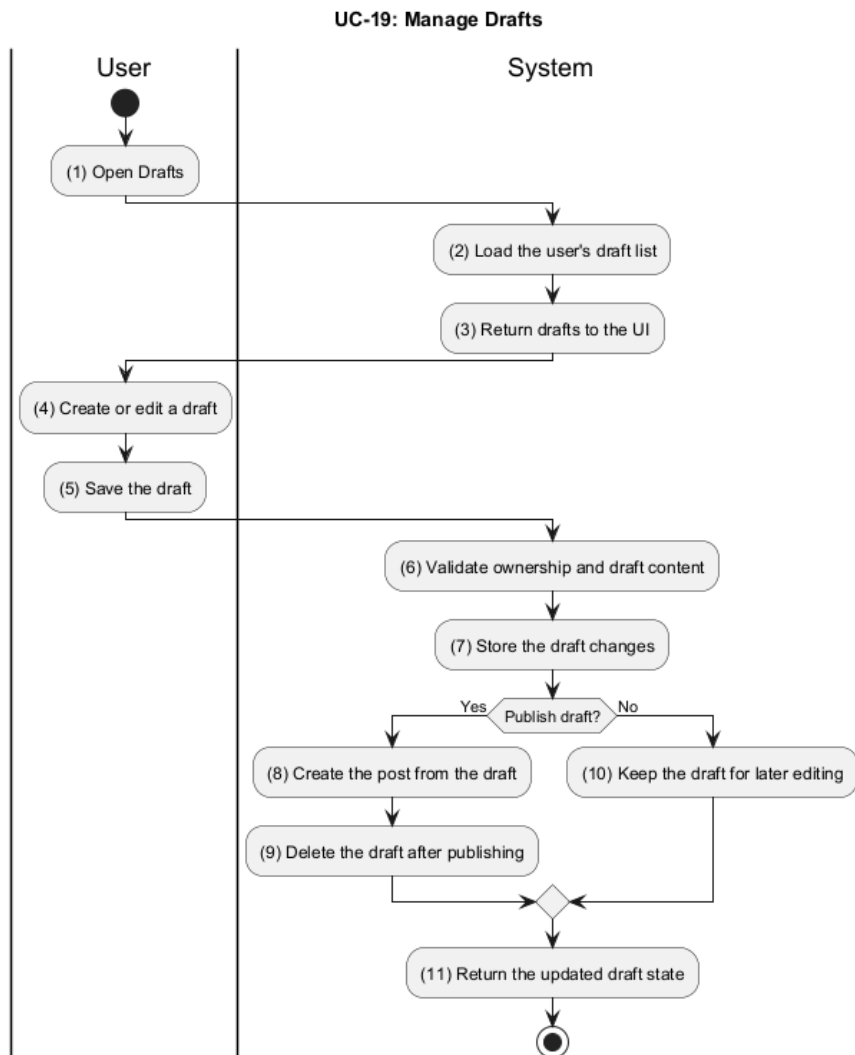


Figure 3.20: Activity Diagram for UC-19

| ID         | Activity | Rule Type           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------|----------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC19-01 | (2–3)    | Workflow Rules      | <b>Load the User’s Draft List:</b> <ul style="list-style-type: none"> <li>The service loads drafts by author using <i>GetDraftsByAuthor()</i>.</li> <li>The service returns the draft summaries with pagination metadata.</li> </ul>                                                                                                                                                                                                                                                                                                                                 |
| BR-UC19-02 | (6)      | Authorization Rules | <b>Validate Draft Ownership and Payload:</b> <ul style="list-style-type: none"> <li>The controller requires an authenticated user before draft operations.</li> <li>if <code>authUser == null</code>, then the controller returns a forbidden response.</li> <li>The service parses the author ID and draft ID to ObjectIDs.</li> <li>if either ID cannot be parsed, then the service returns an invalid-ID error.</li> <li>The service loads the draft and if the draft author does not match the requester, then the service returns a forbidden error.</li> </ul> |
| BR-UC19-03 | (6–7)    | Workflow Rules      | <b>Store Draft Changes:</b> <ul style="list-style-type: none"> <li>The service creates or updates the draft record with community, type, title, text, tags, images, videos, and poll data.</li> <li>The draft record updates <b>UpdatedAt</b> whenever changes are saved.</li> <li>if the draft does not exist or the author is invalid, then the operation stops before saving.</li> </ul>                                                                                                                                                                          |

Table 3.39: Business Rules for UC-19 - Manage Drafts

| ID         | Activ-ity | Rule Type      | Description                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------|-----------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC19-04 | (8–10)    | Workflow Rules | <b>Publish or Keep the Draft:</b> <ul style="list-style-type: none"> <li>if publish is requested, then the service validates that title, community, and type are present.</li> <li>The service creates a post from the draft using <i>CreatePost()</i>.</li> <li>After a successful publish, the service deletes the draft record.</li> <li>if publish is not requested, then the service keeps the draft for later editing.</li> </ul> |
| BR-UC19-05 | (11)      | Message Rules  | <b>Return Updated Draft State:</b> <ul style="list-style-type: none"> <li>Draft actions shall use MSG-31, MSG-16, MSG-32, and MSG-86 where applicable.</li> <li>Successful publish or update responses may return the updated draft or post inline without a dedicated additional toast.</li> </ul>                                                                                                                                     |
| BR-UC19-06 | (11)      | Message Rules  | <b>Return Draft Error Message:</b> <ul style="list-style-type: none"> <li>Draft validation or lookup failures shall return ERR-58, ERR-26, ERR-10, ERR-39, or ERR-46.</li> </ul>                                                                                                                                                                                                                                                        |

Table 3.39: Business Rules for UC-19 - Manage Drafts

### 3.1.20 UC-20 - Private Messaging

|                       |                                                                                                      |
|-----------------------|------------------------------------------------------------------------------------------------------|
| <b>Name</b>           | Private Messaging                                                                                    |
| <b>Description</b>    | This use case describes how users exchange instant one-to-one messages through the messaging system. |
| <b>Actor</b>          | User                                                                                                 |
| <b>Trigger</b>        | When the user opens a conversation and sends a message.                                              |
| <b>Pre-condition</b>  | The user is logged in to the system.<br>The conversation channel exists and is accessible.           |
| <b>Post-condition</b> | The message is delivered to the conversation members.<br>The channel read state is updated.          |

Table 3.40: Use Case Specification - Private Messaging

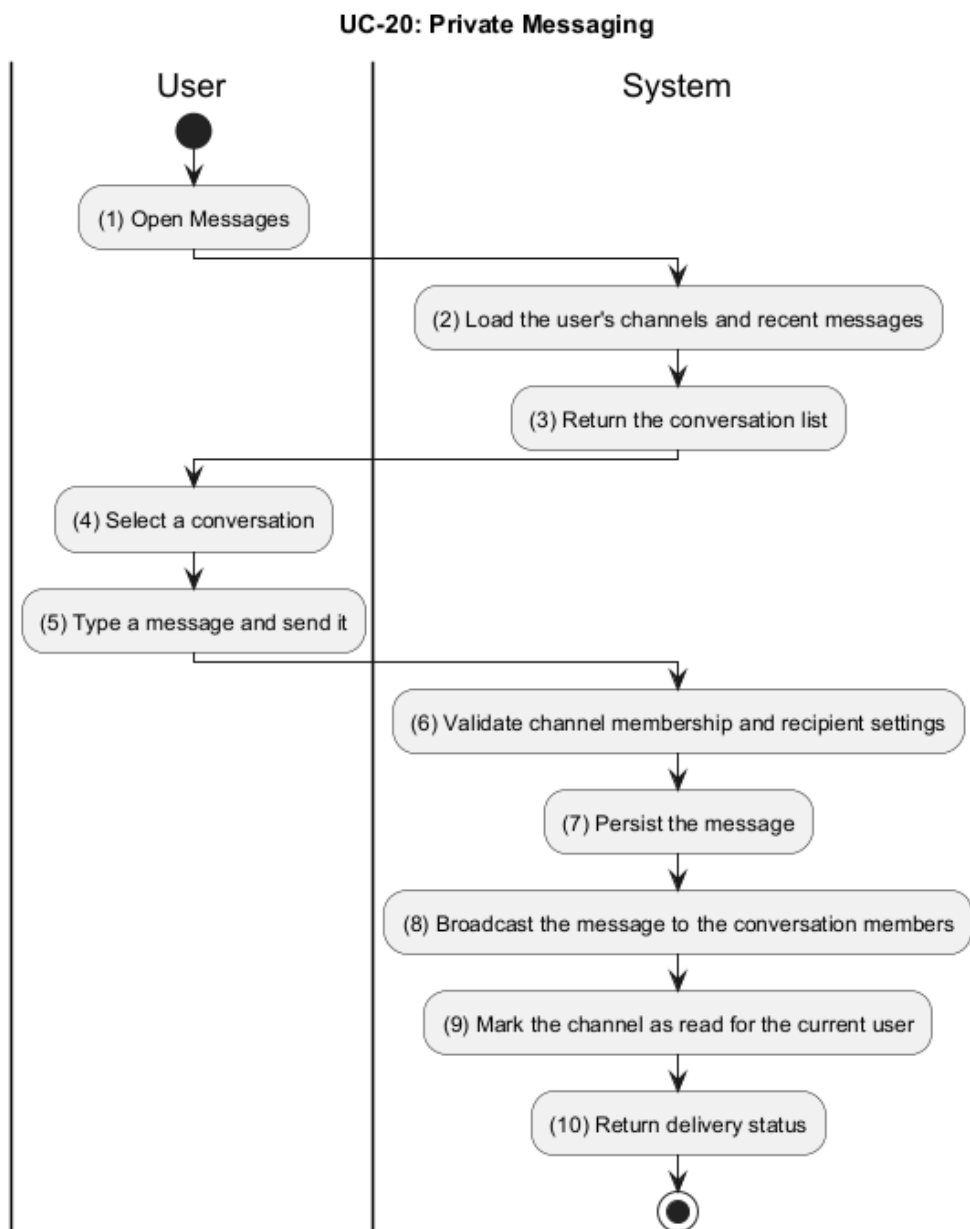


Figure 3.21: Activity Diagram for UC-20

| ID         | Activ-ity | Rule Type           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------|-----------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC20-01 | (2)       | Authorization Rules | <b>Load the User's Conversations:</b> <ul style="list-style-type: none"> <li>The controller requires an authenticated user before loading conversation data.</li> <li>if <b>authUser == null</b>, then the controller returns a forbidden response.</li> <li>The channel service only returns channels when the requester matches the target user ID.</li> <li>if the requester is not the same as the target user, then the service returns a forbidden error.</li> </ul>                                                                                                                                                                                                                   |
| BR-UC20-02 | (2-3)     | Workflow Rules      | <b>Return Conversation List and Recent Message State:</b> <ul style="list-style-type: none"> <li>The service loads the user's channels using <i>GetChannelsByUserID()</i>.</li> <li>The service returns the conversation list together with pagination data.</li> <li>The message filter endpoint returns the user's messages only when the requester is a member of the channel.</li> </ul>                                                                                                                                                                                                                                                                                                 |
| BR-UC20-03 | (6)       | Validation Rules    | <b>Validate New Message Payload and Channel Access:</b> <ul style="list-style-type: none"> <li>The websocket hub decodes the <b>new_message</b> payload before publishing the event.</li> <li>if the payload is malformed or required fields are missing, then the hub sends an error message back to the sender.</li> <li>The message service validates that <b>temp_message_id</b>, <b>channel_id</b>, <b>sender_id</b>, <b>sender_username</b>, and <b>content</b> are present.</li> <li>The service loads the target channel and verifies that the sender is a member of that channel.</li> <li>The service checks each recipient's direct-message setting before continuing.</li> </ul> |

Table 3.41: Business Rules for UC-20 - Private Messaging

| ID         | Activ-<br>ity | Rule Type      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------|---------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC20-04 | (7-8)         | Workflow Rules | <b>Persist and Broadcast the Message:</b> <ul style="list-style-type: none"> <li>The service creates a <code>Message</code> with <code>IsSend = false</code>, <code>IsRead = false</code>, and <code>IsDeleted = false</code>.</li> <li>The repository stores the message and returns the created record.</li> <li>The message service publishes a broadcast event so the websocket hub can send an ACK to the sender and a delivery event to the recipients.</li> <li>if persistence fails, then the service publishes a message error event instead of broadcasting success.</li> </ul> |
| BR-UC20-05 | (9)           | Workflow Rules | <b>Mark the Channel as Read:</b> <ul style="list-style-type: none"> <li>The message service marks the channel as read only for channel members.</li> <li>The repository updates unread messages to <code>is_read = true</code> for messages not sent by the current user.</li> </ul>                                                                                                                                                                                                                                                                                                      |
| BR-UC20-06 | (10)          | Message Rules  | <b>Return Delivery Status:</b> <ul style="list-style-type: none"> <li>Successful message delivery returns the stored payload inline through the websocket ACK and broadcast flow; this use case does not define a dedicated success toast.</li> </ul>                                                                                                                                                                                                                                                                                                                                     |
| BR-UC20-07 | (10)          | Message Rules  | <b>Return Messaging Error:</b> <ul style="list-style-type: none"> <li>Message-delivery failures shall use MSG-45, MSG-81, or MSG-82.</li> <li>Channel and access failures shall return ERR-62, ERR-63, ERR-13, ERR-26, or ERR-10.</li> </ul>                                                                                                                                                                                                                                                                                                                                              |

Table 3.41: Business Rules for UC-20 - Private Messaging



### 3.1.21 UC-21 - Manage Notifications

|                       |                                                                                                  |
|-----------------------|--------------------------------------------------------------------------------------------------|
| <b>Name</b>           | Manage Notifications                                                                             |
| <b>Description</b>    | This use case describes how a user views and clears system notifications.                        |
| <b>Actor</b>          | User                                                                                             |
| <b>Trigger</b>        | When the user opens the notifications page or marks notifications as read.                       |
| <b>Pre-condition</b>  | The user is logged in to the system.<br>Notifications exist for the account.                     |
| <b>Post-condition</b> | The selected notifications are marked as read or deleted.<br>The notification list is refreshed. |

Table 3.42: Use Case Specification - Manage Notifications

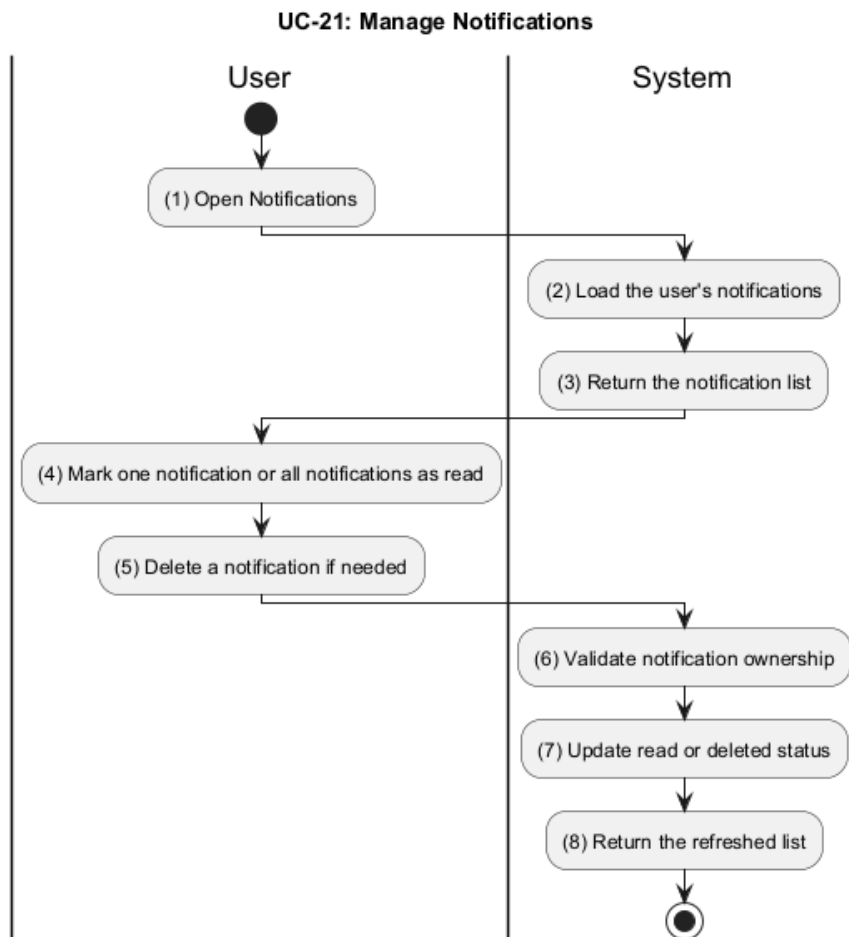


Figure 3.22: Activity Diagram for UC-21

| ID                | Activ-<br>ity | Rule Type           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------|---------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>BR-UC21-01</b> | (2–3)         | Workflow Rules      | <b>Load Notifications:</b> <ul style="list-style-type: none"> <li>The system loads notifications for <b>recipientID</b> using <i>GetNotifications()</i>.</li> <li>The system returns the notification list with pagination data.</li> </ul>                                                                                                                                                                                                                                               |
| <b>BR-UC21-02</b> | (6)           | Authorization Rules | <b>Validate Notification Ownership:</b> <ul style="list-style-type: none"> <li>The notification routes require authentication before any operation is processed.</li> <li>The system only updates or deletes notifications that belong to the authenticated user.</li> <li>If the notification ID is missing or the current user does not own the record, the operation is rejected.</li> </ul>                                                                                           |
| <b>BR-UC21-03</b> | (6)           | Validation Rules    | <b>Validate Notification ID for Single-Item Actions:</b> <ul style="list-style-type: none"> <li>If the user marks one notification as read or deletes one notification, the system validates <b>notificationID</b>.</li> <li>The service converts <b>notificationID</b> to an object ID before calling the repository.</li> </ul>                                                                                                                                                         |
| <b>BR-UC21-04</b> | (7)           | Workflow Rules      | <b>Update Notification State:</b> <ul style="list-style-type: none"> <li>If the user marks all notifications as read, the system updates them using <i>MarkAllAsRead()</i>.</li> <li>If the user marks a single notification as read, the system updates it using <i>MarkAsRead()</i>.</li> <li>If the user deletes a notification, the system removes it using <i>DeleteNotification()</i>.</li> <li>The repository filters updates by both notification ID and recipient ID.</li> </ul> |

Table 3.43: Business Rules for UC-21 - Manage Notifications

| ID                | Activ-ity | Rule Type     | Description                                                                                                                                                                                                                                                                                                                                    |
|-------------------|-----------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>BR-UC21-05</b> | (8)       | Message Rules | <b>Return the Refreshed Notification List:</b> <ul style="list-style-type: none"> <li>Notification-related UI feedback shall use MSG-80, MSG-46, or MSG-47 when the notification action is an invitation.</li> <li>Successful notification-state changes may return the refreshed list inline without a dedicated additional toast.</li> </ul> |
| <b>BR-UC21-06</b> | (8)       | Message Rules | <b>Return Notification Error Message:</b> <ul style="list-style-type: none"> <li>Notification access or lookup failures shall return ERR-26, ERR-13, ERR-10, or ERR-2.</li> </ul>                                                                                                                                                              |

Table 3.43: Business Rules for UC-21 - Manage Notifications

### 3.1.22 UC-22 - Report Violation

|                       |                                                                                                      |
|-----------------------|------------------------------------------------------------------------------------------------------|
| <b>Name</b>           | Report Violation                                                                                     |
| <b>Description</b>    | This use case describes how a user reports inappropriate content or behavior for review.             |
| <b>Actor</b>          | User                                                                                                 |
| <b>Trigger</b>        | When the user submits a report form.                                                                 |
| <b>Pre-condition</b>  | The user is logged in to the system.<br>The reported target is available.                            |
| <b>Post-condition</b> | The report is stored in the moderation queue.<br>The report can be reviewed by moderators or admins. |

Table 3.44: Use Case Specification - Report Violation

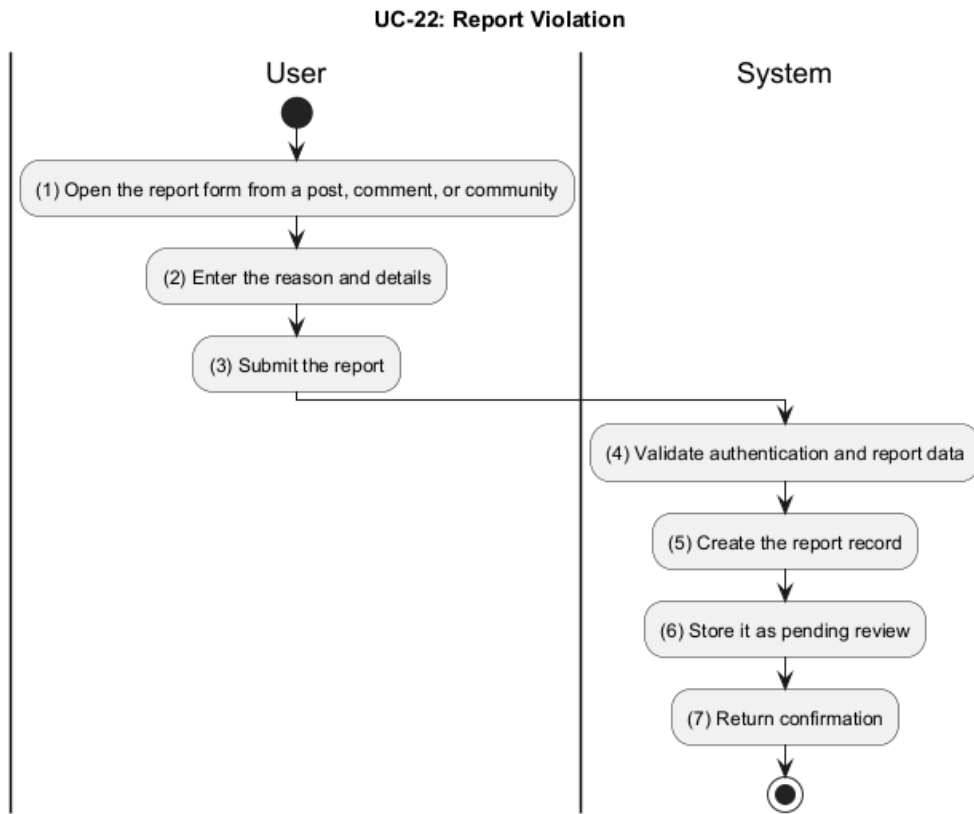


Figure 3.23: Activity Diagram for UC-22

| ID         | Activ-ity | Rule Type                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------|-----------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC22-01 | (4)       | Authorization / Validation Rules | <b>Submit Report Request:</b> <ul style="list-style-type: none"> <li>The report action requires an authenticated user.</li> <li>The report form is opened from a post detail or post card action.</li> <li>The controller binds the JSON body to <code>ReportPostRequest</code>.</li> <li>The submitted <b>reason</b> is required.</li> <li>The submitted <b>description</b> is optional.</li> <li>if <code>authUser == null</code>, then return a forbidden response.</li> <li>if the payload is invalid, then return a bad request response.</li> </ul> |

Table 3.45: Business Rules for UC-22 - Report Violation

| ID         | Activ-<br>ity | Rule Type        | Description                                                                                                                                                                                                                                                                                                                                             |
|------------|---------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC22-02 | (4)           | Validation Rules | <b>Validate the Target Post:</b> <ul style="list-style-type: none"> <li>The service parses <b>reporterID</b> and <b>postID</b> to object IDs.</li> <li>The service loads the target post before creating the report.</li> <li>if the post does not exist or the IDs are invalid, then stop the operation and return the corresponding error.</li> </ul> |
| BR-UC22-03 | (4)           | Constraint Rules | <b>Prevent Invalid Reports:</b> <ul style="list-style-type: none"> <li>The system rejects reports submitted by the author of the target post.</li> <li>The system rejects duplicate reports from the same user for the same post.</li> </ul>                                                                                                            |
| BR-UC22-04 | (5–6)         | Workflow Rules   | <b>Create and Store the Report:</b> <ul style="list-style-type: none"> <li>The service creates a <b>Report</b> with <b>TargetType = post</b>.</li> <li>The repository stores the report with <b>IsDeleted = false</b>.</li> <li>The report remains pending until reviewed by moderators or administrators.</li> </ul>                                   |
| BR-UC22-05 | (7)           | Message Rules    | <b>Return Operation Result:</b> <ul style="list-style-type: none"> <li>Report submission shall use MSG-23 on success and MSG-24 on failure.</li> <li>Report validation failures shall return ERR-57, ERR-53, ERR-59, ERR-28, ERR-26, or ERR-10.</li> </ul>                                                                                              |

Table 3.45: Business Rules for UC-22 - Report Violation

### 3.1.23 UC-23 - Create Community

|                       |                                                                                           |
|-----------------------|-------------------------------------------------------------------------------------------|
| <b>Name</b>           | Create Community                                                                          |
| <b>Description</b>    | This use case describes how a user creates a new community and becomes its owner.         |
| <b>Actor</b>          | User                                                                                      |
| <b>Trigger</b>        | When the user submits the create community form.                                          |
| <b>Pre-condition</b>  | The user is logged in to the system.<br>The community name is available and valid.        |
| <b>Post-condition</b> | The community is created successfully.<br>The creator becomes the owner and first member. |

Table 3.46: Use Case Specification - Create Community

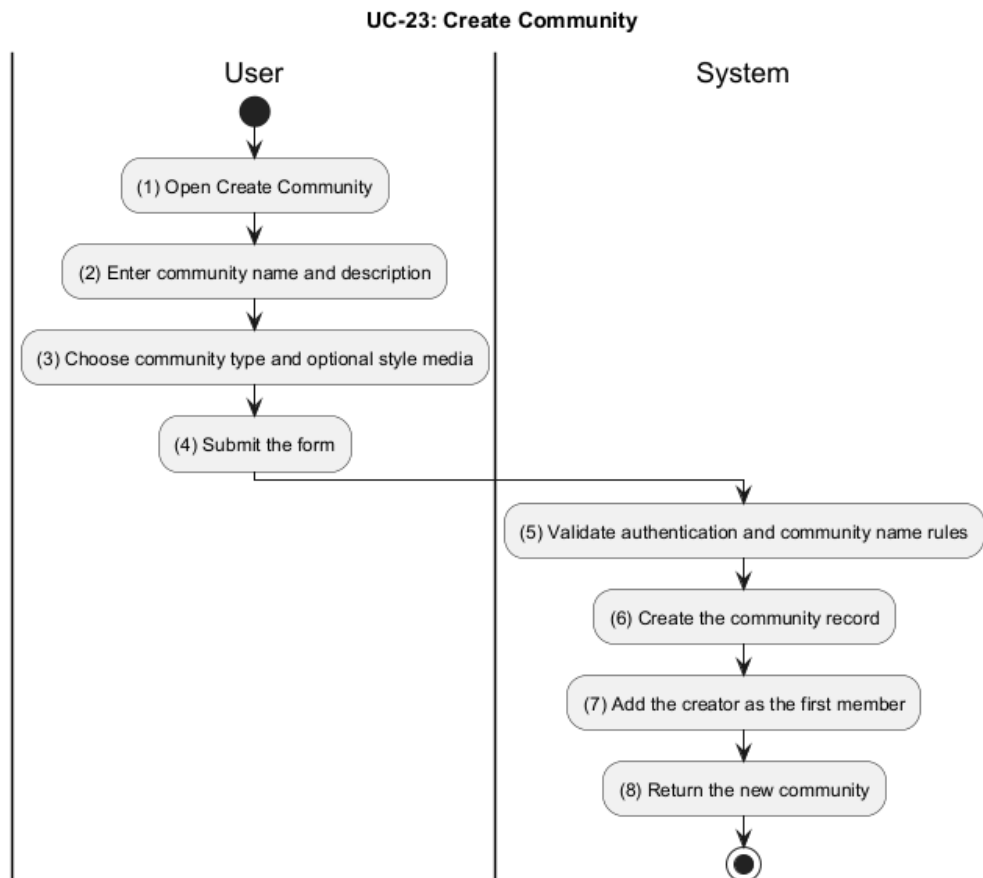


Figure 3.24: Activity Diagram for UC-23

| ID         | Activ-<br>ity | Rule Type           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------|---------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC23-01 | (5)           | Authorization Rules | <b>Open and Submit the Create Community Form:</b> <ul style="list-style-type: none"> <li>The create-community action requires an authenticated user.</li> <li>if <b>authUser == null</b>, then the controller returns a forbidden response.</li> <li>The controller passes the authenticated user's ID to the service when creating the community.</li> </ul>                                                                                                |
| BR-UC23-02 | (5)           | Validation Rules    | <b>Validate Community Input:</b> <ul style="list-style-type: none"> <li>The controller binds the request to <b>CreateCommunityRequest</b>.</li> <li>The community name is required and must satisfy the configured length rule.</li> <li>The community description, when provided, must respect its length limit.</li> <li>The request may include optional avatar, banner, setting, rules, moderators, creator metadata, and adult-content flag.</li> </ul> |
| BR-UC23-03 | (5)           | Constraint Rules    | <b>Prevent Duplicate Community Names:</b> <ul style="list-style-type: none"> <li>The repository enforces a unique index on community name.</li> <li>if the same community name already exists, then the service returns a community-name-exists error.</li> </ul>                                                                                                                                                                                            |
| BR-UC23-04 | (6)           | Workflow Rules      | <b>Create the Community Record:</b> <ul style="list-style-type: none"> <li>The service creates a community with the submitted fields and the creator ID.</li> <li>The new community is stored with <b>IsDeleted = false</b> and <b>IsBanned = false</b>.</li> <li>The service persists the community through <b>CreateCommunity()</b>.</li> </ul>                                                                                                            |

Table 3.47: Business Rules for UC-23 - Create Community

| ID         | Activity | Rule Type      | Description                                                                                                                                                                                                                                          |
|------------|----------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC23-05 | (7)      | Workflow Rules | <b>Create the Initial Membership:</b> <ul style="list-style-type: none"> <li>After the community is created, the service creates a membership record for the creator.</li> <li>The creator becomes the first member of the new community.</li> </ul> |
| BR-UC23-06 | (8)      | Message Rules  | <b>Return the New Community:</b> <ul style="list-style-type: none"> <li>Successful community creation shall use MSG-62.</li> </ul>                                                                                                                   |
| BR-UC23-07 | (8)      | Message Rules  | <b>Return Community Creation Error:</b> <ul style="list-style-type: none"> <li>Failed community creation shall use MSG-63.</li> <li>Validation failures shall return ERR-40, ERR-26, ERR-10, or ERR-14.</li> </ul>                                   |

Table 3.47: Business Rules for UC-23 - Create Community

### 3.1.24 UC-24 - Approve / Reject Post

|                       |                                                                                                              |
|-----------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Name</b>           | Approve / Reject Post                                                                                        |
| <b>Description</b>    | This use case describes how a moderator reviews pending or edited posts and decides whether to publish them. |
| <b>Actor</b>          | Moderator                                                                                                    |
| <b>Trigger</b>        | When the moderator opens the moderation queue and chooses approve or reject.                                 |
| <b>Pre-condition</b>  | The moderator is logged in to the system.<br>The target post is pending review or edited.                    |
| <b>Post-condition</b> | The post is approved or rejected.<br>The moderation result is stored and reflected in the community.         |

Table 3.48: Use Case Specification - Approve / Reject Post



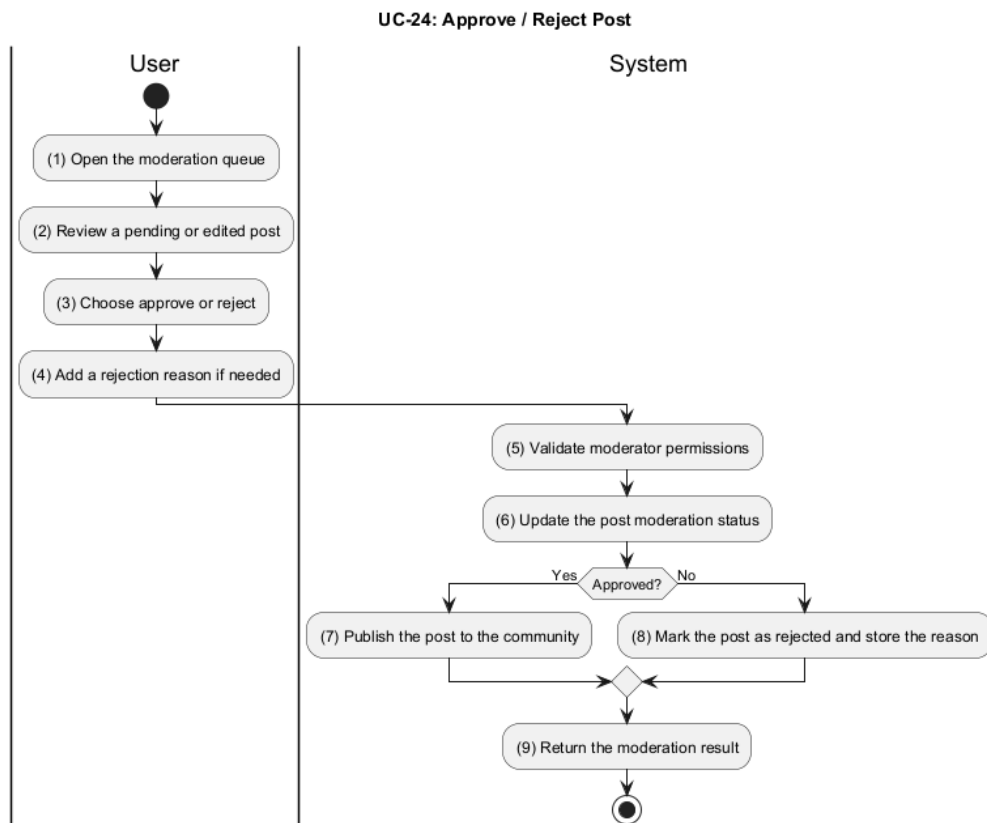


Figure 3.25: Activity Diagram for UC-24

| ID         | Activ-ity | Rule Type           | Description                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------|-----------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC24-01 | (5)       | Authorization Rules | <b>Open the Moderation Queue and Validate Moderator Permissions:</b> <ul style="list-style-type: none"> <li>The moderation queue endpoints require authentication.</li> <li>The service checks that the requester is a moderator of the target community before loading or moderating posts.</li> <li>if <b>authUser == null</b> or the user is not a moderator, then the system returns a forbidden response.</li> </ul> |

Table 3.49: Business Rules for UC-24 - Approve / Reject Post

| ID         | Activ-<br>ity | Rule Type        | Description                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------|---------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC24-02 | (2)           | Workflow Rules   | <b>Load Pending or Edited Posts:</b> <ul style="list-style-type: none"> <li>The system loads pending posts with <i>GetPendingPosts()</i>.</li> <li>The system loads edited posts with <i>GetEditedPosts()</i>.</li> <li>The queued posts are filtered by community ID and moderation status before being returned.</li> </ul>                                                                                                          |
| BR-UC24-03 | (5)           | Validation Rules | <b>Validate the Moderation Decision:</b> <ul style="list-style-type: none"> <li>The controller binds the request to <code>ModeratePostRequest</code>.</li> <li>The request must include the community ID, post ID, and approve flag.</li> <li>if the moderation route parameters are missing or the request body is invalid, then the controller returns a bad request response.</li> </ul>                                            |
| BR-UC24-04 | (5)           | Validation Rules | <b>Validate the Target Post:</b> <ul style="list-style-type: none"> <li>The service loads the target post before applying the decision.</li> <li>The post must belong to the same community as the moderation request.</li> <li>The post must still be in <b>moderation_status = pending</b>.</li> <li>if the post is missing, in another community, or not pending, then the operation stops with the corresponding error.</li> </ul> |
| BR-UC24-05 | (7)           | Workflow Rules   | <b>Approve the Post:</b> <ul style="list-style-type: none"> <li>if <b>approve = true</b>, then the service sets <b>moderation_status = approved</b>.</li> <li>The service updates <b>moderated_at</b> for the post.</li> <li>The service publishes a <code>PostApprovedEvent</code> so the post becomes visible to the community flow.</li> </ul>                                                                                      |

Table 3.49: Business Rules for UC-24 - Approve / Reject Post (Continued)

| ID         | Activ-ity | Rule Type      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------|-----------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC24-06 | (8)       | Workflow Rules | <b>Reject the Post:</b> <ul style="list-style-type: none"> <li>if <b>approve</b> = <b>false</b>, then the service sets <b>moderation_status</b> = <b>rejected</b>.</li> <li>The service stores a <b>ModerationResult</b> with <b>IsViolation</b> = <b>true</b> and the rejection reason.</li> <li>if no rejection reason is supplied, then the service uses the default moderator rejection reason.</li> <li>The service publishes a <b>PostRejectedEvent</b> for downstream notifications.</li> </ul> |
| BR-UC24-07 | (9)       | Message Rules  | <b>Return the Moderation Result:</b> <ul style="list-style-type: none"> <li>Moderator approval shall use MSG-66; approval failures shall use MSG-67.</li> <li>Rejection or removal flows may reuse MSG-87 and MSG-27 when the post is removed from the queue.</li> </ul>                                                                                                                                                                                                                               |
| BR-UC24-08 | (9)       | Message Rules  | <b>Return Moderation Error Message:</b> <ul style="list-style-type: none"> <li>Moderation failures shall return ERR-53, ERR-13, ERR-39, or ERR-26.</li> </ul>                                                                                                                                                                                                                                                                                                                                          |

Table 3.49: Business Rules for UC-24 - Approve / Reject Post

### 3.1.25 UC-25 - Ban / Mute User

|                       |                                                                                                         |
|-----------------------|---------------------------------------------------------------------------------------------------------|
| <b>Name</b>           | Ban / Mute User                                                                                         |
| <b>Description</b>    | This use case describes how a moderator restricts a user's participation in a community.                |
| <b>Actor</b>          | Moderator                                                                                               |
| <b>Trigger</b>        | When the moderator chooses to ban or mute a user.                                                       |
| <b>Pre-condition</b>  | The moderator is logged in to the system.<br>The target user belongs to the community.                  |
| <b>Post-condition</b> | The ban or mute restriction is stored.<br>The user is prevented from performing the restricted actions. |

Table 3.50: Use Case Specification - Ban / Mute User

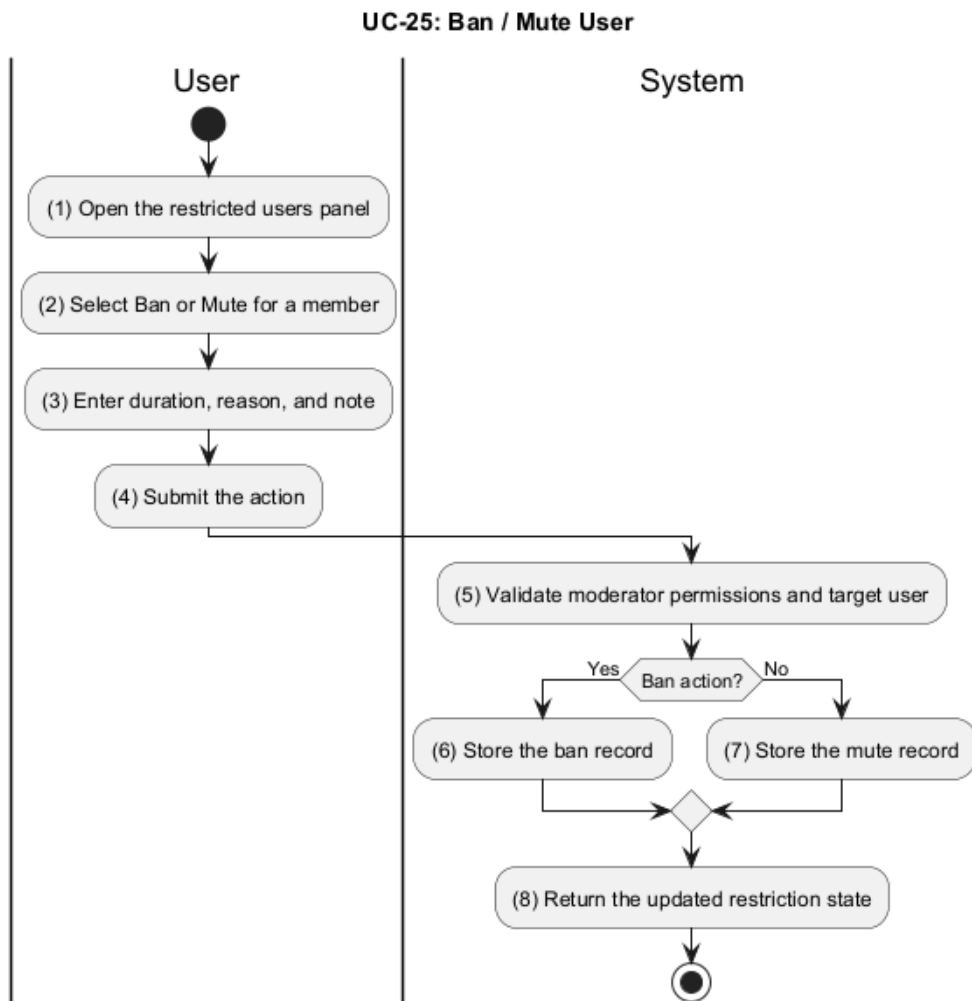


Figure 3.26: Activity Diagram for UC-25

| ID         | Activ-ity | Rule Type           | Description                                                                                                                                                                                                                                                                                                                                                                            |
|------------|-----------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC25-01 | (5)       | Authorization Rules | <b>Open the Restricted Users Panel and Validate Moderator Access:</b> <ul style="list-style-type: none"> <li>The ban and mute actions require authentication.</li> <li>The service checks that the requester is a moderator of the target community.</li> <li>if <b>authUser == null</b> or the requester is not a moderator, then the system returns a forbidden response.</li> </ul> |

Table 3.51: Business Rules for UC-25 - Ban / Mute User

| ID         | Activ-<br>ity | Rule Type        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------|---------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC25-02 | (5)           | Validation Rules | <b>Validate the Restriction Request:</b> <ul style="list-style-type: none"> <li>The system binds the request to <b>CommunityBanUserRequest</b>.</li> <li>The request must include <b>communityID</b>, <b>userID</b>, <b>type</b>, and <b>lengthDays</b>.</li> <li>The restriction type must be either <b>banned</b> or <b>muted</b>.</li> <li>The reason field is accepted by the API and is stored with the restriction record.</li> </ul>                                                                                       |
| BR-UC25-03 | (5)           | Validation Rules | <b>Validate the Target User and Community:</b> <ul style="list-style-type: none"> <li>The service converts the community ID, user ID, and requester ID to object IDs.</li> <li>if any ID is invalid, then the operation stops with the corresponding error.</li> </ul>                                                                                                                                                                                                                                                            |
| BR-UC25-04 | (6)           | Workflow Rules   | <b>Store the Ban Record:</b> <ul style="list-style-type: none"> <li>if <b>type = banned</b>, then the service creates a <b>CommunityBan</b> record with <b>Type = banned</b>.</li> <li>The service calculates <b>ExpiresAt</b> from <b>lengthDays</b>.</li> <li>The repository clears any existing active restriction for the same user and community before inserting the new ban.</li> <li>The ban record stores the target user, community, reason, requester, banned time, expiration time, and soft-delete flags.</li> </ul> |

Table 3.51: Business Rules for UC-25 - Ban / Mute User (Continued)

| ID         | Activ-ity | Rule Type      | Description                                                                                                                                                                                                                                                                                                                                                                                      |
|------------|-----------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC25-05 | (7)       | Workflow Rules | <b>Store the Mute Record:</b> <ul style="list-style-type: none"> <li>if <b>type = muted</b>, then the service creates a <b>CommunityBan</b> record with <b>Type = muted</b>.</li> <li>The same expiration and audit fields are stored for mute actions.</li> <li>The repository clears any existing active restriction for the same user and community before inserting the new mute.</li> </ul> |
| BR-UC25-06 | (8)       | Message Rules  | <b>Return the Updated Restriction State:</b> <ul style="list-style-type: none"> <li>Restriction actions shall use MSG-68, MSG-69, MSG-70, MSG-71, MSG-72, or MSG-73 as applicable.</li> </ul>                                                                                                                                                                                                    |
| BR-UC25-07 | (8)       | Message Rules  | <b>Return Restriction Error Message:</b> <ul style="list-style-type: none"> <li>Validation or authorization failures shall return ERR-28, ERR-39, ERR-47, ERR-13, or ERR-26.</li> </ul>                                                                                                                                                                                                          |

Table 3.51: Business Rules for UC-25 - Ban / Mute User

### 3.1.26 UC-26 - Configure Community

|                       |                                                                                                |
|-----------------------|------------------------------------------------------------------------------------------------|
| <b>Name</b>           | Configure Community                                                                            |
| <b>Description</b>    | This use case describes how a community owner updates rules, privacy, and appearance settings. |
| <b>Actor</b>          | Owner                                                                                          |
| <b>Trigger</b>        | When the owner opens the community settings page and saves changes.                            |
| <b>Pre-condition</b>  | The owner is logged in to the system.<br>The owner has permission to manage the community.     |
| <b>Post-condition</b> | The community configuration is updated.<br>The new settings are applied to the community.      |

Table 3.52: Use Case Specification - Configure Community

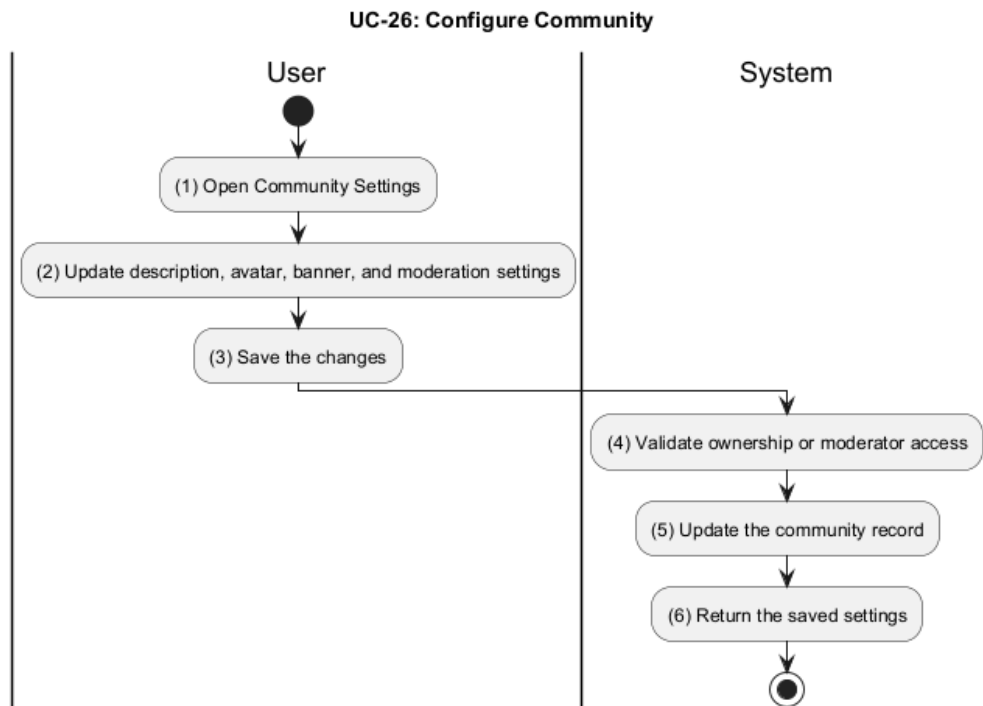


Figure 3.27: Activity Diagram for UC-26

| ID         | Activ-ity | Rule Type           | Description                                                                                                                                                                                                                                                                                                                                                                                             |
|------------|-----------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC26-01 | (4)       | Authorization Rules | <b>Open Community Settings and Validate Access:</b> <ul style="list-style-type: none"> <li>The community settings page requires a logged-in user.</li> <li>The service allows the update only when the requester is the community creator or an active moderator.</li> <li>if <b>authUser == null</b> or the requester lacks moderator access, then the system returns a forbidden response.</li> </ul> |

Table 3.53: Business Rules for UC-26 - Configure Community

| ID         | Activ-<br>ity | Rule Type        | Description                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------|---------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC26-02 | (4)           | Validation Rules | <b>Validate the Settings Request:</b> <ul style="list-style-type: none"> <li>The controller binds the request to <code>UpdateCommunityRequest</code>.</li> <li>The request must include the community ID.</li> <li>The request may include description, avatar, banner, moderation settings, and rules.</li> <li>if the payload cannot be bound, then the controller returns a bad request response.</li> </ul> |
| BR-UC26-03 | (5)           | Workflow Rules   | <b>Load the Community Record:</b> <ul style="list-style-type: none"> <li>The service loads the target community before applying changes.</li> <li>if the community does not exist, then the operation stops with the corresponding error.</li> </ul>                                                                                                                                                            |
| BR-UC26-04 | (5)           | Workflow Rules   | <b>Apply Community Field Updates:</b> <ul style="list-style-type: none"> <li>The service updates only the non-nil fields from the request.</li> <li>The configurable fields are description, avatar, banner, settings, and rules.</li> <li>The service rejects save attempts that do not change any field.</li> </ul>                                                                                           |
| BR-UC26-05 | (5)           | Constraint Rules | <b>Require at Least One Change:</b> <ul style="list-style-type: none"> <li>if no update fields are provided, then the service returns <code>ErrNoFieldsToUpdate</code>.</li> </ul>                                                                                                                                                                                                                              |
| BR-UC26-06 | (5)           | Workflow Rules   | <b>Persist the Community Settings:</b> <ul style="list-style-type: none"> <li>The service saves the updated community through <i><code>UpdateCommunity()</code></i>.</li> <li>The repository replaces the full community record after the in-memory updates are applied.</li> </ul>                                                                                                                             |

Table 3.53: Business Rules for UC-26 - Configure Community (Continued)



| ID         | Activ-ity | Rule Type     | Description                                                                                                                                                                                                     |
|------------|-----------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC26-07 | (6)       | Message Rules | <b>Return the Saved Community Settings:</b> <ul style="list-style-type: none"> <li>Successful community-configuration changes shall use MSG-74.</li> </ul>                                                      |
| BR-UC26-08 | (6)       | Message Rules | <b>Return Community Update Error:</b> <ul style="list-style-type: none"> <li>Failed updates shall use MSG-75.</li> <li>Access or validation failures shall return ERR-39, ERR-13, ERR-26, or ERR-14.</li> </ul> |

Table 3.53: Business Rules for UC-26 - Configure Community

### 3.1.27 UC-27 - Assign Roles

|                       |                                                                                               |
|-----------------------|-----------------------------------------------------------------------------------------------|
| <b>Name</b>           | Assign Roles                                                                                  |
| <b>Description</b>    | This use case describes how a community owner appoints moderators or other roles for members. |
| <b>Actor</b>          | Owner                                                                                         |
| <b>Trigger</b>        | When the owner invites or activates a role for another member.                                |
| <b>Pre-condition</b>  | The owner is logged in to the system.<br>The target member belongs to the community.          |
| <b>Post-condition</b> | The role assignment is stored successfully.<br>The target member gains the new permissions.   |

Table 3.54: Use Case Specification - Assign Roles

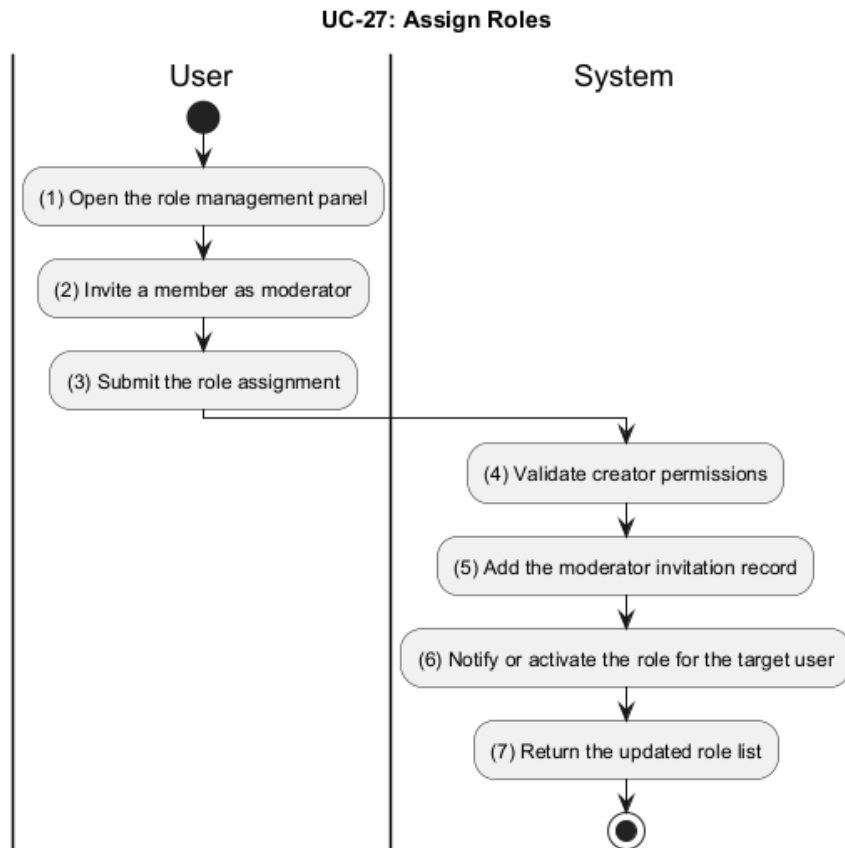


Figure 3.28: Activity Diagram for UC-27

| ID         | Activ-ity | Rule Type           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------|-----------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC27-01 | (4)       | Authorization Rules | <b>Open the Role Management Panel and Validate Creator Access:</b> <ul style="list-style-type: none"> <li>The role assignment actions require authentication.</li> <li>The service allows adding moderators only when the requester is the community creator.</li> <li>The target user must be invited from the creator's community role management flow.</li> <li>if <b>authUser == null</b> or the requester is not the creator, then the system returns a forbidden response.</li> </ul> |

Table 3.55: Business Rules for UC-27 - Assign Roles

| ID         | Activ-<br>ity | Rule Type        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------|---------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC27-02 | (4)           | Validation Rules | <b>Validate the Moderator Invitation Request:</b> <ul style="list-style-type: none"> <li>The controller binds the request to <code>AddModeratorRequest</code>.</li> <li>The request must include the community ID and at least one moderator user ID.</li> <li>The target user IDs are resolved from the invited usernames in the current UI flow.</li> <li>if the request payload cannot be bound or contains no target users, then the controller returns a bad request response.</li> </ul> |
| BR-UC27-03 | (5)           | Constraint Rules | <b>Prevent Duplicate Active Moderators:</b> <ul style="list-style-type: none"> <li>The service rejects a user who is already an active moderator.</li> <li>Existing inactive invitations for the same user are removed before a new invite is stored.</li> <li>if the moderator already exists as active, then the service returns <code>ErrModeratorAlreadyExists</code>.</li> </ul>                                                                                                          |
| BR-UC27-04 | (5)           | Workflow Rules   | <b>Add the Moderator Invitation Record:</b> <ul style="list-style-type: none"> <li>The service loads the community and the invited user before storing the role change.</li> <li>The service creates a <code>Moderator</code> entry with <code>IsActive = false</code> and the current assignment time.</li> <li>The community record is replaced after the invitation is appended.</li> <li>The service publishes a <code>ModeratorAddedEvent</code> after persistence succeeds.</li> </ul>   |

Table 3.55: Business Rules for UC-27 - Assign Roles (Continued)

| ID         | Activ-ity | Rule Type      | Description                                                                                                                                                                                                                                                                                                                                                                         |
|------------|-----------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC27-05 | (6)       | Workflow Rules | <b>Activate the Moderator Role:</b> <ul style="list-style-type: none"> <li>The invited user can activate the moderator role through the activation endpoint.</li> <li>The repository marks the matching moderator entry as active and refreshes the assignment timestamp.</li> <li>if the invitation cannot be matched, then the repository returns a bad request error.</li> </ul> |
| BR-UC27-06 | (7)       | Message Rules  | <b>Return the Updated Role List:</b> <ul style="list-style-type: none"> <li>Role-assignment feedback shall use MSG-60 on success and MSG-61 on failure.</li> </ul>                                                                                                                                                                                                                  |
| BR-UC27-07 | (7)       | Message Rules  | <b>Return Role Assignment Error:</b> <ul style="list-style-type: none"> <li>Validation failures shall return ERR-28, ERR-39, ERR-42, ERR-46, ERR-43, ERR-44, or ERR-26.</li> </ul>                                                                                                                                                                                                  |

Table 3.55: Business Rules for UC-27 - Assign Roles

### 3.1.28 UC-28 - Handle Global Reports

|                       |                                                                                                 |
|-----------------------|-------------------------------------------------------------------------------------------------|
| <b>Name</b>           | Handle Global Reports                                                                           |
| <b>Description</b>    | This use case describes how a system admin reviews and acts on platform-wide violation reports. |
| <b>Actor</b>          | System Admin                                                                                    |
| <b>Trigger</b>        | When the admin opens the global reports dashboard.                                              |
| <b>Pre-condition</b>  | The admin is logged in to the system.<br>There are reports available for review.                |
| <b>Post-condition</b> | The selected reports are reviewed or deleted.<br>The report queue is updated.                   |

Table 3.56: Use Case Specification - Handle Global Reports

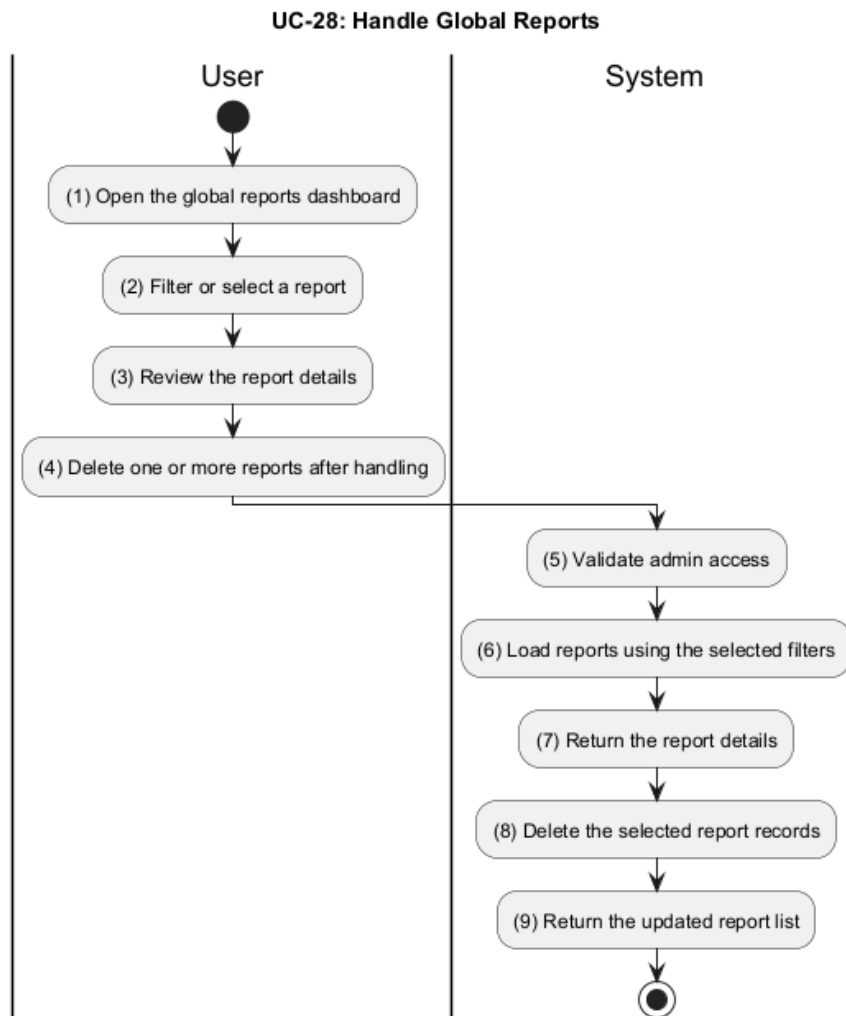


Figure 3.29: Activity Diagram for UC-28

| ID         | Activ-ity | Rule Type           | Description                                                                                                                                                                                                                                                                                                                                                                                 |
|------------|-----------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC28-01 | (5)       | Authorization Rules | <b>Open the Global Reports Dashboard and Validate Admin Access:</b> <ul style="list-style-type: none"> <li>The global reports routes require authentication and admin privileges.</li> <li>if the requester is not an admin, then the system returns a forbidden response.</li> <li>The dashboard and report detail actions are only available through the admin reports routes.</li> </ul> |

Table 3.57: Business Rules for UC-28 - Handle Global Reports

| ID         | Activ-<br>ity | Rule Type        | Description                                                                                                                                                                                                                                                                                                                                                        |
|------------|---------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC28-02 | (6)           | Validation Rules | <b>Validate the Report Filter Request:</b> <ul style="list-style-type: none"> <li>The system binds the query to <i>GetReportsFilterQuery</i>.</li> <li>The request may include reporter ID, target ID, target type, reason, date range, page, and page size.</li> <li>if the query cannot be bound, then the controller returns a bad request response.</li> </ul> |
| BR-UC28-03 | (7)           | Workflow Rules   | <b>Load Report Details and Filtered Results:</b> <ul style="list-style-type: none"> <li>The service loads reports using <i>GetReportsFilter()</i>.</li> <li>The admin can open a single report using <i>GetReportByIDAdmin()</i>.</li> </ul>                                                                                                                       |
| BR-UC28-04 | (8)           | Validation Rules | <b>Validate Selected Report IDs for Deletion:</b> <ul style="list-style-type: none"> <li>The controller binds batch deletion requests to <i>DeleteReportsRequest</i>.</li> <li>The request must contain at least one report ID.</li> <li>if the batch list is empty or the request is invalid, then the controller returns a bad request response.</li> </ul>      |
| BR-UC28-05 | (8)           | Workflow Rules   | <b>Delete Single or Multiple Reports:</b> <ul style="list-style-type: none"> <li>The admin can delete one report using <i>DeleteReportByIDAdmin()</i>.</li> <li>The admin can delete multiple reports using <i>DeleteReportsByIDAdmin()</i>.</li> <li>The service deletes the selected report records after validation succeeds.</li> </ul>                        |
| BR-UC28-06 | (9)           | Message Rules    | <b>Return the Updated Report List:</b> <ul style="list-style-type: none"> <li>Report-administration UI feedback shall use MSG-76, MSG-18, MSG-43, or MSG-44.</li> </ul>                                                                                                                                                                                            |

Table 3.57: Business Rules for UC-28 - Handle Global Reports (Continued)

| ID         | Activ-ity | Rule Type     | Description                                                                                                                                                              |
|------------|-----------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC28-07 | (9)       | Message Rules | <b>Return Report Management Error:</b> <ul style="list-style-type: none"> <li>Administrative failures shall return ERR-12, ERR-64, ERR-26, ERR-14, or ERR-13.</li> </ul> |

Table 3.57: Business Rules for UC-28 - Handle Global Reports

### 3.1.29 UC-29 - Manage Platform

|                       |                                                                                                     |
|-----------------------|-----------------------------------------------------------------------------------------------------|
| <b>Name</b>           | Manage Platform                                                                                     |
| <b>Description</b>    | This use case describes how a system admin manages users and communities globally.                  |
| <b>Actor</b>          | System Admin                                                                                        |
| <b>Trigger</b>        | When the admin opens the platform management area and selects a target entity.                      |
| <b>Pre-condition</b>  | The admin is logged in to the system.<br>The admin has global management privileges.                |
| <b>Post-condition</b> | The selected user or community is banned or unbanned.<br>The platform state is updated accordingly. |

Table 3.58: Use Case Specification - Manage Platform

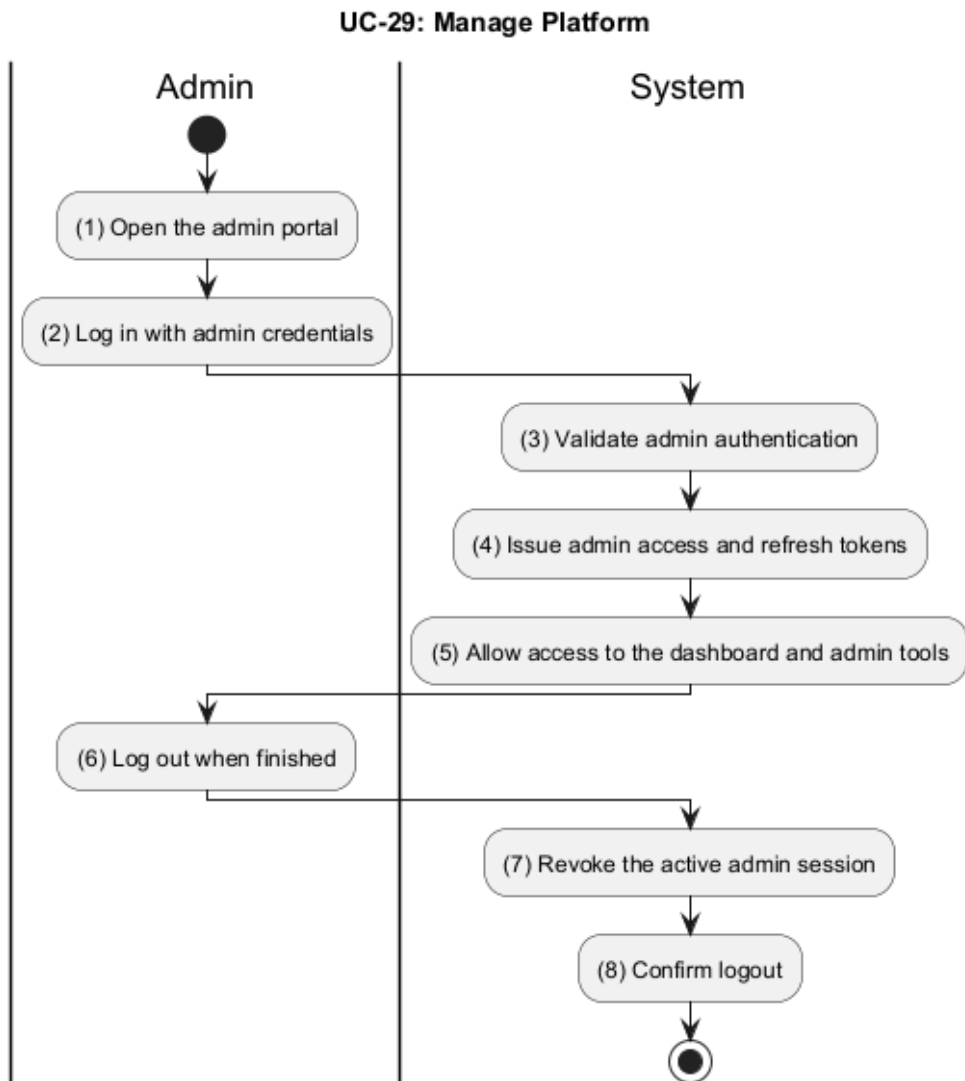


Figure 3.30: Activity Diagram for UC-29



| ID         | Activ-<br>ity | Rule Type           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------|---------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC29-01 | (3)           | Authorization Rules | <b>Validate Admin Login and Access Rights:</b> <ul style="list-style-type: none"> <li>The login endpoint accepts an identifier and password.</li> <li>The service loads the account by email or username and requires the <b>admin</b> role.</li> <li>The service rejects banned, deleted, or non-admin accounts before issuing tokens.</li> <li>The service verifies the password for local accounts in <i><b>AdminLogin()</b></i>.</li> </ul>                                                                                                                            |
| BR-UC29-02 | (3)           | Validation Rules    | <b>Validate the Admin Login Request:</b> <ul style="list-style-type: none"> <li>The system binds the login payload to <b>UserLoginRequest</b>.</li> <li>The request must include both identifier and password fields.</li> <li>if the payload cannot be bound, then the system returns a bad request response.</li> <li>if the credentials are invalid, then the service returns an invalid credentials response.</li> </ul>                                                                                                                                               |
| BR-UC29-03 | (3-5)         | Workflow Rules      | <b>Issue Admin Tokens and Allow Protected Access:</b> <ul style="list-style-type: none"> <li>The service generates access and refresh tokens with <i><b>AdminLogin()</b></i> after successful authentication.</li> <li>The login system returns the authenticated user together with both tokens.</li> <li>Admin dashboard and tool routes require <b>RequireAuth()</b> and <b>RequireAdmin()</b> before they can be reached.</li> <li>The protected admin stats routes expose the dashboard overview and related admin tools only after the token is accepted.</li> </ul> |

Table 3.59: Business Rules for UC-29 - Manage Platform

| ID         | Activ-ity | Rule Type      | Description                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------|-----------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC29-04 | (7)       | Workflow Rules | <b>Revoke the Active Admin Session:</b> <ul style="list-style-type: none"> <li>The logout endpoint reads the bearer access token and optional refresh token from the request.</li> <li>The service extracts the token JTIs and invalidates them when token revocation is available in <i>AdminLogout()</i>.</li> <li>The logout call completes even when token invalidation is unavailable, so the admin session ends cleanly.</li> </ul> |
| BR-UC29-05 | (8)       | Message Rules  | <b>Return the Admin Auth Result:</b> <ul style="list-style-type: none"> <li>Successful admin sign-in shall use MSG-3; logout completion has no dedicated message code in the current code.</li> <li>Authentication and access failures shall return ERR-1, ERR-12, ERR-2, ERR-31, or ERR-28.</li> </ul>                                                                                                                                   |

Table 3.59: Business Rules for UC-29 - Manage Platform

### 3.1.30 UC-30 - View Analytics

|                       |                                                                                                                              |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>           | View Analytics                                                                                                               |
| <b>Description</b>    | This use case describes how a system admin reviews platform usage and performance metrics.                                   |
| <b>Actor</b>          | System Admin                                                                                                                 |
| <b>Trigger</b>        | When the admin opens the analytics dashboard.                                                                                |
| <b>Pre-condition</b>  | The admin is logged in to the system.<br>Analytics data is available for the selected period.                                |
| <b>Post-condition</b> | The analytics dashboard is displayed with current metrics.<br>The admin can review user, content, and moderation statistics. |

Table 3.60: Use Case Specification - View Analytics

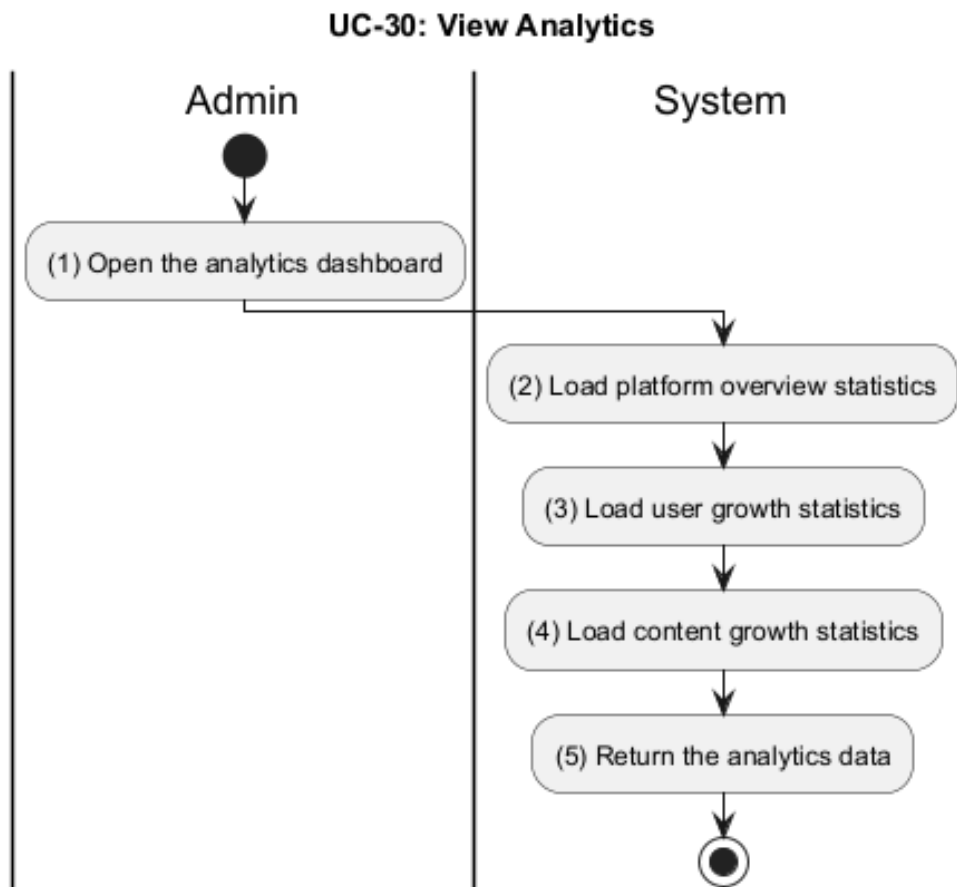


Figure 3.31: Activity Diagram for UC-30

| ID         | Activ-ity | Rule Type           | Description                                                                                                                                                                                                                                                                                                                                                                                         |
|------------|-----------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC30-01 | (2)       | Authorization Rules | <b>Open the Analytics Dashboard with Admin Access:</b> <ul style="list-style-type: none"> <li>The analytics routes require authentication and the <code>admin</code> role.</li> <li>The protected admin stats routes are guarded by <code>RequireAuth()</code> and <code>RequireAdmin()</code>.</li> <li>if the requester is not an admin, then the system returns a forbidden response.</li> </ul> |

Table 3.61: Business Rules for UC-30 - View Analytics

| ID         | Activ-<br>ity | Rule Type        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------|---------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC30-02 | (2)           | Validation Rules | <b>Validate the Analytics Query Parameters:</b> <ul style="list-style-type: none"> <li>The controller binds the user statistics query to <code>GetUserStatsQuery</code>.</li> <li>The controller binds the content statistics query to <code>GetContentStatsQuery</code>.</li> <li>The <code>period</code> field is optional and defaults to <code>week</code> when omitted.</li> <li>if the query cannot be bound, then the controller returns a bad request response.</li> </ul>           |
| BR-UC30-03 | (2)           | Workflow Rules   | <b>Load the Platform Overview Snapshot:</b> <ul style="list-style-type: none"> <li>The service loads the platform overview using <i><code>GetPlatformOverview()</code></i>.</li> <li>The overview aggregates user, community, content, and moderation statistics from the repositories.</li> <li>The overview response includes the current snapshot timestamp in <code>UpdatedAt</code>.</li> </ul>                                                                                         |
| BR-UC30-04 | (3–5)         | Workflow Rules   | <b>Load User and Content Growth Statistics:</b> <ul style="list-style-type: none"> <li>The service loads user statistics using <i><code>GetUserStats()</code></i> and content statistics using <i><code>GetContentStats()</code></i>.</li> <li>The service selects the growth window from the requested period and falls back to <code>week</code> when needed.</li> <li>The response includes the overview and the corresponding growth series for the requested analytics view.</li> </ul> |

Table 3.61: Business Rules for UC-30 - View Analytics (Continued)

| ID         | Activ-ity | Rule Type     | Description                                                                                                                                                                                                                                                                                                                                              |
|------------|-----------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-UC30-05 | (5)       | Message Rules | <b>Return the Analytics Result or Error:</b> <ul style="list-style-type: none"> <li>Analytics loading failures shall use MSG-79.</li> <li>Successful analytics requests return dashboard data inline and do not define a dedicated success toast in the current code.</li> <li>Access failures shall return ERR-12, ERR-10, ERR-2, or ERR-13.</li> </ul> |

Table 3.61: Business Rules for UC-30 - View Analytics

## 3.2 Data Requirements

### 3.2.1 Logical Data Model

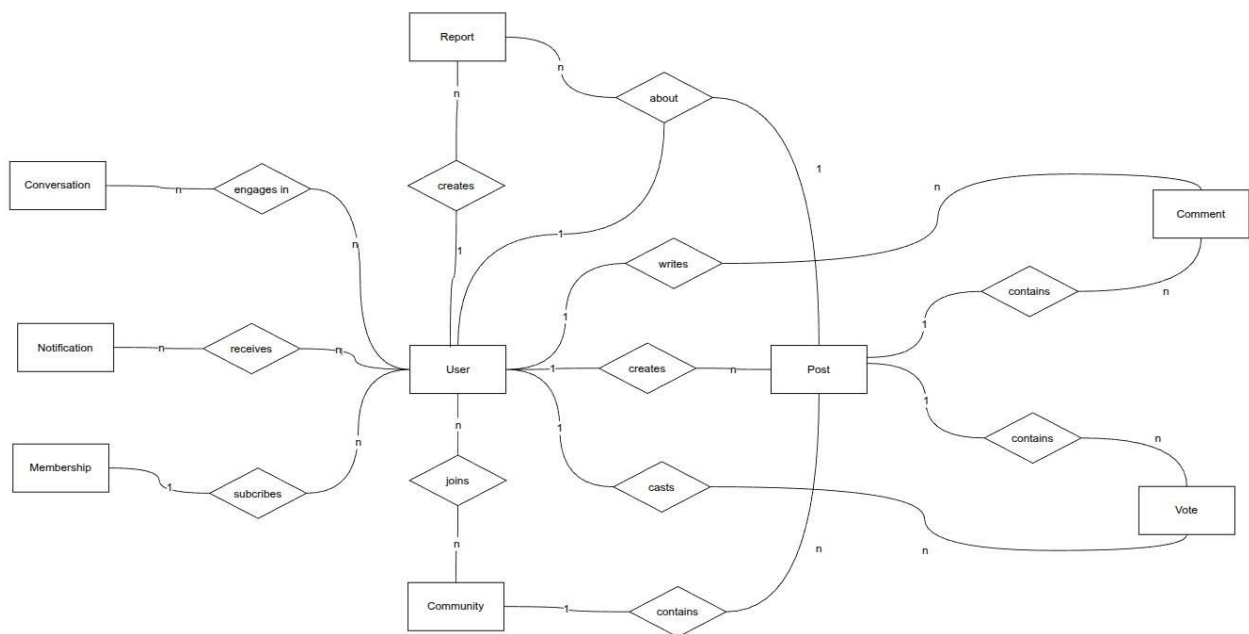


Figure 3.32: Logical Data Model for LKForum

### 3.2.2 Data Dictionary

#### Authentication and User Profile Models

| Data Element | Description                                           | Composition / Data Type | Length | Values                                                            |
|--------------|-------------------------------------------------------|-------------------------|--------|-------------------------------------------------------------------|
| ID           | Unique identifier of a user.                          | objectId                | 24     | System-generated, unique.                                         |
| Username     | Display name used to identify a user account.         | string                  | –      | Must be unique within the platform.                               |
| Email        | Email address used for authentication and contact.    | string                  | –      | Must be unique and valid.                                         |
| Reputation   | User reputation score.                                | integer                 | –      | System-maintained counter.                                        |
| Password     | Hashed password for a local account.                  | string                  | –      | Stored only for local providers.                                  |
| Provider     | Authentication source for a user account.             | enum                    | –      | local, google                                                     |
| ProviderID   | External account identifier returned by the provider. | string                  | –      | Used for third-party login linkage.                               |
| Role         | Platform-wide role of the user.                       | enum                    | –      | user, admin                                                       |
| RoleContent  | Role-specific data attached to the user account.      | object                  | –      | Contains user or admin sub-document data.                         |
| Settings     | User preference configuration.                        | object                  | –      | Contains appearance, notification, privacy, and content settings. |
| IsVerified   | Indicates whether the account has been verified.      | boolean                 | –      | true, false                                                       |
| IsBanned     | Indicates whether the account is currently banned.    | boolean                 | –      | true, false                                                       |
| BanUntil     | Timestamp when a ban expires, if any.                 | timestamp               | –      | Null means permanent ban.                                         |
| BanReason    | Reason recorded for a ban action.                     | string                  | –      | Optional free text.                                               |
| CreatedAt    | Timestamp when the record was created.                | timestamp               | –      | System-generated.                                                 |
| DeletedAt    | Timestamp for soft-deleted records.                   | timestamp               | –      | Null when not deleted.                                            |

Table 3.62: User

| Data Element | Description                  | Composition / Data Type | Length | Values                      |
|--------------|------------------------------|-------------------------|--------|-----------------------------|
| AsUser       | User-specific profile data.  | object                  | –      | See UserRoleContent table.  |
| AsAdmin      | Admin-specific profile data. | object                  | –      | See AdminRoleContent table. |

Table 3.63: RoleContent

| Data Element | Description                           | Composition / Data Type | Length | Values                                             |
|--------------|---------------------------------------|-------------------------|--------|----------------------------------------------------|
| Avatar       | Profile image.                        | object                  | –      | See Image table.                                   |
| Cover        | Cover image.                          | object                  | –      | See Image table.                                   |
| Bio          | Short biography.                      | string                  | –      | User-generated text.                               |
| Gender       | Declared gender.                      | enum                    | –      | male, female, prefer_not_to_say                    |
| DateOfBirth  | User date of birth.                   | timestamp               | –      | Optional date value.                               |
| Location     | Province or city.                     | enum                    | –      | One of the provinces defined in model/location.go. |
| Interests    | List of user interests.               | array[string]           | –      | Values defined in model/interest.go.               |
| SocialLinks  | Collection of public social profiles. | object                  | –      | Website, Facebook, YouTube, GitHub links.          |
| Stats        | User activity statistics.             | object                  | –      | Post, comment, and vote counters plus timestamps.  |

Table 3.64: UserRoleContent

| Data Element | Description           | Composition / Data Type | Length | Values        |
|--------------|-----------------------|-------------------------|--------|---------------|
| Website      | Personal website URL. | string                  | –      | Optional URL. |
| Facebook     | Facebook profile URL. | string                  | –      | Optional URL. |
| YouTube      | YouTube channel URL.  | string                  | –      | Optional URL. |
| GitHub       | GitHub profile URL.   | string                  | –      | Optional URL. |

Table 3.65: SocialLinks

| Data Element | Description                     | Composition / Data Type | Length | Values                     |
|--------------|---------------------------------|-------------------------|--------|----------------------------|
| PostCount    | Number of posts created.        | integer                 | –      | System-maintained counter. |
| CommentCount | Number of comments created.     | integer                 | –      | System-maintained counter. |
| TotalUpvotes | Total upvotes received.         | integer                 | –      | System-maintained counter. |
| JoinedAt     | Timestamp when the user joined. | timestamp               | –      | System-generated.          |
| LastActiveAt | Timestamp of the last activity. | timestamp               | –      | System-generated.          |

Table 3.66: ActivityStats

| Data Element | Description            | Composition / Data Type | Length | Values                                 |
|--------------|------------------------|-------------------------|--------|----------------------------------------|
| Permissions  | Admin capability list. | array[string]           | –      | Permission names granted to the admin. |

Table 3.67: AdminRoleContent

| Data Element  | Description                    | Composition / Data Type | Length | Values                          |
|---------------|--------------------------------|-------------------------|--------|---------------------------------|
| Appearance    | Visual preferences.            | object                  | –      | See AppearanceSettings table.   |
| Notifications | Notification preferences.      | object                  | –      | See NotificationSettings table. |
| Privacy       | Privacy preferences.           | object                  | –      | See PrivacySettings table.      |
| Content       | Content filtering preferences. | object                  | –      | See ContentSettings table.      |

Table 3.68: UserSettings

| Data Element | Description              | Composition / Data Type | Length | Values               |
|--------------|--------------------------|-------------------------|--------|----------------------|
| Theme        | UI theme preference.     | enum                    | –      | light, dark, auto    |
| FontSize     | UI font size preference. | enum                    | –      | small, medium, large |

Table 3.69: AppearanceSettings



| Data Element     | Description                 | Composition / Data Type | Length | Values      |
|------------------|-----------------------------|-------------------------|--------|-------------|
| InAppEnabled     | In-app notification toggle. | boolean                 | –      | true, false |
| EmailEnabled     | Email notification toggle.  | boolean                 | –      | true, false |
| NotifyOn-Comment | Notify on comments.         | boolean                 | –      | true, false |
| NotifyOn-Mention | Notify on mentions.         | boolean                 | –      | true, false |
| NotifyOnUpvote   | Notify on upvotes.          | boolean                 | –      | true, false |
| NotifyOnMessage  | Notify on private messages. | boolean                 | –      | true, false |

Table 3.70: NotificationSettings

| Data Element         | Description                                       | Composition / Data Type | Length | Values      |
|----------------------|---------------------------------------------------|-------------------------|--------|-------------|
| ShowProfile          | Controls whether a profile is public.             | boolean                 | –      | true, false |
| ShowEmail            | Controls whether email is visible on the profile. | boolean                 | –      | true, false |
| ShowPostHistory      | Controls whether post history is visible.         | boolean                 | –      | true, false |
| AllowDirect-Messages | Controls whether direct messages are allowed.     | boolean                 | –      | true, false |
| AllowMentions        | Controls whether mentions are allowed.            | boolean                 | –      | true, false |

Table 3.71: PrivacySettings

| Data Element | Description                               | Composition / Data Type | Length | Values      |
|--------------|-------------------------------------------|-------------------------|--------|-------------|
| AllowNSFW    | Controls whether NSFW content is allowed. | boolean                 | –      | true, false |

Table 3.72: ContentSettings

| Data Element         | Description                                        | Composition / Data Type | Length | Values                    |
|----------------------|----------------------------------------------------|-------------------------|--------|---------------------------|
| EmailVerification-ID | Unique identifier of an email verification record. | objectId                | 24     | System-generated, unique. |

Table 3.73: Verification Records

|                                     |                                                      |           |   |                                |
|-------------------------------------|------------------------------------------------------|-----------|---|--------------------------------|
| EmailVerification-<br>.Email        | Email being verified.                                | string    | – | Must be unique and valid.      |
| EmailVerification-<br>.OTP          | One-time password used for verification.             | string    | 6 | Six-digit numeric code.        |
| EmailVerification-<br>.OTPExpiresAt | Expiration timestamp for the OTP.                    | timestamp | – | System-generated.              |
| EmailVerification-<br>.IsVerified   | Indicates whether the OTP has already been verified. | boolean   | – | true, false                    |
| EmailVerification-<br>.Nonce        | Anti-replay token.                                   | string    | – | System-generated random value. |
| EmailVerification-<br>.CreatedAt    | Timestamp when the record was created.               | timestamp | – | System-generated.              |

Table 3.73: Verification Records

| Data Element                    | Description                                          | Composition / Data Type | Length | Values                         |
|---------------------------------|------------------------------------------------------|-------------------------|--------|--------------------------------|
| PasswordReset-<br>.ID           | Unique identifier of a password reset record.        | objectId                | 24     | System-generated, unique.      |
| PasswordReset-<br>.Email        | Email being reset.                                   | string                  | –      | Must be unique and valid.      |
| PasswordReset-<br>.OTP          | One-time password used for reset.                    | string                  | 6      | Six-digit numeric code.        |
| PasswordReset-<br>.OTPExpiresAt | Expiration timestamp for the OTP.                    | timestamp               | –      | System-generated.              |
| PasswordReset-<br>.IsVerified   | Indicates whether the OTP has already been verified. | boolean                 | –      | true, false                    |
| PasswordReset-<br>.Nonce        | Anti-replay token.                                   | string                  | –      | System-generated random value. |
| PasswordReset-<br>.CreatedAt    | Timestamp when the record was created.               | timestamp               | –      | System-generated.              |

Table 3.74: PasswordReset

| Data Element | Description                   | Composition / Data Type | Length | Values                              |
|--------------|-------------------------------|-------------------------|--------|-------------------------------------|
| URL          | Public URL of a stored image. | string                  | –      | Cloudinary or equivalent media URL. |

Table 3.75: Image

|            |                                        |           |   |                                    |
|------------|----------------------------------------|-----------|---|------------------------------------|
| PublicID   | Provider-specific image identifier.    | string    | – | Used for delete/update operations. |
| UploadedAt | Timestamp when the image was uploaded. | timestamp | – | System-generated.                  |

Table 3.75: Image

| Data Element | Description                            | Composition / Data Type | Length | Values                              |
|--------------|----------------------------------------|-------------------------|--------|-------------------------------------|
| URL          | Public URL of a stored video.          | string                  | –      | Cloudinary or equivalent media URL. |
| PublicID     | Provider-specific video identifier.    | string                  | –      | Used for delete/update operations.  |
| UploadedAt   | Timestamp when the video was uploaded. | timestamp               | –      | System-generated.                   |

Table 3.76: Video

## Community and Membership Models

| Data Element | Description                            | Composition / Data Type | Length | Values                                 |
|--------------|----------------------------------------|-------------------------|--------|----------------------------------------|
| ID           | Unique identifier of a community.      | objectId                | 24     | System-generated, unique.              |
| Name         | Public name of a community.            | string                  | –      | Must be unique.                        |
| Description  | Text describing a community.           | string                  | –      | Optional free text.                    |
| Avatar       | Community avatar image URL.            | string                  | –      | Optional media URL.                    |
| Banner       | Banner image for a community.          | string                  | –      | Optional media URL.                    |
| Setting      | Community configuration settings.      | object                  | –      | Contains privacy and moderation flags. |
| Rules        | Published rules of a community.        | array[object]           | –      | Title and description entries.         |
| Moderators   | List of assigned community moderators. | array[object]           | –      | Moderator sub-documents.               |
| MemberCount  | Current number of community members.   | integer                 | –      | System-maintained counter.             |

Table 3.77: Community

|                |                                                  |           |    |                                         |
|----------------|--------------------------------------------------|-----------|----|-----------------------------------------|
| PostCount      | Current number of community posts.               | integer   | –  | System-maintained counter.              |
| CreateAt       | Timestamp when the community was created.        | timestamp | –  | System-generated.                       |
| CreateByID     | Identifier of the creator.                       | objectId  | 24 | Must correspond to an existing user id. |
| CreateByName   | Username of the creator.                         | string    | –  | Copied from the creator account.        |
| CreateByAvatar | Avatar URL of the creator.                       | string    | –  | Copied from the creator profile.        |
| Is18Plus       | Marks the community as adult content.            | boolean   | –  | true, false                             |
| IsDeleted      | Indicates whether the community is soft deleted. | boolean   | –  | true, false                             |
| IsBanned       | Indicates whether the community is banned.       | boolean   | –  | true, false                             |
| BanReason      | Reason recorded for a community ban.             | string    | –  | Optional free text.                     |

Table 3.77: Community

| Data Element         | Description                                                | Composition / Data Type | Length | Values            |
|----------------------|------------------------------------------------------------|-------------------------|--------|-------------------|
| IsPrivate            | Controls whether the community is visible only to members. | boolean                 | –      | true, false       |
| PostRequire-Approval | Controls whether new posts require moderator approval.     | boolean                 | –      | true, false       |
| JoinRequire-Approval | Controls whether join requests require approval.           | boolean                 | –      | true, false       |
| MaxPostLength        | Maximum allowed post length in characters.                 | integer                 | –      | Positive integer. |

Table 3.78: CommunitySetting

| Data Element | Description                       | Composition / Data Type | Length | Values              |
|--------------|-----------------------------------|-------------------------|--------|---------------------|
| Title        | Short title for a community rule. | string                  | –      | User-defined label. |

Table 3.79: CommunityRule

|             |                                           |        |   |                    |
|-------------|-------------------------------------------|--------|---|--------------------|
| Description | Detailed explanation of a community rule. | string | – | User-defined text. |
|-------------|-------------------------------------------|--------|---|--------------------|

Table 3.79: CommunityRule

| Data Element | Description                                     | Composition / Data Type | Length | Values                                  |
|--------------|-------------------------------------------------|-------------------------|--------|-----------------------------------------|
| UserID       | Identifier of the moderator account.            | objectId                | 24     | Must correspond to an existing user id. |
| Username     | Moderator username.                             | string                  | –      | Copied from the user account.           |
| Avatar       | Moderator avatar.                               | object                  | –      | See Image table.                        |
| IsActive     | Indicates whether the moderator role is active. | boolean                 | –      | true, false                             |
| AssignedAt   | Timestamp when the moderator role was assigned. | timestamp               | –      | System-generated.                       |

Table 3.80: Moderator

| Data Element | Description                                 | Composition / Data Type | Length | Values                                       |
|--------------|---------------------------------------------|-------------------------|--------|----------------------------------------------|
| ID           | Unique identifier of a membership record.   | objectId                | 24     | System-generated, unique.                    |
| UserID       | Identifier of the member user.              | objectId                | 24     | Must correspond to an existing user id.      |
| CommunityID  | Identifier of the target community.         | objectId                | 24     | Must correspond to an existing community id. |
| Status       | Membership review status.                   | enum                    | –      | pending, approved, rejected                  |
| CreatedAt    | Timestamp when the membership was created.  | timestamp               | –      | System-generated.                            |
| User         | Cached user summary for a membership entry. | object                  | –      | Contains id, username, and avatar.           |

Table 3.81: Membership

| Data Element | Description | Composition / Data Type | Length | Values |
|--------------|-------------|-------------------------|--------|--------|
|--------------|-------------|-------------------------|--------|--------|

Table 3.82: UserInfo

|          |                                            |          |    |                                         |
|----------|--------------------------------------------|----------|----|-----------------------------------------|
| ID       | Identifier of the membership user summary. | objectId | 24 | Must correspond to an existing user id. |
| Username | Username of the membership user summary.   | string   | –  | Copied from the user account.           |
| Avatar   | Avatar of the membership user summary.     | object   | –  | See Image table.                        |

Table 3.82: UserInfo

| Data Element | Description                                              | Composition / Data Type | Length | Values                                       |
|--------------|----------------------------------------------------------|-------------------------|--------|----------------------------------------------|
| CommunityID  | Identifier of the banned community.                      | objectId                | 24     | Must correspond to an existing community id. |
| UserID       | Identifier of the banned user.                           | objectId                | 24     | Must correspond to an existing user id.      |
| Type         | Ban category applied by the community.                   | enum                    | –      | banned, muted                                |
| Reason       | Reason recorded for the ban or mute.                     | string                  | –      | Optional free text.                          |
| BannedAt     | Timestamp when the restriction was applied.              | timestamp               | –      | System-generated.                            |
| BannedBy     | Identifier of the moderator who applied the restriction. | objectId                | 24     | Must correspond to an existing user id.      |
| ExpiresAt    | Timestamp when the restriction expires.                  | timestamp               | –      | Null for permanent restrictions.             |

Table 3.83: CommunityBan

## Content and Moderation Models

| Data Element | Description                                     | Composition / Data Type | Length | Values                                       |
|--------------|-------------------------------------------------|-------------------------|--------|----------------------------------------------|
| ID           | Unique identifier of a post.                    | objectId                | 24     | System-generated, unique.                    |
| AuthorID     | Identifier of the post author.                  | objectId                | 24     | Must correspond to an existing user id.      |
| CommunityID  | Identifier of the community that owns the post. | objectId                | 24     | Must correspond to an existing community id. |

Table 3.84: Post

|                  |                                                     |               |   |                                                              |
|------------------|-----------------------------------------------------|---------------|---|--------------------------------------------------------------|
| Type             | Type of post content.                               | enum          | – | text, poll, video, image                                     |
| Title            | Title of a post.                                    | string        | – | Text entered by the author.                                  |
| Content          | Main post payload.                                  | object        | – | Contains text, images, videos, or poll data.                 |
| VotesCount       | Vote summary for a post or comment.                 | object        | – | Up and down counts.                                          |
| CommentsCount    | Number of comments on a post.                       | integer       | – | System-maintained counter.                                   |
| HotScore         | Ranking score used for best sorting.                | float         | – | Calculated from votes and time decay.                        |
| CreatedAt        | Timestamp when the post was created.                | timestamp     | – | System-generated.                                            |
| UpdatedAt        | Timestamp of the last content update.               | timestamp     | – | Null until updated.                                          |
| IsDeleted        | Indicates whether the record is soft deleted.       | boolean       | – | true, false                                                  |
| IsHidden         | Indicates whether the post is hidden by its author. | boolean       | – | true, false                                                  |
| IsEdited         | Indicates whether the post has been edited.         | boolean       | – | true, false                                                  |
| Tags             | Tags attached to a post.                            | array[string] | – | Free-form keywords.                                          |
| IsDraft          | Indicates whether the post is saved as a draft.     | boolean       | – | true, false                                                  |
| IsBan            | Indicates whether a post is banned.                 | boolean       | – | true, false                                                  |
| BanReason        | Reason recorded for a post ban.                     | string        | – | Optional free text.                                          |
| ModerationStatus | Current AI or moderator moderation state.           | enum          | – | pending, approved, rejected, skipped                         |
| ModerationResult | Moderation analysis details.                        | object        | – | Contains violation flag, confidence, categories, and reason. |
| ModeratedAt      | Timestamp when moderation happened.                 | timestamp     | – | Null until moderated.                                        |

Table 3.84: Post

| Data Element | Description | Composition / Data Type | Length | Values |
|--------------|-------------|-------------------------|--------|--------|
|--------------|-------------|-------------------------|--------|--------|

Table 3.85: PostContent

|        |                                  |               |   |                                                          |
|--------|----------------------------------|---------------|---|----------------------------------------------------------|
| Text   | Text body of a post.             | string        | – | Optional for non-text posts.                             |
| Images | Attached images in a post.       | array[object] | – | Array of image metadata.                                 |
| Videos | Attached videos in a post.       | array[object] | – | Array of video metadata.                                 |
| Poll   | Poll payload attached to a post. | object        | – | Question, options, votes, expiry, and multi-select flag. |

Table 3.85: PostContent

| Data Element  | Description                                              | Composition / Data Type | Length | Values                               |
|---------------|----------------------------------------------------------|-------------------------|--------|--------------------------------------|
| Question      | Question presented to poll voters.                       | string                  | –      | Poll text.                           |
| Options       | Available poll options.                                  | array[object]           | –      | Each option has id, text, and votes. |
| TotalVotes    | Total number of poll votes.                              | integer                 | –      | System-maintained counter.           |
| ExpiresAt     | Poll expiry timestamp.                                   | timestamp               | –      | Null when the poll never expires.    |
| AllowMultiple | Controls whether voters may choose more than one option. | boolean                 | –      | true, false                          |

Table 3.86: Poll

| Data Element | Description                    | Composition / Data Type | Length | Values                     |
|--------------|--------------------------------|-------------------------|--------|----------------------------|
| ID           | Identifier of a poll option.   | string                  | –      | Unique within the poll.    |
| Text         | Display text of a poll option. | string                  | –      | Option label.              |
| Votes        | Vote count for a poll option.  | integer                 | –      | System-maintained counter. |

Table 3.87: PollOption

| Data Element | Description | Composition / Data Type | Length | Values |
|--------------|-------------|-------------------------|--------|--------|
|--------------|-------------|-------------------------|--------|--------|

Table 3.88: VotesCount



|      |                      |         |   |                            |
|------|----------------------|---------|---|----------------------------|
| Up   | Number of upvotes.   | integer | – | System-maintained counter. |
| Down | Number of downvotes. | integer | – | System-maintained counter. |

Table 3.88: VotesCount

| Data Element | Description                                    | Composition / Data Type | Length | Values                                        |
|--------------|------------------------------------------------|-------------------------|--------|-----------------------------------------------|
| IsViolation  | Indicates whether the content violates policy. | boolean                 | –      | true, false                                   |
| Confidence   | Confidence score returned by moderation.       | float                   | –      | 0.0 to 1.0                                    |
| Categories   | Categories detected by moderation.             | array[string]           | –      | Issue labels such as hate speech or violence. |
| Reason       | Human-readable moderation explanation.         | string                  | –      | Localized explanation text.                   |
| CheckedText  | Indicates whether text was analyzed.           | boolean                 | –      | true, false                                   |
| CheckedMedia | Indicates whether media was analyzed.          | boolean                 | –      | true, false                                   |

Table 3.89: ModerationResult

| Data Element | Description                               | Composition / Data Type | Length | Values                                  |
|--------------|-------------------------------------------|-------------------------|--------|-----------------------------------------|
| ID           | Unique identifier of a comment.           | objectId                | 24     | System-generated, unique.               |
| Author       | Author details attached to a comment.     | object                  | –      | Contains id, username, and avatar.      |
| PostID       | Identifier of the parent post.            | objectId                | 24     | Must correspond to an existing post id. |
| ParentID     | Identifier of the parent comment, if any. | objectId                | 24     | Null for top-level comments.            |
| Content      | Comment body text.                        | string                  | –      | User-generated text.                    |
| VotesCount   | Vote summary for a comment.               | object                  | –      | Up and down counts.                     |
| CreatedAt    | Timestamp when the comment was created.   | timestamp               | –      | System-generated.                       |
| DeletedAt    | Timestamp for soft-deleted comments.      | timestamp               | –      | Null when not deleted.                  |

Table 3.90: Comment

|           |                                              |         |   |             |
|-----------|----------------------------------------------|---------|---|-------------|
| IsDeleted | Indicates whether a comment is soft deleted. | boolean | – | true, false |
|-----------|----------------------------------------------|---------|---|-------------|

Table 3.90: Comment

| Data Element | Description                       | Composition / Data Type | Length | Values                                  |
|--------------|-----------------------------------|-------------------------|--------|-----------------------------------------|
| ID           | Identifier of the comment author. | objectId                | 24     | Must correspond to an existing user id. |
| Username     | Username of the comment author.   | string                  | –      | Copied from the user account.           |
| Avatar       | Avatar for the comment author.    | object                  | –      | Image metadata.                         |

Table 3.91: CommentAuthor

| Data Element | Description                                | Composition / Data Type | Length | Values                                  |
|--------------|--------------------------------------------|-------------------------|--------|-----------------------------------------|
| ID           | Unique identifier of a draft.              | objectId                | 24     | System-generated, unique.               |
| AuthorID     | Identifier of the draft author.            | objectId                | 24     | Must correspond to an existing user id. |
| CommunityID  | Identifier of the draft community, if any. | string                  | –      | Optional community id string.           |
| Type         | Post type stored in a draft.               | enum                    | –      | text, poll, video, image                |
| Title        | Draft title.                               | string                  | –      | Optional until publish time.            |
| Content      | Draft content payload.                     | object                  | –      | Same shape as post content.             |
| Tags         | Tags saved with the draft.                 | array[string]           | –      | Optional free-form keywords.            |
| CreatedAt    | Timestamp when the draft was created.      | timestamp               | –      | System-generated.                       |
| UpdatedAt    | Timestamp when the draft was last updated. | timestamp               | –      | System-generated.                       |

Table 3.92: Draft

| Data Element | Description                            | Composition / Data Type | Length | Values                              |
|--------------|----------------------------------------|-------------------------|--------|-------------------------------------|
| URL          | Public URL of a stored image.          | string                  | –      | Cloudinary or equivalent media URL. |
| PublicID     | Provider-specific image identifier.    | string                  | –      | Used for delete/update operations.  |
| UploadedAt   | Timestamp when the image was uploaded. | timestamp               | –      | System-generated.                   |

Table 3.93: Image

| Data Element | Description                            | Composition / Data Type | Length | Values                              |
|--------------|----------------------------------------|-------------------------|--------|-------------------------------------|
| URL          | Public URL of a stored video.          | string                  | –      | Cloudinary or equivalent media URL. |
| PublicID     | Provider-specific video identifier.    | string                  | –      | Used for delete/update operations.  |
| UploadedAt   | Timestamp when the video was uploaded. | timestamp               | –      | System-generated.                   |

Table 3.94: Video

## Engagement and Activity Models

| Data Element | Description                                | Composition / Data Type | Length | Values                                  |
|--------------|--------------------------------------------|-------------------------|--------|-----------------------------------------|
| ID           | Unique identifier of a saved-post record.  | objectId                | 24     | System-generated, unique.               |
| UserID       | Identifier of the user who saved the post. | objectId                | 24     | Must correspond to an existing user id. |
| PostID       | Identifier of the saved post.              | objectId                | 24     | Must correspond to an existing post id. |
| SavedAt      | Timestamp when a post was saved.           | timestamp               | –      | System-generated.                       |

Table 3.95: SavedPost

| Data Element | Description                               | Composition / Data Type | Length | Values                    |
|--------------|-------------------------------------------|-------------------------|--------|---------------------------|
| ID           | Unique identifier of a liked-post record. | objectId                | 24     | System-generated, unique. |

Table 3.96: LikedPost

|         |                                            |           |    |                                         |
|---------|--------------------------------------------|-----------|----|-----------------------------------------|
| UserID  | Identifier of the user who liked the post. | objectId  | 24 | Must correspond to an existing user id. |
| PostID  | Identifier of the liked post.              | objectId  | 24 | Must correspond to an existing post id. |
| LikedAt | Timestamp when a post was liked.           | timestamp | –  | System-generated.                       |

Table 3.96: LikedPost

| Data Element | Description                                 | Composition / Data Type | Length | Values                                  |
|--------------|---------------------------------------------|-------------------------|--------|-----------------------------------------|
| ID           | Unique identifier of a post history record. | objectId                | 24     | System-generated, unique.               |
| UserID       | Identifier of the user who viewed the post. | objectId                | 24     | Must correspond to an existing user id. |
| PostID       | Identifier of the viewed post.              | objectId                | 24     | Must correspond to an existing post id. |
| ViewedAt     | Timestamp when a post was viewed.           | timestamp               | –      | System-generated.                       |

Table 3.97: PostHistory

## Communication Models

| Data Element | Description                                            | Composition / Data Type | Length | Values                                              |
|--------------|--------------------------------------------------------|-------------------------|--------|-----------------------------------------------------|
| ID           | Unique identifier of a notification.                   | objectId                | 24     | System-generated, unique.                           |
| RecipientID  | Identifier of the notification recipient.              | objectId                | 24     | Must correspond to an existing user id.             |
| ActorID      | Identifier of the user who triggered the notification. | objectId                | 24     | Optional when the system is the actor.              |
| Type         | Type of notification.                                  | enum                    | –      | comment, like, follow, mention, new_message, system |
| Message      | Notification text shown to the recipient.              | string                  | –      | System-generated or templated text.                 |
| Link         | URL or route associated with the notification.         | string                  | –      | Navigation target.                                  |
| IsRead       | Indicates whether a notification has been read.        | boolean                 | –      | true, false                                         |

Table 3.98: Notification

|           |                                              |            |   |                           |
|-----------|----------------------------------------------|------------|---|---------------------------|
| Metadata  | Additional notification payload.             | Dictionary | – | Arbitrary key-value data. |
| CreatedAt | Timestamp when the notification was created. | timestamp  | – | System-generated.         |

Table 3.98: Notification

| Data Element | Description                                     | Composition / Data Type | Length | Values                                 |
|--------------|-------------------------------------------------|-------------------------|--------|----------------------------------------|
| ID           | Unique identifier of a private message channel. | objectId                | 24     | System-generated, unique.              |
| Members      | Channel participants.                           | array[object]           | –      | Member summaries.                      |
| Settings     | Per-user channel settings.                      | array[object]           | –      | Nickname and notification preferences. |
| Status       | Channel lifecycle status.                       | enum                    | –      | active, block                          |
| CreatedAt    | Timestamp when the channel was created.         | timestamp               | –      | System-generated.                      |
| UpdatedAt    | Timestamp when the channel was last updated.    | timestamp               | –      | System-generated.                      |

Table 3.99: Channel

| Data Element    | Description                                         | Composition / Data Type | Length | Values                               |
|-----------------|-----------------------------------------------------|-------------------------|--------|--------------------------------------|
| UserID          | Identifier of the user owning the channel settings. | objectId                | 24     | Must correspond to a channel member. |
| Nickname        | Custom nickname for the other user in the channel.  | string                  | –      | Optional display label.              |
| Notification    | Controls notification delivery for the channel.     | boolean                 | –      | true, false                          |
| TypingIndicator | Controls typing indicator visibility.               | boolean                 | –      | true, false                          |
| IsDeleted       | Indicates whether the channel setting is hidden.    | boolean                 | –      | true, false                          |

Table 3.100: ChannelSetting

| Data Element | Description | Composition / Data Type | Length | Values |
|--------------|-------------|-------------------------|--------|--------|
|--------------|-------------|-------------------------|--------|--------|

Table 3.101: ChannelMember

|          |                                   |          |    |                                         |
|----------|-----------------------------------|----------|----|-----------------------------------------|
| UserID   | Identifier of the channel member. | objectId | 24 | Must correspond to an existing user id. |
| Username | Username of the channel member.   | string   | –  | Copied from the user account.           |
| Avatar   | Avatar URL of the channel member. | string   | –  | Optional media URL.                     |

Table 3.101: ChannelMember

| Data Element   | Description                                               | Composition / Data Type | Length | Values                                     |
|----------------|-----------------------------------------------------------|-------------------------|--------|--------------------------------------------|
| ID             | Unique identifier of a private message.                   | objectId                | 24     | System-generated, unique.                  |
| ChannelID      | Identifier of the private message channel.                | objectId                | 24     | Must correspond to an existing channel id. |
| SenderID       | Identifier of the message sender.                         | objectId                | 24     | Null for system messages.                  |
| SenderUsername | Cached sender username.                                   | string                  | –      | Copied from the sender account.            |
| Type           | Message category.                                         | enum                    | –      | user, system                               |
| Content        | Message body text.                                        | string                  | –      | User-generated or system-generated text.   |
| IsSend         | Indicates whether the message was delivered successfully. | boolean                 | –      | true, false                                |
| IsDeleted      | Indicates whether the message is soft deleted.            | boolean                 | –      | true, false                                |
| DeletedAt      | Timestamp when the message was deleted.                   | timestamp               | –      | Null when not deleted.                     |

Table 3.102: Message

## Reporting and Administration Models

| Data Element | Description                                        | Composition / Data Type | Length | Values                                    |
|--------------|----------------------------------------------------|-------------------------|--------|-------------------------------------------|
| ID           | Unique identifier of a report.                     | objectId                | 24     | System-generated, unique.                 |
| ReporterID   | Identifier of the user who submitted the report.   | objectId                | 24     | Must correspond to an existing user id.   |
| TargetID     | Identifier of the reported user, post, or comment. | objectId                | 24     | Must correspond to an existing target id. |

Table 3.103: Report

|             |                                                 |           |   |                                              |
|-------------|-------------------------------------------------|-----------|---|----------------------------------------------|
| TargetType  | Type of reported target.                        | enum      | – | user, post, comment                          |
| Reason      | Short reason for a report or moderation action. | string    | – | Free text provided by the user or moderator. |
| Description | Longer report details or explanation.           | string    | – | Optional free text.                          |
| IsDeleted   | Indicates whether the report is soft deleted.   | boolean   | – | true, false                                  |
| DeletedAt   | Timestamp for deleted reports.                  | timestamp | – | Null when not deleted.                       |
| CreatedAt   | Timestamp when the report was created.          | timestamp | – | System-generated.                            |

Table 3.103: Report

## Enum and Value

| Data Element      | Description                 | Composition / Data Type | Length | Values                                              |
|-------------------|-----------------------------|-------------------------|--------|-----------------------------------------------------|
| AuthProvider      | User authentication source. | enum                    | –      | local, google                                       |
| Role              | Platform-wide user role.    | enum                    | –      | user, admin                                         |
| Gender            | Declared gender.            | enum                    | –      | male, female, prefer_not_to_say                     |
| VNProvince        | User location value.        | enum                    | –      | One of the provinces defined in model/location.go.  |
| Interest          | User interest label.        | enum                    | –      | Values defined in model/interest.go.                |
| PostType          | Post content type.          | enum                    | –      | text, poll, video, image                            |
| Moderation-Status | Moderation state.           | enum                    | –      | pending, approved, rejected, skipped                |
| Membership-Status | Membership review state.    | enum                    | –      | pending, approved, rejected                         |
| Notification-Type | Notification type.          | enum                    | –      | comment, like, follow, mention, new_message, system |
| ChannelStatus     | Channel lifecycle status.   | enum                    | –      | active, block                                       |
| MessageType       | Message category.           | enum                    | –      | user, system                                        |
| ReportTarget-Type | Reported target type.       | enum                    | –      | user, post, comment                                 |
| VoteType          | Vote direction type.        | enum                    | –      | up, down                                            |

Table 3.104: Enum and Value Classes

|                |                   |      |   |               |
|----------------|-------------------|------|---|---------------|
| VoteTargetType | Vote target type. | enum | – | post, comment |
|----------------|-------------------|------|---|---------------|

Table 3.104: Enum and Value Classes

### 3.3 Data Integrity, Retention, and Disposal

- DI-1:** The system shall retain user-deleted content (such as posts and comments) in a deactivated state for 30 days. After this period, the content will be permanently purged from the system. This allows users a grace period to recover accidentally deleted items.
- DI-2:** When a user requests to delete their account, the account shall be immediately deactivated and scheduled for permanent deletion. All personally identifiable information and associated content will be permanently deleted or anonymized after 90 days, unless required to be kept for legal reasons.
- DI-3:** Moderation data, including content reports submitted by users and action logs taken by moderators, shall be retained for one year after the report is closed to monitor for patterns of abuse.
- DI-4:** Core content such as active posts and comments will be retained indefinitely as long as the author’s account and the community they belong to exist.



## 4 Other Nonfunctional Requirements

### 4.1 Performance Requirements

- PR-1:** For normal operations, page navigation and standard read requests such as browsing posts, opening a post, loading a profile, and retrieving notifications should return the first visible result within 3 seconds under normal network conditions.
- PR-2:** Search requests across posts, users, and communities should return results within 5 seconds for typical keyword queries, with pagination applied to large result sets.
- PR-3:** Write operations such as creating posts, comments, votes, reports, and direct messages should return a confirmation response within 3 seconds after server validation completes, excluding third-party network delays.
- PR-4:** Time-consuming external tasks, including OTP email delivery and media upload processing, shall be handled asynchronously whenever possible so the user is not blocked longer than necessary.
- PR-5:** The system shall support concurrent use by end users, moderators, and administrators without data corruption in shared resources such as post counts, membership records, reports, notifications, and message channels.

### 4.2 Security Requirements

- SR-1:** All authenticated operations shall require valid access credentials, and protected endpoints shall reject requests with missing, expired, invalid, or revoked tokens.
- SR-2:** The system shall enforce role-based authorization for user, moderator, community owner, and administrator functions, including moderation queues, role assignment, report handling, and platform management.
- SR-3:** Sensitive account recovery flows shall use time-limited OTP values and verification tokens, and the system shall reject expired or mismatched OTP submissions.
- SR-4:** The system shall verify resource ownership before allowing updates or deletion of posts, comments, drafts, notifications, reports, channels, and user-specific settings.
- SR-5:** User passwords shall never be stored or transmitted in plain text, and password reset operations shall require successful OTP verification before a new password is accepted.
- SR-6:** Uploaded content such as avatars, covers, post images, and post videos shall be validated before storage to reduce the risk of invalid or malicious files entering the system.
- SR-7:** Communication between browser clients, frontend applications, backend services, and third-party identity providers shall be protected using secure transport protocols in production deployment.

## 4.3 Software Quality Attributes

- QA-1: Availability** – LKForum should remain available for continuous daily use, including account access, browsing, community interaction, moderation, and private messaging features.
- QA-2: Reliability** – The system shall preserve data consistency for community membership, moderation status, notifications, votes, reports, and message delivery even when operations are retried or partially fail.
- QA-3: Usability** – The user interface shall provide clear validation, confirmation, and error messages for common actions such as sign up, login, posting, reporting, moderation decisions, and password recovery.
- QA-4: Maintainability** – The software should remain modular across frontend, backend, and supporting services so that authentication, community management, moderation, messaging, and analytics can evolve independently.
- QA-5: Scalability** – The architecture should support growth in users, communities, posts, comments, notifications, and direct messages by relying on pagination, soft deletion, and external media storage services.
- QA-6: Compatibility** – The web application shall operate correctly on current versions of Chrome, Firefox, Safari, and Edge as defined in the operating environment.

## 5 Other Requirements

- OR-1:** The system shall integrate with third-party services required by release 1.0, including Google OAuth for social login and external email delivery for OTP-based verification and password recovery.
- OR-2:** Media assets uploaded by users shall be stored through an external media service so that application servers are not responsible for long-term binary file storage.
- OR-3:** The system shall maintain audit-relevant records for moderation actions, reports, bans, mutes, and administrative changes in accordance with the retention requirements defined in Section 3.5.
- OR-4:** The application shall continue to support both regular end-user workflows and administrative workflows through separate interfaces while using the same platform data and authorization model.
- OR-5:** The specification, user-facing messages, and operational behavior should remain consistent across future releases so business rules, activity diagrams, and test cases can be traced back to stable identifiers.

## 6 Appendix

### 6.1 Messages

| Message Code | Message Content                                                           | Button                         |
|--------------|---------------------------------------------------------------------------|--------------------------------|
| MSG-1        | Verification code sent to your email. Please check your inbox.            | –                              |
| MSG-2        | A new verification code has been sent to your email.                      | –                              |
| MSG-3        | Login successful                                                          | –                              |
| MSG-4        | Email verified successfully. You can now complete your registration.      | –                              |
| MSG-5        | Registration completed successfully. You are now logged in.               | –                              |
| MSG-6        | Password reset code sent to your email. Please check your inbox.          | –                              |
| MSG-7        | OTP verified successfully. You can now reset your password.               | –                              |
| MSG-8        | Password reset successfully. You can now login with your new password.    | –                              |
| MSG-9        | Mã OTP đã được gửi đến email của bạn!                                     | –                              |
| MSG-10       | Mã OTP mới đã được gửi đến email của bạn!                                 | –                              |
| MSG-11       | Xác thực email thành công!                                                | –                              |
| MSG-12       | Đăng ký thành công! Chào mừng bạn đến với LKForum.                        | –                              |
| MSG-13       | Đổi mật khẩu thành công! Vui lòng đăng nhập lại.                          | –                              |
| MSG-14       | Bạn có chắc chắn muốn xóa bài viết này? Hành động này không thể hoàn tác. | Xóa/Hủy                        |
| MSG-15       | Bạn có chắc muốn xóa bình luận này? Hành động này không thể hoàn tác.     | Xóa/Hủy                        |
| MSG-16       | Bạn có chắc muốn xóa bản nháp này?                                        | Xóa/Hủy                        |
| MSG-17       | Bạn có chắc muốn xóa người dùng này? Hành động này không thể hoàn tác.    | Xóa/Hủy                        |
| MSG-18       | Bạn có chắc muốn xóa báo cáo này?                                         | Xóa/Hủy                        |
| MSG-19       | Bạn có chắc chắn muốn rời khỏi lk/<community-name>?                       | Rời/Hủy                        |
| MSG-20       | Bạn có chắc muốn bỏ các thay đổi? Các thay đổi chưa lưu sẽ bị mất.        | Bỏ thay đổi/Tiếp tục chỉnh sửa |

| Message Code | Message Content                                | Button |
|--------------|------------------------------------------------|--------|
| MSG-21       | Đã sao chép liên kết!                          | —      |
| MSG-22       | Không thể sao chép liên kết                    | —      |
| MSG-23       | Đã gửi báo cáo thành công                      | —      |
| MSG-24       | Không thể gửi báo cáo. Vui lòng thử lại.       | —      |
| MSG-25       | Không thể gửi phiếu bầu. Vui lòng thử lại.     | —      |
| MSG-26       | Không thể hủy phiếu bầu. Vui lòng thử lại.     | —      |
| MSG-27       | Không thể xóa bài viết. Vui lòng thử lại.      | —      |
| MSG-28       | Không thể xóa bình luận. Vui lòng thử lại.     | —      |
| MSG-29       | Cập nhật bài viết thành công                   | —      |
| MSG-30       | Không thể cập nhật bài viết. Vui lòng thử lại. | —      |
| MSG-31       | Đã lưu bản nháp!                               | —      |
| MSG-32       | Không thể xóa bản nháp. Vui lòng thử lại.      | —      |
| MSG-33       | Đã cấm người dùng                              | —      |
| MSG-34       | Không thể cấm người dùng                       | —      |
| MSG-35       | Đã gỡ cấm người dùng                           | —      |
| MSG-36       | Không thể gỡ cấm người dùng                    | —      |
| MSG-37       | Đã xóa người dùng                              | —      |
| MSG-38       | Không thể xóa người dùng                       | —      |
| MSG-39       | Đã cấm cộng đồng                               | —      |
| MSG-40       | Không thể cấm cộng đồng                        | —      |
| MSG-41       | Đã gỡ cấm cộng đồng                            | —      |
| MSG-42       | Không thể gỡ cấm cộng đồng                     | —      |
| MSG-43       | Đã xóa báo cáo                                 | —      |
| MSG-44       | Không thể xóa báo cáo                          | —      |
| MSG-45       | Không thể gửi tin nhắn. Vui lòng thử lại.      | —      |
| MSG-46       | Bạn đã chấp nhận làm moderator!                | —      |
| MSG-47       | Không thể chấp nhận lời mời                    | —      |
| MSG-48       | Đã cập nhật hồ sơ!                             | —      |

| Message Code | Message Content                                                            | Button  |
|--------------|----------------------------------------------------------------------------|---------|
| MSG-49       | Không thể cập nhật hồ sơ.                                                  | —       |
| MSG-50       | Không thể đổi mật khẩu.                                                    | —       |
| MSG-51       | Đã lưu                                                                     | —       |
| MSG-52       | Không thể lưu cài đặt.                                                     | —       |
| MSG-53       | Đã tải ảnh đại diện lên!                                                   | —       |
| MSG-54       | Không thể tải ảnh đại diện.                                                | —       |
| MSG-55       | Đã xóa ảnh đại diện!                                                       | —       |
| MSG-56       | Không thể xóa ảnh đại diện.                                                | —       |
| MSG-57       | Bạn có chắc muốn xóa ảnh đại diện?                                         | Xóa/Hủy |
| MSG-58       | Không thể tải cộng đồng                                                    | —       |
| MSG-59       | Đã tham gia cộng đồng <community-name>!                                    | —       |
| MSG-60       | Đã thêm <username> làm quản trị viên!                                      | —       |
| MSG-61       | Không thể thêm quản trị viên. Vui lòng kiểm tra tên người dùng và thử lại. | —       |
| MSG-62       | Đã tạo cộng đồng "lk/<community-name>" thành công!                         | —       |
| MSG-63       | Không thể tạo cộng đồng. Vui lòng thử lại.                                 | —       |
| MSG-64       | Không thể đăng bình luận. Vui lòng thử lại.                                | —       |
| MSG-65       | Không thể gửi trả lời. Vui lòng thử lại.                                   | —       |
| MSG-66       | Post approved successfully!                                                | —       |
| MSG-67       | Failed to approve post. Please try again.                                  | —       |
| MSG-68       | User banned successfully!                                                  | —       |
| MSG-69       | Đã tắt tiếng người dùng thành công!                                        | —       |
| MSG-70       | Đã bỏ cấm người dùng thành công!                                           | —       |
| MSG-71       | Không thể bỏ cấm người dùng. Vui lòng thử lại.                             | —       |
| MSG-72       | Đã bỏ tắt tiếng người dùng thành công!                                     | —       |
| MSG-73       | Không thể bỏ tắt tiếng người dùng. Vui lòng thử lại.                       | —       |
| MSG-74       | Đã lưu cài đặt thành công!                                                 | —       |
| MSG-75       | Không thể lưu cài đặt                                                      | —       |

| Message Code | Message Content                                  | Button |
|--------------|--------------------------------------------------|--------|
| MSG-76       | Không thể tải danh sách báo cáo                  | –      |
| MSG-77       | Không thể tải danh sách người dùng               | –      |
| MSG-78       | Không thể tải danh sách cộng đồng                | –      |
| MSG-79       | Không thể tải thống kê                           | –      |
| MSG-80       | Không thể tải thông báo                          | –      |
| MSG-81       | Failed to create channel: <error>                | –      |
| MSG-82       | Không thể mở trò chuyện. Vui lòng thử lại.       | –      |
| MSG-83       | Không thể tải ảnh đại diện. Vui lòng thử lại.    | –      |
| MSG-84       | Không thể tải bài viết. Vui lòng thử lại.        | –      |
| MSG-85       | Không thể tải bài viết đã lưu. Vui lòng thử lại. | –      |
| MSG-86       | Failed to load draft. Please try again.          | –      |
| MSG-87       | Đã xóa bài viết thành công                       | –      |

Table 6.1: Application Messages

## 6.2 Error Codes

| Error ID | Code                     | Message                                     |
|----------|--------------------------|---------------------------------------------|
| ERR-1    | INVALID_CREDENTIALS      | Email hoặc mật khẩu không đúng              |
| ERR-2    | INVALID_TOKEN            | Token không hợp lệ hoặc đã hết hạn          |
| ERR-3    | INVALID_CLAIMS           | Thông tin token không hợp lệ                |
| ERR-4    | INVALID_ISSUER           | Nguồn phát hành token không hợp lệ          |
| ERR-5    | INVALID_AUDIENCE         | Đối tượng token không hợp lệ                |
| ERR-6    | TOKEN_INVALIDATED        | Token đã bị vô hiệu hóa                     |
| ERR-7    | MISSING_AUTH_-<br>HEADER | Thiếu Authorization header                  |
| ERR-8    | INVALID_AUTH_-<br>HEADER | Định dạng Authorization header không hợp lệ |
| ERR-9    | MISSING_TOKEN            | Thiếu token xác thực                        |
| ERR-10   | NOT_AUTHENTICATED        | Chưa xác thực                               |

| Error ID | Code                        | Message                                                                                                    |
|----------|-----------------------------|------------------------------------------------------------------------------------------------------------|
| ERR-11   | INVALID_AUTH_-<br>CONTEXT   | Ngữ cảnh xác thực không hợp lệ                                                                             |
| ERR-12   | ADMIN_ACCESS_-<br>REQUIRED  | Cần quyền quản trị viên                                                                                    |
| ERR-13   | FORBIDDEN                   | Bạn không có quyền thực hiện hành động này                                                                 |
| ERR-14   | BAD_REQUEST                 | Yêu cầu không hợp lệ                                                                                       |
| ERR-15   | EMAIL_NOT_VERIFIED          | Email chưa được xác thực                                                                                   |
| ERR-16   | EMAIL_ALREADY_-<br>VERIFIED | Email đã được xác thực                                                                                     |
| ERR-17   | INVALID_OTP                 | Mã xác thực không đúng                                                                                     |
| ERR-18   | OTP_EXPIRED                 | Mã xác thực đã hết hạn                                                                                     |
| ERR-19   | LOGIN_METHOD_-<br>MISMATCH  | Tài khoản này đã đăng ký bằng Google. Vui lòng đăng nhập bằng Google hoặc liên hệ quản trị viên để hỗ trợ. |
| ERR-20   | EMAIL_NOT_-<br>REGISTERED   | Email không tồn tại trong hệ thống                                                                         |
| ERR-21   | INVALID_USERNAME            | Username phải từ 3-20 ký tự                                                                                |
| ERR-22   | PASSWORD_TOO_SHORT          | Mật khẩu phải có ít nhất 8 ký tự                                                                           |
| ERR-23   | PASSWORD_TOO_WEAK           | Mật khẩu phải chứa ít nhất 1 chữ hoa, 1 chữ thường, 1 số và 1 ký tự đặc biệt (@\$!*%?&)                    |
| ERR-24   | INTERNAL_ERROR              | Lỗi hệ thống                                                                                               |
| ERR-25   | NO_FIELDS_TO_-<br>UPDATE    | Không có trường nào để cập nhật                                                                            |
| ERR-26   | INVALID_ID                  | Định dạng ID không hợp lệ                                                                                  |
| ERR-27   | PAGINATION_INVALID          | Số trang hoặc kích thước trang không hợp lệ. Kích thước trang phải nhỏ hơn 500.                            |
| ERR-28   | USER_NOT_FOUND              | Không tìm thấy người dùng                                                                                  |
| ERR-29   | USERNAME_EXISTS             | Tên người dùng đã tồn tại                                                                                  |
| ERR-30   | EMAIL_EXISTS                | Email đã được sử dụng                                                                                      |
| ERR-31   | USER_INACTIVE               | Tài khoản người dùng đã bị vô hiệu hóa                                                                     |
| ERR-32   | INVALID_GENDER              | Giá trị giới tính không hợp lệ                                                                             |
| ERR-33   | INVALID_DATE_-<br>FORMAT    | Định dạng ngày không hợp lệ, sử dụng YYYY-MM-DD                                                            |
| ERR-34   | AGE_TOO_YOUNG               | Phải từ 13 tuổi trở lên                                                                                    |
| ERR-35   | INVALID_BIRTH_DATE          | Ngày sinh không hợp lệ                                                                                     |
| ERR-36   | INVALID_PROVINCE            | Tỉnh/thành phố không hợp lệ                                                                                |



| Error ID | Code                         | Message                                                     |
|----------|------------------------------|-------------------------------------------------------------|
| ERR-37   | TOO_MANY_INTERESTS           | Tối đa 10 sở thích                                          |
| ERR-38   | INVALID_INTEREST             | Sở thích không hợp lệ                                       |
| ERR-39   | COMMUNITY_NOT_FOUND          | Không tìm thấy cộng đồng                                    |
| ERR-40   | COMMUNITY_NAME_EXISTS        | Tên cộng đồng đã tồn tại                                    |
| ERR-41   | COMMUNITY_DELETED            | Cộng đồng đã bị xóa                                         |
| ERR-42   | MODERATOR_ALREADY_EXISTS     | Người dùng đã là moderator của cộng đồng                    |
| ERR-43   | CANNOT_REMOVE_MODERATOR      | Không thể xóa moderator này                                 |
| ERR-44   | CANNOT_REMOVE_CREATOR        | Không thể xóa người tạo cộng đồng khỏi danh sách moderator  |
| ERR-45   | BANNED_COMMUNITY             | Người dùng đã bị ban khỏi cộng đồng                         |
| ERR-46   | USER_NOT_MEMBER              | Bạn chưa tham gia cộng đồng này                             |
| ERR-47   | MEMBERSHIP_NOT_FOUND         | Không tìm thấy thành viên                                   |
| ERR-48   | ALREADY_MEMBER               | Bạn đã là thành viên của cộng đồng này                      |
| ERR-49   | MEMBERSHIP_CREATE_FAILED     | Không thể tạo thành viên                                    |
| ERR-50   | MEMBERSHIP_DELETE_FAILED     | Không thể xóa thành viên                                    |
| ERR-51   | INVALID_MEMBERSHIP_DATA      | Dữ liệu thành viên không hợp lệ                             |
| ERR-52   | MEMBERSHIP_ALREADY_PROCESSED | Yêu cầu tham gia đã được xử lý                              |
| ERR-53   | POST_NOT_FOUND               | Không tìm thấy bài viết                                     |
| ERR-54   | VOTE_NOT_FOUND               | Không tìm thấy bình chọn                                    |
| ERR-55   | POLL_ALREADY_VOTED           | Bạn đã bình chọn lựa chọn này rồi                           |
| ERR-56   | POLL_CANNOT_EDIT             | Không thể chỉnh sửa bình chọn sau khi đã có người bình chọn |
| ERR-57   | ALREADY_REPORTED             | Bạn đã báo cáo nội dung này rồi                             |
| ERR-58   | DRAFT_NOT_FOUND              | Không tìm thấy bản nháp                                     |
| ERR-59   | COMMENT_NOT_FOUND            | Không tìm thấy bình luận                                    |
| ERR-60   | DEPTH_TOO_HIGH               | Độ sâu phải từ 0 đến 2                                      |
| ERR-61   | USER_MUTED                   | Người dùng đã bị cấm bình luận                              |
| ERR-62   | CHANNEL_NOT_FOUND            | Không tìm thấy kênh                                         |

| Error ID | Code             | Message                 |
|----------|------------------|-------------------------|
| ERR-63   | NO_MESSAGE_FOUND | Không tìm thấy tin nhắn |
| ERR-64   | REPORT_NOT_FOUND | Không tìm thấy report   |

Table 6.2: Application Error Codes